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**Data-Driven Simulation of Financial Scenarios:
The Adaptation of GANs to Overcome the Heavy-Tail Problem**

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塵も積もれば山となる

Chiri mo tsumoreba yama to naru

“Even specks of dust, when piled up, become a mountain”

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Introduction

The accurate simulation of temporal scenarios represents a foundational challenge across applied mathematics, quantitative finance, risk management, and insurance. The mathematical formulations developed for financial markets naturally extend to other critical domains characterized by complex dynamical systems and high-impact anomalies, such as seismology, climatology, and quantitative epidemiology. In each of these settings, the ultimate objective is identical: generating synthetic yet statistically authentic trajectories capable of stress-testing systems against catastrophic losses or systemic failures.

Historically, scenario generation has relied on traditional econometrics and classical stochastic calculus, including geometric Brownian motions, jump diffusion processes, and parametric autoregressive models (such as ARCH/GARCH families). However, these conventional frameworks inherently depend on rigid distributional assumptions, most notably the hypothesis of normality or bounded variance. When confronted with real-world complexities, these parametric simplifications have repeatedly generated severe modeling misspecifications. A prominent historical illustration is the 2008 global financial crisis, where standard risk and pricing engines systematically underestimated tail correlations and extreme co-movements, treating severe market contractions as negligible statistical impossibilities.

Indeed, modeling financial time series is an exceptionally arduous task. Empirical returns exhibit well-documented non-stationary properties and universal regularities known as *stylized facts*.

The advent of modern, data-driven methodology marks a significant paradigm shift. By learning the underlying probability distributions directly from historical trajectories without restrictive structural priors, non-parametric machine learning frameworks have demonstrated the capacity to reproduce these intricate stylized facts with unprecedented fidelity, thereby reducing model risk and providing quantitative practitioners with far more reliable decision-making tools.

Nevertheless, reproducing extreme tail scenarios, the low-frequency and high-impact events situated in the deep distribution margins far from the mean, remains the most persistent and mathematically intricate open challenge. This fundamental limitation is referred to throughout this work as the **heavy-tail problem**. Because tail risk governs the catastrophic breakdown of physical, environmental, and financial architectures alike, resolving this distributional truncation transcends academic curiosity: in natural hazard forecasting, reliable extreme-scenario modeling directly informs disaster mitigation and public safety, while in capital markets, it safeguards institutional solvency and prevents broader systemic contagion.

Within this conceptual framework, this thesis investigates one of the most transformative advances in contemporary deep learning: the Generative Adversarial Network (GAN) paradigm. Methodologically, the work unfolds as an evolutionary, step-by-step exploration of how adversarial architectures have been progressively adapted, regularized, and structurally redesigned to confront and resolve the heavy-tail problem in quantitative finance.

The first chapter establishes the theoretical groundwork of generative modeling, beginning with the original minimax formulation. It examines the pathological failure modes inherent to vanilla GANs; principally vanishing gradients, training instability, and severe mode collapse driven by the properties of the Jensen-Shannon divergence. To resolve these convergence dilemmas, the discussion transitions toward optimal transport principles, detailing the Kantorovich-Rubinstein duality and the implementation of the Wasserstein

GAN with Gradient Penalty (WGAN-GP) as a mathematically principled approach to enforce the required 1-Lipschitz continuity.

Building upon these foundational principles, the second chapter focuses on domain-specific applications by deploying baseline GAN architectures on historical financial market series. It provides an initial empirical demonstration of how standard generative networks replicate classical stylized facts, including volatility clustering, non-linear dependencies, and fat tails. At the same time, this chapter delineates the operational boundaries of early models, illustrating how unspecialized networks inevitably begin to attenuate or distort extreme quantile behavior as scenario complexity increases.

The third chapter introduces and formalizes the central challenge of this research: the systematic underestimation of deep tail risk and empirical fourth moments in generative models. It examines three prominent architectures from the literature designed to tackle this pathology. This analysis explores both univariate dynamics and higher-dimensional multi-asset settings, shedding light on how explicit tail loss functions and moment penalties attempt to alleviate Gaussian mode collapse.

Finally, the fourth chapter presents an extensive implementation and empirical benchmarking of these architectures to identify the definitive framework for extreme scenario generation. The investigation demonstrates that the superior solution does not lie in conventional joint neural networks, but rather in a novel hybrid synthesis: the integration of Temporal Convolutional Networks within a WGAN-GP framework augmented by Extreme Value Theory. At the univariate level, this architecture captures temporal memory and tail asymptotes with minimal error. Extending this methodology to the multivariate regime does not require computationally expensive joint adversarial training; by coupling the calibrated univariate margins through an empirical copula, the resulting portfolio model accurately reproduces joint tail risk, excess kurtosis, and cross-asset rank dependencies within a fraction of the computational runtime.

Ultimately, this work aims to demonstrate that addressing the heavy-tail problem does not necessarily require increasingly complex models. By combining modern generative architectures with classical mathematical tools, it is possible to construct a simulation framework that is computationally efficient, transparent, and accessible. In doing so, we show that data-driven scenario simulation can move beyond theoretical experimentation, providing a reliable and practical methodology for understanding extreme events and managing risk in real-world applications.

Chapter 1

Useful architectures

The promise of deep learning lies in discovering rich, hierarchical models that represent probability distributions over data encountered in applications such as image detection and financial modeling.

Historically, discriminative models have been the most successful, largely due to the backpropagation algorithm, dropout regularization, and the use of piecewise linear activation functions that possess well-behaved gradients. In contrast, generative models have struggled to benefit from these advancements, facing significant challenges in approximating the intractable probabilistic computations that arise in maximum likelihood estimation.

The computational inefficiency associated with the need to approximate probability distributions or rely on Markov chains remained a significant limitation for these models until the introduction of Generative Adversarial Networks (GANs) in 2014 by Goodfellow et al[6].

1.1 GAN

The main idea behind *Generative Adversarial Networks* is to pit a generative model against an adversary: a discriminative model that learns to determine whether a given sample is drawn from the generative model or from the true data distribution.

This framework can be viewed as a game where the generative model acts as a counterfeiter, trying to produce fake currency and use it without detection, while the discriminative model acts as the police, trying to detect the counterfeit currency. The competition in this game drives both models to continuously improve their methods until the counterfeits become indistinguishable from the genuine articles.

In game theory, this type of interaction is formulated as a minimax two-player game, which allows us to establish its formal mathematical definition.

1.1.1 Definition and first observations

Both the generative model G and the discriminative model D are *multilayer perceptrons*. To learn the generator's distribution p_g over data $\mathbf{x}$, a prior distribution on the input noise variables $p_z(\mathbf{z})$ is first defined. Then, a mapping to the data space is represented as $G(\mathbf{z}; \theta_g)$, where G is a differentiable function defined by a multilayer perceptron with parameters θ_g . Additionally, a second multilayer perceptron $D(\mathbf{x}; \theta_d)$ is defined to output a single scalar. The value $D(\mathbf{x})$ represents the probability that $\mathbf{x}$ came from the data distribution rather than from p_g . The models D and G are trained simultaneously to play the following two-player minimax game, governed by the value function $V(D, G)$ defined below:

Definition 1.1.1 (GAN - Objective function).

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]$$

From a mathematical perspective, this objective function works because it frames the learning process as a two-player zero-sum game whose equilibrium corresponds to matching the data distribution. The first term, $\mathbb{E}_{\mathbf{x}}[\log D(\mathbf{x})]$, acts as a cross-entropy loss that forces the discriminator to assign high probabilities to real data, while the second term, $\mathbb{E}_{\mathbf{z}}[\log(1 - D(G(\mathbf{z})))]$, drives it to classify synthetic samples as fake. Conversely, the generator seeks to minimize this second term to produce samples that fool D . In the non-parametric limit, maximizing over D yields an optimal critic $D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$, which effectively measures the density ratio between real and generated data. Substituting D_G^* into the objective turns the problem for G into minimizing the Jensen-Shannon divergence, which achieves its unique global minimum if and only if the generator perfectly recovers the true data distribution ($p_g = p_{\text{data}}$).

In practice, the game must be implemented using an iterative numerical approach. Optimizing D to completion within the inner loop of training is computationally prohibitive and would lead to overfitting on finite datasets. Instead, the training process alternates between k steps of optimizing D and one step of optimizing G . This strategy ensures that D is maintained near its optimal solution, provided that G changes slowly enough.

Furthermore, 1.1.1 may not provide a sufficient gradient for G to learn effectively. Early in the learning process, D can reject generated samples with high confidence because they are clearly distinct from the true training data. Consequently, the gradient of the term $\log(1 - D(G(\mathbf{z})))$ tends to zero. Rather than training G to minimize $\log(1 - D(G(\mathbf{z})))$ the generator could be trained to maximize $\log D(G(\mathbf{z}))$. This alternative objective function yields the same fixed point for the dynamics while providing much stronger gradients during the early stages of training.

This saturation issue represents one of the primary challenges addressed by subsequent generations of GANs. The complete training procedure is outlined in the algorithm below:

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets.

- 1: **for** number of training iterations **do**
- 2: **for** k steps **do**
- 3: Sample minibatch of m noise samples $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$ from noise prior $p_z(\mathbf{z})$.
- 4: Sample minibatch of m examples $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\mathbf{x})$.
- 5: Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(\mathbf{x}^{(i)}) + \log(1 - D(G(\mathbf{z}^{(i)}))) \right]$$

- 6: **end for**
- 7: Sample minibatch of m noise samples $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$ from noise prior $p_z(\mathbf{z})$.
- 8: Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log(1 - D(G(\mathbf{z}^{(i)})))$$

- 9: **end for**
-

For gradient-based updates, any standard learning rule can be applied. The number of optimization steps, k , applied to the discriminator is a hyperparameter. It is typically set to 1, as this represents the least computationally expensive option.

1.1.2 Theoretical results

The generator G implicitly defines a probability distribution p_g as the distribution of the samples $G(z)$ obtained when $z \sim p_z$. The objective of algorithm 1 is to converge to a reliable estimator of p_{data} . To represent a model with infinite capacity, convergence is studied directly within the space of probability density functions. This approach yields several important theoretical results.

Before presenting the theoretical results is necessary to remark on the definitions of Kullback–Leibler (KL) and Jensen–Shannon (JS) divergences.

Let X be a compact metric set and let Σ denote the set of all the Borel subsets of X . Let $\text{Prob}(X)$ denote the space of probability measures defined on X . The divergences between two distributions $\mathbb{P}_r, \mathbb{P}_g \in \text{Prob}(X)$ can be defined as follow.

Definition 1.1.2 (Kullback–Leibler divergence). *$\mathbb{P}_r$ and $\mathbb{P}_g$ are assumed to be absolutely continuous, and therefore admit densities P_r and P_g , with respect to the same measure μ defined on X , the KL divergence from $\mathbb{P}_r$ to $\mathbb{P}_g$ is:*

$$KL(\mathbb{P}_r \parallel \mathbb{P}_g) = \int \log \left(\frac{P_r(x)}{P_g(x)} \right) P_r(x) d\mu(x),$$

The KL divergence is asymmetric and possibly infinite when there are points such that $P_g(x) = 0$ and $P_r(x) > 0$.

Definition 1.1.3 (Jensen–Shannon divergence). *Consider $\mathbb{P}_m$ as the mixture $(\mathbb{P}_r + \mathbb{P}_g)/2$ and the JS divergence is:*

$$JS(\mathbb{P}_r, \mathbb{P}_g) = KL(\mathbb{P}_r \parallel \mathbb{P}_m) + KL(\mathbb{P}_g \parallel \mathbb{P}_m),$$

The reason the Jensen–Shannon divergence is always well-defined (and finite) lies in the absolute continuity of both distributions with respect to the mixture measure $\mathbb{P}_m = \frac{\mathbb{P}_r + \mathbb{P}_g}{2}$. Unlike the standard Kullback–Leibler divergence $KL(\mathbb{P}_r \parallel \mathbb{P}_g)$, which diverges to $+\infty$ whenever $\mathbb{P}_r$ is not absolutely continuous with respect to $\mathbb{P}_g$ (e.g., when their supports do not fully overlap), both $\mathbb{P}_r$ and $\mathbb{P}_g$ are naturally absolutely continuous with respect to $\mathbb{P}_m$ ($\mathbb{P}_r \ll \mathbb{P}_m$ and $\mathbb{P}_g \ll \mathbb{P}_m$). Indeed, $\mathbb{P}_m(A) = 0$ directly implies $\mathbb{P}_r(A) = 0$ and $\mathbb{P}_g(A) = 0$ for any measurable set A . Consequently, by taking $\mu = \mathbb{P}_m$ as the reference dominating measure, the Radon–Nikodym derivatives $\frac{d\mathbb{P}_r}{d\mathbb{P}_m}$ and $\frac{d\mathbb{P}_g}{d\mathbb{P}_m}$ always exist and are bounded from above by 2. This ensures that both $KL(\mathbb{P}_r \parallel \mathbb{P}_m)$ and $KL(\mathbb{P}_g \parallel \mathbb{P}_m)$ are always finite, guaranteeing that $0 \leq JS(\mathbb{P}_r, \mathbb{P}_g) \leq \log 2$ even when the supports of $\mathbb{P}_r$ and $\mathbb{P}_g$ are disjoint. The theoretical analysis begins by establishing the optimal discriminator D for any given generator G .

Theorem 1.1.1. *For G fixed, the optimal discriminator D is*

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \quad (1.1)$$

Proof. The training criterion for the discriminator D , given any fixed generator G , is to maximize the functional $V(G, D)$:

$$V(G, D) = \int_{\mathcal{X}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathcal{Z}} p_z(z) \log(1 - D(G(z))) dz. \quad (1.2)$$

Here, the two integrals are defined over distinct domains:

- $\mathcal{Z} \subseteq \mathbb{R}^{d_z}$ denotes the latent space governed by the known, tractable noise distribution p_z ;
- $\mathcal{X} \subseteq \mathbb{R}^d$ denotes the data space, where $\text{Supp}(p_{\text{data}}) \subset \mathcal{X}$ contains the real data trajectories.

The generative network $G : \mathcal{Z} \rightarrow \mathcal{X}$ defines a measurable mapping that pushes forward the latent probability measure p_z onto the data space $\mathcal{X}$, yielding the induced generative distribution $p_g = G_{\#}p_z$. Applying the change of variables theorem (or the law of the unconscious statistician), the expectation over the latent variable z in (1.2) can be expressed directly as an integral over the data space $\mathcal{X}$ under the induced density p_g :

$$\int_{\mathcal{Z}} p_z(z) \log(1 - D(G(z))) dz = \int_{\mathcal{X}} p_g(x) \log(1 - D(x)) dx. \quad (1.3)$$

Consequently, both expectations can be combined under a single integral over $\mathcal{X}$:

$$V(G, D) = \int_{\mathcal{X}} [p_{\text{data}}(x) \log(D(x)) + p_g(x) \log(1 - D(x))] dx. \quad (1.4)$$

Notice that the integrand vanishes almost everywhere outside the union of the supports $\mathcal{X}_{\text{supp}} := \text{Supp}(p_{\text{data}}) \cup \text{Supp}(p_g) \subseteq \mathcal{X}$. Thus, the value of $D(x)$ outside $\mathcal{X}_{\text{supp}}$ does not affect $V(G, D)$.

For any fixed $x \in \mathcal{X}_{\text{supp}}$, maximizing the integral with respect to D reduces to maximizing the integrand pointwise. Setting $a = p_{\text{data}}(x)$, $b = p_g(x)$, and $y = D(x) \in (0, 1)$, we consider the scalar objective:

$$f(y) = a \log(y) + b \log(1 - y).$$

Since $(a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, f is strictly concave on $(0, 1)$. Taking the first derivative:

$$f'(y) = \frac{a}{y} - \frac{b}{1-y} = 0 \implies a(1-y) = by \implies y^* = \frac{a}{a+b}.$$

Because $f''(y) = -\frac{a}{y^2} - \frac{b}{(1-y)^2} < 0$, this stationary point is the unique global maximum. We conclude that for any given generator G , the optimal discriminator is uniquely determined on $\mathcal{X}_{\text{supp}}$ by:

$$D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}, \quad (1.5)$$

which concludes the proof. $\square$

It should be noted that the training objective for D can be interpreted as maximizing the log-likelihood for estimating the conditional probability $P(Y = y|x)$. Here, the binary variable Y indicates whether the sample x originates from p_{data} (with $y = 1$) or from p_g (with $y = 0$). Consequently, the minimax game introduced in definition 1.1.1 can be reformulated as follows:

$$\begin{aligned} C(G) &= \max_D V(G, D) \\ &= \mathbb{E}_{x \sim p_{\text{data}}} [\log D_G^*(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D_G^*(G(z)))] \\ &= \mathbb{E}_{x \sim p_{\text{data}}} [\log D_G^*(x)] + \mathbb{E}_{x \sim p_g} [\log(1 - D_G^*(x))] \\ &= \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)} \right] + \mathbb{E}_{x \sim p_g} \left[\log \frac{p_g(x)}{p_{\text{data}}(x) + p_g(x)} \right] \end{aligned} \quad (1.6)$$

Theorem 1.1.2. *The global minimum of the virtual training criterion $C(G)$ is achieved if and only if $p_g = p_{\text{data}}$. At that point, $C(G)$ achieves the value $-\log 4$.*

Proof. For $p_g = p_{\text{data}}$, $D_G^*(x) = \frac{1}{2}$, (consider Eq.1.1). Hence, by inspecting Eq. 1.6 at $D_G^*(x) = \frac{1}{2}$, we find $C(G) = \log \frac{1}{2} + \log \frac{1}{2} = -\log 4$. To see that this is the best possible value of $C(G)$, reached only for $p_g = p_{\text{data}}$, observe that

$$\mathbb{E}_{x \sim p_{\text{data}}} [-\log 2] + \mathbb{E}_{x \sim p_g} [-\log 2] = -\log 4$$

and that by subtracting this expression from $C(G) = V(D_G^*, G)$, we obtain:

$$C(G) = -\log(4) + KL\left(p_{\text{data}} \left\| \frac{p_{\text{data}} + p_g}{2} \right\| \right) + KL\left(p_g \left\| \frac{p_{\text{data}} + p_g}{2} \right\| \right) \quad (1.7)$$

where KL is the Kullback–Leibler divergence (def 1.1.2). In the previous expression is recognize the Jensen–Shannon divergence (def 1.1.3) between the model’s distribution and the data generating process:

$$C(G) = -\log(4) + 2 \cdot JS(p_{\text{data}} \parallel p_g) \quad (1.8)$$

Since the Jensen–Shannon divergence between two distributions is always non-negative and zero only when they are equal $C^* = -\log(4)$ is the global minimum of $C(G)$ and $p_g = p_{\text{data}}$ is the only solution. Consequently, the generative model perfectly replicates the data-generating process. $\square$

Finally, the convergence of Algorithm 1 can be analyzed.

Theorem 1.1.3. *If G and D have enough capacity, and at each step of Algorithm 1, the discriminator is allowed to reach its optimum given G , and p_g is updated so as to improve the criterion*

$$\mathbb{E}_{\mathbf{x} \sim p_{\text{data}}}[\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g}[\log(1 - D_G^*(\mathbf{x}))]$$

then p_g converges to p_{data} .

Proof. Consider $V(G, D) = U(p_g, D)$ as a function of p_g as done in the above criterion. Note that $U(p_g, D)$ is convex in p_g . The subderivatives of a supremum of convex functions include the derivative of the function at the point where the maximum is attained. In other words, if $f(x) = \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$ and $f_\alpha(x)$ is convex in x for every α , then $\partial f_\beta(x) \in \partial f$ if $\beta = \arg \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$. This is equivalent to computing a *gradient descent* update for p_g at the optimal D given the corresponding G . The $\sup_D U(p_g, D)$ is convex in p_g with a unique global optimum, as proven in Theorem 1.1.2, therefore with sufficiently small updates of p_g , p_g converges to p_{data} , concluding the proof. $\square$

In practice, GAN represent a limited family of probability distributions p_g via the parameterized mapping $G(\mathbf{z}; \theta_g)$. Consequently, the optimization process is performed better with respect to the parameter θ_g rather than the distribution p_g itself. Furthermore, the use of a multilayer perceptron to define G introduces multiple critical points in the parameter space, which complicates the theoretical analysis. Nevertheless, the excellent empirical performance of these networks justifies their application, despite the absence of strong theoretical guarantees for global convergence.

1.1.3 Advantages and disadvantages

Regarding the disadvantages, aside from the previously discussed saturation of $\log(1 - D(G(\mathbf{z})))$, several other limitations must be addressed. First, the representation of the distribution $p_g(\mathbf{x})$ isn’t explicit, which complicates its direct evaluation.

More critically, the training process requires careful synchronization between D and G . If the generator is trained extensively without adequately updating D , a phenomenon known as the “Helvetica scenario” (often also referred to as “mode collapse”) can occur. In this situation, G maps too many values of $\mathbf{z}$ to the same value of $\mathbf{x}$, thereby failing to capture sufficient diversity to accurately model p_{data} .

In contrast, the advantages of this framework are primarily computational. Markov chains are not required; gradients are obtained solely via the backpropagation algorithm, and no complex inference processes

are needed during training. Furthermore, GANs offer distinct statistical benefits: the generator is never updated directly using the training data examples, but only through gradients flowing from the discriminator. This characteristic allows the network to represent degenerate distributions that cannot be handled by traditional Markov chain-dependent methods.

Given these significant advantages, GAN prove to be highly effective across a wide range of practical scenarios, owing to their ability to approximate a vast array of complex functions. The subsequent chapter will explore the application of these generative models within the financial domain.

1.1.4 Conditioned models

In an unconditioned model, such as GAN, there is no explicit control over the modes of the generated data. However, directing the data generation process is often highly desirable. To achieve this level of control, conditioning the model becomes necessary. Consequently, introducing the formal definition of *Conditional Generative Adversarial Nets*, as introduced by Mirza and Osindero [10], is essential.

Generative Adversarial Networks can be extended to a conditional framework if both the generator and the discriminator are conditioned on some supplementary information, denoted as y . The variable y can represent any form of auxiliary information, such as class labels or specific data features. In practice, this conditioning is achieved by feeding y into both the discriminator and the generator as an additional input layer.

Within the generator, the prior input noise $p_z(z)$ and the conditional variable y are combined into a joint hidden representation. The adversarial training framework allows for considerable flexibility regarding how this hidden representation is constructed. While the conditioning input and the prior noise can simply serve as joint inputs to a single layer of a multilayer perceptron, more complex architectures may also employ higher order interactions.

Similarly, in the discriminator, both the data sample x and the conditioning variable y are provided as inputs to a discriminative function, which is again represented by a multilayer perceptron.

The objective function from Definition 1.1.1 can be rewritten as follows:

Definition 1.1.4 (CGAN - Objective function).

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x|y)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z|y)))]$$

The conditioning variable y can be either categorical, such as normal and stressed market periods in finance, or any form of continuous variable.

Consequently, CGANs prove to be highly effective in numerous applications that require conditional generation, as they retain all the inherent advantages, alongside the inevitable disadvantages, of standard GANs.

1.2 Wasserstein–GAN

As discussed in subsection 1.1.3, standard GANs suffer from several notable limitations, such as vanishing gradients and mode collapse. Arjovsky et al [1] introduced a solution in 2017.

These issues are primarily attributed to the use of the Kullback–Leibler (KL) and Jensen–Shannon (JS) divergences. Indeed, optimization under these metrics can be highly problematic, often leading to slow learning or severe difficulties in convergence under specific conditions.

Consequently, it is essential to identify a weaker distance metric that provides better convergence properties, thereby mitigating the fundamental flaws inherent in the original GAN formulation.

The objective of this section is to analyze various approaches for measuring the proximity between the generated model distribution and the true data distribution. Equivalently, it examines different mathematical formulations for defining a distance or divergence, denoted as $\rho(\mathbb{P}_\theta, \mathbb{P}_r)$, with the ultimate goal of identifying the most suitable metric for constructing a robust Generative Adversarial Network.

1.2.1 Wasserstein distance

Here, $\theta \in \mathbb{R}^p$ denotes the trainable parameter vector of the generator network characterizing the synthetic measure $\mathbb{P}_\theta$, while Σ represents the Borel σ -algebra of measurable subsets on the data space $\mathcal{X}$.

The most fundamental distinction among these statistical distances lies in their impact on the convergence properties of sequences of probability distributions.

Formally, a sequence of distributions $(\mathbb{P}_t)_{t \in N}$ converges if and only if there is a limit distribution $\mathbb{P}_\infty$ such that $\rho(\mathbb{P}_t, \mathbb{P}_\infty)$ tends to zero. This behavior strictly depends on how exactly the distance ρ is defined. Informally, a distance ρ induces a weaker topology when it makes it easier for a sequence of distributions to converge.

To effectively optimize the parameter θ , it is desirable to define the model distribution $\mathbb{P}_\theta$ such that the mapping $\theta \rightarrow \mathbb{P}_\theta$ is continuous. Indeed, by continuity when a sequence of parameters θ_t converges to θ also the distributions $\mathbb{P}_{\theta_t}$ converges to $\mathbb{P}_\theta$.

The main reason for requiring this continuity is that, given a distance ρ between distributions, it is essential for the loss function $\theta \mapsto \rho(\mathbb{P}_\theta, \mathbb{P}_r)$ to be continuous. Equivalently, the mapping $\theta \mapsto \mathbb{P}_\theta$ must be continuous with respect to the chosen distance ρ .

The weaker the chosen distance metric, the easier it becomes to define a continuous mapping $\theta \mapsto \mathbb{P}_\theta$ and consequently for the distributions to converge.

Now it is necessary to introduce the most used distances and divergences between two distributions $\mathbb{P}_r, \mathbb{P}_g \in \text{Prob}(X)$. In the previous section, Kullback–Leibler 1.1.2 and Jensen–Shannon 1.1.3 divergences have already been defined.

Definition 1.2.1 (Total Variation). *The TV distance is defined as:*

$$\delta(\mathbb{P}_r, \mathbb{P}_g) = \sup_{A \in \Sigma} |\mathbb{P}_r(A) - \mathbb{P}_g(A)|.$$

Definition 1.2.2 (Wasserstein–1 distance). *The Earth–Mover (EM) distance or Wasserstein–1 is defined as:*

$$W(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|]$$

where $\Pi(\mathbb{P}_r, \mathbb{P}_g)$ denotes the set of all joint distributions $\gamma(x, y)$ whose marginals are respectively $\mathbb{P}_r$ and $\mathbb{P}_g$.

Intuitively, the Earth Mover’s (EM) distance models probability distributions as piles of physical mass distributed over the space $\mathcal{X}$. In this analogy, transforming the real distribution $\mathbb{P}_r$ into the generated distribution $\mathbb{P}_g$ requires moving mass from locations $x \sim \mathbb{P}_r$ to target locations $y \sim \mathbb{P}_g$.

A transport plan $\gamma(x, y) \in \Pi(\mathbb{P}_r, \mathbb{P}_g)$ represents a joint probability distribution whose marginals coincide with $\mathbb{P}_r$ and $\mathbb{P}_g$, respectively:

$$\int_{\mathcal{X}} \gamma(x, y) dy = \mathbb{P}_r(x) \quad \text{and} \quad \int_{\mathcal{X}} \gamma(x, y) dx = \mathbb{P}_g(y).$$

Under this constraint, $\gamma(x, y)$ explicitly quantifies the infinitesimal fraction of mass transported from position x to position y . Moving an amount of mass $\gamma(x, y)$ across a metric distance $\|x - y\|$ incurs a transport work proportional to $\|x - y\| d\gamma(x, y)$.

The total operational cost of a given transport plan γ is obtained by integrating this quantity over all pairs $(x, y) \in \mathcal{X} \times \mathcal{X}$, yielding $\mathbb{E}_{(x,y) \sim \gamma}[\|x - y\|]$. The Wasserstein distance $W_1(\mathbb{P}_r, \mathbb{P}_g)$ is then defined as the infimum over all admissible transport plans:

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma}[\|x - y\|].$$

It corresponds to the minimum possible cost required to morph $\mathbb{P}_r$ into $\mathbb{P}_g$ under the most efficient transport schedule. Intuitively, $\gamma(x, y)$ indicates how much "mass" must be transported from x to y in order to transform the distribution $\mathbb{P}_r$ into the distribution $\mathbb{P}_g$.

It is now pertinent to determine which of the aforementioned distances or divergences is the weakest.

The following example illustrates how apparently simple sequences of probability distributions converge under the EM distance but do not converge under the other distances and divergences defined above.

Example. Let $Z \sim U[0, 1]$ be the uniform distribution on the unit interval. Let $\mathbb{P}_0$ be the distribution of $(0, Z) \in \mathbb{R}^2$ (a 0 on the x-axis and the random variable Z on the y-axis), uniform on a straight vertical line passing through the origin. Now, let $g_\theta(z) = (\theta, z)$ with θ a single real parameter. Below is the proof in detail.

- $W(\mathbb{P}_0, \mathbb{P}_\theta) = |\theta|,$
- $JS(\mathbb{P}_0, \mathbb{P}_\theta) = \begin{cases} \log 2 & \text{if } \theta \neq 0, \\ 0 & \text{if } \theta = 0, \end{cases}$
- $KL(\mathbb{P}_\theta \parallel \mathbb{P}_0) = KL(\mathbb{P}_0 \parallel \mathbb{P}_\theta) = \begin{cases} +\infty & \text{if } \theta \neq 0, \\ 0 & \text{if } \theta = 0, \end{cases}$
- and $\delta(\mathbb{P}_0, \mathbb{P}_\theta) = \begin{cases} 1 & \text{if } \theta \neq 0, \\ 0 & \text{if } \theta = 0. \end{cases}$

Derivation of the distances and divergences:

- **Wasserstein distance (W_1):** Let $X = (0, Z) \sim \mathbb{P}_0$ and $Y = (\theta, Z') \sim \mathbb{P}_\theta$, where $Z, Z' \sim \mathcal{U}[0, 1]$. By definition,

$$W_1(\mathbb{P}_0, \mathbb{P}_\theta) = \inf_{\gamma \in \Pi(\mathbb{P}_0, \mathbb{P}_\theta)} \mathbb{E}_{(X,Y) \sim \gamma}[\|X - Y\|].$$

For any realization of $X = (0, z_1)$ and $Y = (\theta, z_2)$, the Euclidean distance in $\mathbb{R}^2$ is bounded from below by the horizontal separation:

$$\|X - Y\| = \sqrt{\theta^2 + (z_1 - z_2)^2} \geq \sqrt{\theta^2} = |\theta|.$$

Taking the expectation under any coupling γ yields:

$$\mathbb{E}_{(X,Y) \sim \gamma}[\|X - Y\|] \geq |\theta| \implies \inf_{\gamma \in \Pi} \mathbb{E}_{(X,Y) \sim \gamma}[\|X - Y\|] \geq |\theta|.$$

Now consider the deterministic coupling γ^* that sets $Z' = Z$ almost surely, corresponding to the joint distribution of $((0, Z), (\theta, Z))$. Under γ^* , the vertical coordinates match exactly ($z_1 = z_2 = Z$), hence:

$$\mathbb{E}_{(X,Y) \sim \gamma^*}[\|X - Y\|] = \mathbb{E}_{Z \sim \mathcal{U}[0,1]} \left[\sqrt{\theta^2 + (Z - Z)^2} \right] = |\theta|.$$

Since the lower bound is attained by $\gamma^* \in \Pi(\mathbb{P}_0, \mathbb{P}_\theta)$, we conclude:

$$W_1(\mathbb{P}_0, \mathbb{P}_\theta) = |\theta|.$$

- **Kullback–Leibler divergence (KL):** If $\theta = 0$, the two distributions coincide ($\mathbb{P}_0 = \mathbb{P}_\theta$), so $\text{KL}(\mathbb{P}_0 \parallel \mathbb{P}_0) = 0$. For $\theta \neq 0$, the supports of the two distributions are distinct parallel vertical segments in $\mathbb{R}^2$:

$$\text{Supp}(\mathbb{P}_0) = \{0\} \times [0, 1] \quad \text{and} \quad \text{Supp}(\mathbb{P}_\theta) = \{\theta\} \times [0, 1].$$

Since $\text{Supp}(\mathbb{P}_0) \cap \text{Supp}(\mathbb{P}_\theta) = \emptyset$, the set $A = \{0\} \times [0, 1]$ satisfies:

$$\mathbb{P}_0(A) = 1 \quad \text{while} \quad \mathbb{P}_\theta(A) = 0.$$

Hence, $\mathbb{P}_0$ is not absolutely continuous with respect to $\mathbb{P}_\theta$ ($\mathbb{P}_0 \not\ll \mathbb{P}_\theta$). By definition of the Kullback–Leibler divergence:

$$\text{KL}(\mathbb{P}_0 \parallel \mathbb{P}_\theta) = \int_{\text{Supp}(\mathbb{P}_0)} \log \left(\frac{d\mathbb{P}_0}{d\mathbb{P}_\theta} \right) d\mathbb{P}_0 = +\infty.$$

By symmetry of the disjoint supports, $\text{KL}(\mathbb{P}_\theta \parallel \mathbb{P}_0) = +\infty$ for any $\theta \neq 0$.

- **Jensen–Shannon divergence (JS):** If $\theta = 0$, $\text{JS}(\mathbb{P}_0, \mathbb{P}_0) = 0$. For $\theta \neq 0$, recall the definition:

$$\text{JS}(\mathbb{P}_0, \mathbb{P}_\theta) = \frac{1}{2} \text{KL}(\mathbb{P}_0 \parallel \mathbb{P}_m) + \frac{1}{2} \text{KL}(\mathbb{P}_\theta \parallel \mathbb{P}_m),$$

where $\mathbb{P}_m = \frac{1}{2}\mathbb{P}_0 + \frac{1}{2}\mathbb{P}_\theta$ is the mixture measure. Because $\text{Supp}(\mathbb{P}_0) \cap \text{Supp}(\mathbb{P}_\theta) = \emptyset$, for any point $\mathbf{x} \in \text{Supp}(\mathbb{P}_0)$, the density of the mixture distribution satisfies:

$$\frac{d\mathbb{P}_0}{d\mathbb{P}_m}(\mathbf{x}) = \frac{p_0(\mathbf{x})}{\frac{1}{2}p_0(\mathbf{x}) + \frac{1}{2}p_\theta(\mathbf{x})} = \frac{p_0(\mathbf{x})}{\frac{1}{2}p_0(\mathbf{x}) + 0} = 2.$$

Substituting this Radon–Nikodym derivative into the first term:

$$\text{KL}(\mathbb{P}_0 \parallel \mathbb{P}_m) = \int_{\text{Supp}(\mathbb{P}_0)} \log \left(\frac{d\mathbb{P}_0}{d\mathbb{P}_m} \right) d\mathbb{P}_0 = \int_{\text{Supp}(\mathbb{P}_0)} \log(2) d\mathbb{P}_0 = \log 2.$$

By identical reasoning on $\text{Supp}(\mathbb{P}_\theta)$, we obtain $\frac{d\mathbb{P}_\theta}{d\mathbb{P}_m}(\mathbf{x}) = 2$ and $\text{KL}(\mathbb{P}_\theta \parallel \mathbb{P}_m) = \log 2$. Therefore:

$$\text{JS}(\mathbb{P}_0, \mathbb{P}_\theta) = \frac{1}{2} \log 2 + \frac{1}{2} \log 2 = \log 2 \quad (\forall \theta \neq 0).$$

- **Total Variation distance (δ):** If $\theta = 0$, $\delta(\mathbb{P}_0, \mathbb{P}_0) = 0$. For $\theta \neq 0$, using the definition over the Borel σ -algebra Σ :

$$\delta(\mathbb{P}_0, \mathbb{P}_\theta) = \sup_{A \in \Sigma} |\mathbb{P}_0(A) - \mathbb{P}_\theta(A)|.$$

Since probability measures are bounded by 1, we have $|\mathbb{P}_0(A) - \mathbb{P}_\theta(A)| \leq 1$ for any $A \in \Sigma$, which implies $\delta(\mathbb{P}_0, \mathbb{P}_\theta) \leq 1$. Choosing the Borel set $A^* = \{0\} \times [0, 1] = \text{Supp}(\mathbb{P}_0)$, we find:

$$\mathbb{P}_0(A^*) = 1 \quad \text{and} \quad \mathbb{P}_\theta(A^*) = 0.$$

Evaluating the difference gives:

$$|\mathbb{P}_0(A^*) - \mathbb{P}_\theta(A^*)| = |1 - 0| = 1.$$

Since the supremum cannot exceed 1 and is attained at A^* , we have $\delta(\mathbb{P}_0, \mathbb{P}_\theta) = 1$ for all $\theta \neq 0$.

When $\theta_t \rightarrow 0$, the sequence $(\mathbb{P}_{\theta_t})_{t \in \mathbb{N}}$ converges to $\mathbb{P}_0$ under the EM distance, but does not converge at all under either the JS, KL, reverse KL, or TV divergences.

This example illustrates a scenario in which it is possible to learn a probability distribution over a low-dimensional manifold via gradient descent using EM distance. Conversely, this optimization is not feasible with the other divergences and distances due to their resulting not continuous loss function.

The same conclusion holds when the supports have non-empty intersection, provided that this intersection is contained in a set of measure zero. This specific condition arises when two low dimensional manifolds intersect in general position.

1.2.2 The Optimal Transport Problem and Wasserstein Distances

In this section, a detailed presentation of the Wasserstein distance will be given using the book written by Villani [14]. The fundamental intuition behind optimal transport is evaluating the most efficient way to transfer a distribution of mass to another distribution. If we define a cost function $c(x, y)$ representing the effort to move one unit of mass from x to y , the optimal transport cost between two probability measures μ and ν is given by the infimum of the total cost over all possible transport plans π . Formally:

$$C(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) d\pi(x, y)$$

When the cost function is defined in terms of a distance metric on a Polish space, this optimal cost formulation naturally induces a rigorous distance between probability distributions.

Definition 1.2.3 (Wasserstein distances). *Let $(\mathcal{X}, d)$ be a Polish metric space, and let $p \in [1, \infty)$. For any two probability measures μ, ν on $\mathcal{X}$, the Wasserstein distance of order p is defined as:*

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p}$$

The mathematical formulation of optimal transport and the Wasserstein distance rigorously requires the underlying space $\mathcal{X}$ to be a *Polish space*, defined as a separable, completely metrizable topological space. From a theoretical standpoint, this topological structure is strictly necessary to ensure that the space is “well-behaved” enough to support tight probability measures and guarantee the validity of key convergence theorems, such as Prokhorov’s theorem.

However, from an applied machine learning perspective, this rigorous topological constraint poses absolutely no limitation to the implementation of Generative Adversarial Networks. In standard deep learning architectures, both the latent noise space and the target feature space (such as multivariate financial time series) are inherently modeled as finite-dimensional Euclidean spaces, typically $\mathcal{X} = \mathbb{R}^d$.

The Euclidean space $\mathbb{R}^d$, equipped with the standard Euclidean metric, is complete (every Cauchy sequence converges within the space) and separable (the subset of points with rational coordinates, $\mathbb{Q}^d$, is countable and dense). Therefore, $\mathbb{R}^d$ is trivially a Polish space. Consequently, the fundamental topological prerequisites for Wasserstein-based optimal transport are intrinsically and universally satisfied by the natural data structures used in neural networks.

Metric Axioms of the Wasserstein Distance

To rigorously employ the Wasserstein formulation as a reliable loss function in generative modeling, it is imperative to verify that it constitutes a true mathematical distance. At a general level, the optimal transport

cost might yield infinite values or fail to meet strict metric criteria. However, by restricting the domain to the Wasserstein space $P_p(\mathcal{X})$ (where distributions possess finite p -th moments), W_p is guaranteed to be finite and strictly satisfies all the fundamental axioms of a metric space.

Specifically, the Wasserstein distance satisfies the following three core metric properties:

1. **Symmetry:** $W_p(\mu, \nu) = W_p(\nu, \mu)$. This follows trivially from the symmetric nature of the underlying distance function $d(x, y)$ and the construction of the joint probability measure in the optimal coupling.
2. **Identity of indiscernibles:** $W_p(\mu, \nu) = 0 \iff \mu = \nu$. If the transport cost evaluates to strictly zero, it mathematically implies that the optimal transference plan is entirely concentrated on the diagonal of the product space $\mathcal{X} \times \mathcal{X}$ (where $x = y$). Consequently, absolutely no mass needs to be displaced, meaning the target distribution is the identical pushforward of the source distribution ($\nu = \text{Id}_\# \mu$).
3. **Triangle Inequality:** $W_p(\mu_1, \mu_3) \leq W_p(\mu_1, \mu_2) + W_p(\mu_2, \mu_3)$. While conceptually intuitive, transferring mass directly from μ_1 to μ_3 cannot cost more than transferring it through an intermediate state μ_2 ; its formal justification relies on the Gluing Lemma. By constructing a joint probability space for the three random variables, the property is rigorously established through the application of the Minkowski inequality in L^p spaces.

Satisfying these axioms ensures that W_p is not merely a heuristic formulation, but a formally topological metric. This structural solidity is precisely what allows the distance to be reliably minimized during the training of complex neural architectures, ensuring that the convergence process is geometrically coherent and mathematically justified.

To ensure the distance yields finite values, it is evaluated within the *Wasserstein space* of order p , denoted as $P_p(\mathcal{X})$. This space comprises all probability measures possessing a finite moment of order p .

Topological Properties and Weak Convergence

The hierarchical structure of these metrics dictates that $p \leq q \implies W_p \leq W_q$. Within this hierarchy, W_1 (commonly known as the Kantorovich-Rubinstein distance) represents the weakest and most mathematically flexible distance, whereas W_2 is highly structured, better reflecting the underlying geometric features of the space and scaling more efficiently with dimensionality.

A pivotal property of the Wasserstein space is its topological completeness: if $\mathcal{X}$ is a complete separable metric space (Polish space), then $P_p(\mathcal{X})$ equipped with W_p is also a Polish space.

Crucially, the Wasserstein distance W_p metrizes the weak convergence in $P_p(\mathcal{X})$. This implies that a sequence of measures μ_k converges to μ in the Wasserstein sense if and only if it converges weakly in the traditional sense and its moments of order p also converge. Consequently, W_p is a continuous distance functional on $P_p(\mathcal{X})$.

Topological Foundations for W_1 : Probability Measures in the Dual Space

The most important advantage of W_1 is that it proves to be weaker than the other evaluated divergences and distances.

To fully grasp why the Wasserstein distance resolves the convergence issues of standard GANs, we must analyze the topological properties of probability distances. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a compact set and $C_b(\mathcal{X})$ be the space of continuous and bounded functions on $\mathcal{X}$, equipped with the supremum norm $\|f\|_\infty$. We can define its dual space, $C_b(\mathcal{X})^*$, as the space of all linear and continuous functionals on $C_b(\mathcal{X})$.

The crucial mathematical step is embedding the space of probability measures, $\text{Prob}(\mathcal{X})$, into this dual space. This is achieved through the linear mapping $\Phi : \text{Prob}(\mathcal{X}) \rightarrow C_b(\mathcal{X})^*$, explicitly defined by the

expectation:

$$\Phi(\mathbb{P})(f) := \mathbb{E}_{x \sim \mathbb{P}}[f(x)]$$

By the Riesz Representation Theorem, Φ acts as an isometric immersion. This allows us to rigorously treat $\text{Prob}(\mathcal{X})$ as a subset of the dual space $C_b(\mathcal{X})^*$.

Because $\text{Prob}(\mathcal{X})$ is embedded in a dual space, it naturally inherits two distinct topologies. The first is the *strong topology* (the norm topology), which is equivalent to the Total Variation (TV) distance. From an optimization perspective, this topology is exceedingly rigid: it renders the mapping from the network's parameters to the generated distribution, $\theta \mapsto \mathbb{P}_\theta$, fundamentally discontinuous, leading directly to the vanishing gradient problem.

The second is the *weak* topology*. By definition, convergence in the weak* topology implies the convergence of the expectations of continuous functions. For the space of probability measures $\text{Prob}(\mathcal{X})$, this weak* topology is precisely the one metrized by the Wasserstein distance.

Therefore, the Wasserstein distance is not an arbitrary metric, but the natural consequence of evaluating probability distributions through the weak* topology of the dual space $C_b(\mathcal{X})^*$. This weaker topology ensures that sequences of distributions can converge smoothly, rendering the mapping $\theta \mapsto \mathbb{P}_\theta$ continuous and providing stable, non-vanishing gradients everywhere during the training process.

Methodological Advantages

While probability theory offers numerous metrics to assess convergence (such as the Prokhorov, bounded Lipschitz, weak-*, or Total Variation distances), the Wasserstein framework provides several profound methodological advantages for advanced modeling:

- **Robustness to Large Distances:** Unlike the weak-* distance, Wasserstein metrics effectively account for large variations and extreme distances within the underlying space $\mathcal{X}$.
- **Geometric Integration:** The mapping inherently incorporates the geometric structure of the original space, making it exceptionally well-adapted for models combining weak convergence with complex spatial geometries.
- **Upper Bound Tractability:** Being defined as an infimum over all possible couplings, W_p is inherently easier to bound from above; constructing any valid transport plan automatically yields an upper bound for the distance.
- **Rich Duality Framework:** The W_1 metric possesses a highly symmetric dual formulation, which is technically convenient for transitioning between primal and dual optimization problems.

Kantorovich Duality

The theoretical cornerstone that makes the Wasserstein distance computationally tractable in modern machine learning applications is its dual formulation. The general Kantorovich Duality Theorem establishes that for a lower semicontinuous cost function $c(x, y)$, the primal minimization problem over transport plans π is strictly equivalent to a dual maximization problem:

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) = \sup_{\phi - \psi \leq c} \left(\int_{\mathcal{Y}} \phi(y) d\nu(y) - \int_{\mathcal{X}} \psi(x) d\mu(x) \right)$$

Here, the supremum is taken over all pairs of integrable functions (ϕ, ψ) , termed Kantorovich potentials, that satisfy the pointwise constraint $\phi(y) - \psi(x) \leq c(x, y)$ for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$. This constraint ensures

that the price difference charged between target and source locations never exceeds the physical transport cost $c(x, y)$, thereby preventing unbounded arbitrage in the dual formulation.

In this dual framework, the supremum can be restricted to functions where ψ is c -convex and ϕ is its c -transform (i.e., $\phi = \psi^c$). While this is a powerful theoretical result, optimizing over general c -convex functions remains computationally challenging.

The crucial mathematical simplification for Generative Adversarial Networks emerges when we focus on the W_1 Wasserstein distance. In this specific scenario, the optimal transport cost function is defined exactly as the metric of the space itself: $c(x, y) = d(x, y)$. Under this specific cost, a profound topological property arises: a c -convex function strictly coincides with a 1-Lipschitz function. Furthermore, a 1-Lipschitz function acts as its own c -transform ($\psi^c = \psi$). This occurs because the defining inequality of a 1-Lipschitz function, $\psi(x) \leq \psi(y) + d(x, y)$, ensures that evaluating the infimum over y simply returns $\psi(x)$.

By combining the general duality theorem with the inherent Lipschitz property above, the dual problem elegantly collapses into the Kantorovich-Rubinstein duality formula:

$$W_1(\mu, \nu) = \sup_{\|\psi\|_{\text{Lip}} \leq 1} \left\{ \int_{\mathcal{X}} \psi(x) d\mu(x) - \int_{\mathcal{X}} \psi(x) d\nu(x) \right\} \quad (1.9)$$

From an algorithmic perspective, this equation is the exact mathematical foundation of the Wasserstein GAN. Instead of searching for an infinitely complex joint probability distribution in the primal space, the optimization is robustly framed as finding a single 1-Lipschitz continuous function that maximizes the difference in expectations between the real and generated distributions.

In practice, this 1-Lipschitz function ψ is parameterized by the neural network's Discriminator. This reveals precisely why regularization techniques, such as weight clipping or gradient penalties, are theoretically mandatory in WGAN architectures: they are the practical mechanisms required to enforce the strictly necessary $\|\psi\|_{\text{Lip}} \leq 1$ constraint.

1.2.3 Continuity of loss function

It is now pertinent to determine whether $W(\mathbb{P}_r, \mathbb{P}_\theta)$ acts as a continuous loss function with respect to θ under mild assumptions, so that it can be used in a neural network.

Assumption 1.2.1. *Let $g : \mathcal{Z} \times \mathbb{R}^d \rightarrow \mathcal{X}$ be locally Lipschitz between finite-dimensional vector spaces. Its evaluation on coordinates (z, θ) is denoted by $g_\theta(z)$. The function g is said to satisfy Assumption 1 for a certain probability distribution p over $\mathcal{Z}$ if there exist local Lipschitz constants $L(\theta, z)$ such that:*

$$\mathbb{E}_{z \sim p}[L(\theta, z)] < +\infty$$

The following theorem establishes this fundamental property.

Theorem 1.2.1. *Let $\mathbb{P}_r$ be a fixed distribution over a space $\mathcal{X}$. Let Z be a random variable (e.g., Gaussian) over another space $\mathcal{Z}$. Let $g : \mathcal{Z} \times \mathbb{R}^d \rightarrow \mathcal{X}$ be a function, denoted by $g_\theta(z)$, where z represents the first coordinate and θ the second. Let $\mathbb{P}_\theta$ denote the distribution of $g_\theta(Z)$. Then:*

1. *If g is continuous in θ , then $W(\mathbb{P}_r, \mathbb{P}_\theta)$ is also continuous in θ .*
2. *If g is locally Lipschitz and satisfies Assumption 1, then $W(\mathbb{P}_r, \mathbb{P}_\theta)$ is continuous everywhere and differentiable almost everywhere.*
3. *Statements 1 and 2 are false for the Jensen-Shannon divergence $JS(\mathbb{P}_r, \mathbb{P}_\theta)$ and all Kullback-Leibler (KL) divergences.*

The subsequent corollary demonstrates that learning by minimizing the Earth-Mover (EM) distance is theoretically sound when utilizing neural networks.

Corollary 1.2.1. *Let g_θ be any feedforward neural network parameterized by θ , and let $p(z)$ be a prior distribution over z such that $\mathbb{E}_{z \sim p(z)}[\|z\|] < \infty$ (e.g., Gaussian, uniform, etc.). Then, Assumption 1 is satisfied, and consequently, $W(\mathbb{P}_r, \mathbb{P}_\theta)$ is continuous everywhere and differentiable almost everywhere with respect to θ .*

The fundamental intuition behind this corollary lies in the architectural nature of neural networks. A standard feedforward neural network is essentially a composition of affine transformations and pointwise nonlinear activation functions (such as Sigmoid or ReLU). Since these individual components are locally Lipschitz continuous, their composition preserves this property. Consequently, the network as a whole satisfies the regularity conditions required by Assumption 1.2.1, ensuring that the Wasserstein distance remains well-behaved during the optimization process.

These findings demonstrate that the Earth-Mover distance serves as a significantly more robust cost function for the optimization problem compared to the Jensen-Shannon divergence. The following theorem formally describes the relative strength of the topologies induced by these distances and divergences, establishing KL as the strongest, followed by JS and TV, with EM being the weakest.

Theorem 1.2.2. *Let $\mathbb{P}$ be a distribution on a compact space $\mathcal{X}$ and $(\mathbb{P}_n)_{n \in \mathbb{N}}$ be a sequence of distributions on $\mathcal{X}$. Then, considering all limits as $n \rightarrow \infty$,*

1. *The following statements are equivalent:*

- $\delta(\mathbb{P}_n, \mathbb{P}) \rightarrow 0$, where δ is the total variation distance.
- $JS(\mathbb{P}_n, \mathbb{P}) \rightarrow 0$, where JS is the Jensen-Shannon divergence.

2. *The following statements are equivalent:*

- $W(\mathbb{P}_n, \mathbb{P}) \rightarrow 0$.
- $\mathbb{P}_n \xrightarrow{D} \mathbb{P}$, where $\xrightarrow{D}$ represents convergence in distribution for random variables.

3. *$KL(\mathbb{P}_n \parallel \mathbb{P}) \rightarrow 0$ or $KL(\mathbb{P} \parallel \mathbb{P}_n) \rightarrow 0$ imply the statements in (1).*

4. *The statements in (1) imply the statements in (2).*

This theorem establishes a fundamental topological hierarchy among probability metrics, revealing why standard GAN training criteria fail when applied to complex, high-dimensional empirical data.

The core takeaways of this result can be understood through the following key dimensions:

Topological Strength and Convergence Hierarchy. In functional analysis, a distance ρ_1 is said to be stronger than another distance ρ_2 if convergence with respect to ρ_1 implies convergence with respect to ρ_2 . Statements (3) and (4) establish the strict ordering:

$$KL \implies (\delta \iff JS) \implies (W_1 \iff \xrightarrow{D}).$$

This implies that convergence under the Kullback–Leibler, Total Variation, or Jensen–Shannon divergences is a much more demanding condition than convergence in Wasserstein distance. Crucially, statement (2) links the Wasserstein distance directly to the weak topology (convergence in distribution, $\mathbb{P}_n \xrightarrow{D} \mathbb{P}$), which requires only that expectations of continuous bounded test functions converge: $\mathbb{E}_{\mathbb{P}_n}[f(x)] \rightarrow \mathbb{E}_{\mathbb{P}}[f(x)]$ for all $f \in C_b(\mathcal{X})$.

The Manifold Hypothesis and Singularity. In generative modeling, real-world data typically concentrate on a lower-dimensional submanifold $\mathcal{M} \subset \mathcal{X}$ of dimension $d_{\mathcal{M}} \ll d$. Simultaneously, the generator maps a low-dimensional latent vector $z \in \mathbb{R}^{d_z}$ through neural layers, causing the synthetic measure $\mathbb{P}_{\theta}$ to also reside on a lower-dimensional manifold $\mathcal{M}_{\theta} \subset \mathcal{X}$. In general position, two lower-dimensional manifolds in a higher-dimensional space intersect on a set of measure zero, or fail to intersect entirely ($\mathcal{M} \cap \mathcal{M}_{\theta} = \emptyset$).

Under this disjoint support condition:

- Total Variation takes its maximal value: $\delta(\mathbb{P}, \mathbb{P}_{\theta}) = 1$.
- Jensen–Shannon divergence remains locally constant at its supremum: $\text{JS}(\mathbb{P}, \mathbb{P}_{\theta}) = \log 2$.
- Kullback–Leibler divergence is undefined or infinite: $\text{KL}(\mathbb{P} \parallel \mathbb{P}_{\theta}) = +\infty$.

Vanishing Gradients versus Continuous Guidance. Because δ , JS, and KL do not account for the underlying metric structure of the ambient space $\mathcal{X}$, they only evaluate whether the supports overlap. As long as the supports are disjoint, JS remains flat at $\log 2$ regardless of how close or far the manifolds are in Euclidean terms. Consequently, the gradient of the loss function with respect to the generator parameters is identically zero:

$$\nabla_{\theta} \text{JS}(\mathbb{P}_r, \mathbb{P}_{\theta}) = \mathbf{0},$$

yielding no informative learning signal to pull $\mathbb{P}_{\theta}$ toward $\mathbb{P}_r$.

In contrast, statement (2) shows that the Earth Mover’s distance W_1 metrizes the weak convergence topology. It leverages the ground metric $\|x - y\|$ of the ambient space $\mathcal{X}$, measuring the physical geometric work required to transport probability mass between the manifolds. Even when $\mathcal{M}$ and $\mathcal{M}_{\theta}$ are completely disjoint, $W_1(\mathbb{P}_r, \mathbb{P}_{\theta})$ varies smoothly and monotonically with respect to the distance between them. This continuity guarantees non-zero, informative gradients almost everywhere, enabling gradient descent to systematically guide the generator toward the true data distribution. This naturally leads to a practical approximation for optimizing the EM distance.

1.2.4 Definition

Theorem 1.2.2 indicates that the Wasserstein distance $W(\mathbb{P}_r, \mathbb{P}_{\theta})$ possesses superior optimization properties compared to the JS divergence. However, calculating the infimum 1.2.2 by its original definition is computationally intractable. To overcome this limitation, the Kantorovich-Rubinstein duality, seen above, is applied, which reformulates the distance as follows:

$$W(\mathbb{P}_r, \mathbb{P}_{\theta}) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim \mathbb{P}_r}[f(x)] - \mathbb{E}_{x \sim \mathbb{P}_{\theta}}[f(x)] \quad (1.10)$$

where the supremum is taken over all 1-Lipschitz functions $f : \mathcal{X} \rightarrow \mathbb{R}$.

If the strict constraint $\|f\|_L \leq 1$ is relaxed to $\|f\|_L \leq K$ (meaning the function is K -Lipschitz for some constant K), the resulting value is simply scaled, yielding $K \cdot W(\mathbb{P}_r, \mathbb{P}_{\theta})$.

Therefore, by introducing a parameterized family of functions $\{f_w\}_{w \in \mathcal{W}}$ that are all K -Lipschitz, the problem can be practically approximated by solving the following maximization:

$$\max_{w \in \mathcal{W}} \mathbb{E}_{x \sim \mathbb{P}_r}[f_w(x)] - \mathbb{E}_{z \sim p(z)}[f_w(g_{\theta}(z))] \quad (1.11)$$

Assuming that the supremum is successfully achieved for some parameters $w \in \mathcal{W}$, this process calculates the Wasserstein distance up to a multiplicative constant. Consequently, it becomes possible to differentiate $W(\mathbb{P}_r, \mathbb{P}_{\theta})$ with respect to the generator’s parameters θ by backpropagating through this objective function, estimating the gradient as $\mathbb{E}_{z \sim p(z)}[\nabla_{\theta} f_w(g_{\theta}(z))]$. This provides a practical and mathematically principled method to train the generative model.

The theoretical justification for this optimization procedure is formally established by the following theorem.

Theorem 1.2.3. Let $\mathbb{P}_r$ be any given data distribution, and let $\mathbb{P}_\theta$ be the distribution of the generated samples $g_\theta(Z)$, where Z is a random variable with probability density p and g_θ satisfies Assumption 1. Then, there exists a solution $f : \mathcal{X} \rightarrow \mathbb{R}$ to the maximization problem:

$$\max_{\|f\|_L \leq 1} \mathbb{E}_{x \sim \mathbb{P}_r}[f(x)] - \mathbb{E}_{x \sim \mathbb{P}_\theta}[f(x)]$$

Furthermore, the gradient of the Wasserstein distance with respect to the generator's parameters is given by:

$$\nabla_\theta W(\mathbb{P}_r, \mathbb{P}_\theta) = -\mathbb{E}_{z \sim p(z)}[\nabla_\theta f(g_\theta(z))]$$

provided that both terms are well-defined.

The remaining challenge lies in identifying the optimal function f that solves the aforementioned maximization problem. To practically approximate this solution, a neural network is trained like a discriminator. This network is parameterized by weights w constrained within a compact space $\mathcal{W}$. The gradients can then be computed via backpropagation through the empirical estimate $\mathbb{E}_{z \sim p(z)}[\nabla_\theta f_w(g_\theta(z))]$, similar to the standard GAN framework.

Crucially, the compactness of the parameter space $\mathcal{W}$ guarantees that all functions f_w within this parameterized family are K -Lipschitz. The Lipschitz constant K depends exclusively on the boundaries of $\mathcal{W}$ and not on the individual weight values. Consequently, this approach successfully approximates the Wasserstein distance up to an irrelevant scaling factor, limited only by the representational capacity of the critic network. To enforce the constraint that the parameters w remain within a compact space, a straightforward and practical method is to apply weight clipping. Specifically, the weights are attached to a fixed hypercube (e.g., $\mathcal{W} = [-c, c]^l$, with $c = 0.01$) following each gradient update step.

This complete optimization process, which defines the standard Wasserstein Generative Adversarial Network (WGAN) framework, is detailed in the following algorithm.

Algorithm 2 Wasserstein GAN (WGAN) training procedure

Require: α , the learning rate. c , the clipping parameter. m , the batch size. n_{critic} , the number of iterations of the critic per generator iteration.

Require: w_0 , initial critic parameters. θ_0 , initial generator parameters.

```

1: while  $\theta$  has not converged do
2:   for  $t = 0, \dots, n_{\text{critic}}$  do
3:     Sample  $\{x^{(i)}\}_{i=1}^m \sim \mathbb{P}_r$  a batch from the real data.
4:     Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
5:      $g_w \leftarrow \nabla_w [\frac{1}{m} \sum_{i=1}^m f_w(x^{(i)}) - \frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))]$ 
6:      $w \leftarrow w + \alpha \cdot \text{RMSProp}(w, g_w)$ 
7:      $w \leftarrow \text{clip}(w, -c, c)$ 
8:   end for
9:   Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
10:   $g_\theta \leftarrow -\nabla_\theta \frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))$ 
11:   $\theta \leftarrow \theta - \alpha \cdot \text{RMSProp}(\theta, g_\theta)$ 
12: end while

```

The default hyperparameter values utilized in the foundational experiments are $\alpha = 0.00005$, $c = 0.01$, $m = 64$, $n_{\text{critic}} = 5$.

An important practical consideration in optimizing WGANs concerns the choice of the optimization algorithm. Empirical observations indicate that WGAN training frequently exhibits instability when employing

momentum-based optimizers, such as Adam (with $\beta_1 > 0$), for the critic network, or when utilizing excessively high learning rates. This instability is primarily attributed to the highly nonstationary nature of the critic’s loss function.

Under such conditions, momentum dynamics can exacerbate optimization difficulties; specifically, during unstable training phases where the loss diverges, the cosine similarity between the Adam update step and the true gradient frequently turns negative. Consequently, the standard practice in WGAN training involves substituting Adam with RMSProp, an algorithm widely recognized for its robust performance in highly non-stationary environments.

Although weight clipping is employed for its inherent simplicity, it is widely acknowledged as a suboptimal method for enforcing the Lipschitz constraint. The primary issue lies in the sensitivity of the clipping parameter. If the threshold is set too high, the critic requires a significantly longer time to reach optimality, leading to slow convergence. Conversely, a small clipping parameter frequently induces vanishing gradients, particularly when the network is deep or lacks batch normalization.

While alternative regularization methods yielded marginal improvements in early studies, clipping remained the initial standard solely due to its ease of implementation.

Consequently, the development of more robust techniques to enforce Lipschitz continuity in neural networks has been studied and for these reasons WGAN-GP was defined (ref. subsection 1.2.5).

1.2.5 Advantages and disadvantages

The most significant advantage in WGAN is the ability to train the discriminator to optimality. This is theoretically achievable because, unlike traditional divergences, the EM distance is continuous and differentiable almost everywhere.

In standard GANs utilizing the Jensen-Shannon divergence, an optimal discriminator leads to local saturation, resulting in a true gradient of zero and directly causing the vanishing gradient problem. Conversely, the WGAN discriminator cannot saturate; the Lipschitz constraint forces it to converge toward a linear function. This property ensures that the generator receives remarkably clean and reliable gradient information across the entire space, regardless of the discriminator’s training level.

Furthermore, the ability to train the discriminator to optimality fundamentally mitigates the mode collapse phenomenon. In the traditional framework, mode collapse often occurs because the optimal strategy for the generator, given a static discriminator, is to collapse its output into a sum of Dirac deltas concentrated at the specific points where the discriminator assigns the highest probability. By continuously training the WGAN discriminator to reliably approximate the Wasserstein distance without facing vanishing gradients, the generator is forced to capture the true diversity of the data distribution rather than exploiting vulnerabilities in a suboptimal discriminative network.

When the discriminator is fully trained, as is possible in WGAN, it provides a reliable loss metric for the generator, allowing it to be optimized in a way similar to a standard neural network. Consequently, the delicate balancing of capacity between the generator and the discriminator, a notoriously difficult and heuristic-driven aspect of standard GAN training, is no longer required. Under the Wasserstein framework, a better-trained discriminator directly yields higher-quality and more informative gradients for the generator.

Furthermore, WGANs demonstrate significantly greater robustness regarding architectural choices. Empirical evaluations establish consistent performance across diverse generator architectures, including standard convolutional networks, architectures lacking batch normalization, and simple multi-layer perceptrons. While non-standard or simplified architectures typically fail or perform exceptionally poorly in traditional GAN setups due to gradient instability, the WGAN framework maintains stable training dynamics regardless of these structural variations.

Ultimately, this combination of theoretical results, architectural flexibility, and the consistent avoidance of mode collapse firmly establishes the Wasserstein approach as a fundamentally more robust, reliable, and

generally preferable alternative to standard GAN formulations.

1.2.6 Extensions of WGAN: WGAN-GP and CWGAN

Following its introduction, the WGAN framework was widely adopted in generative modeling applications due to its enhanced stability. However, as previously discussed, relying on weight clipping to enforce the Lipschitz continuity of the critic introduced significant optimization challenges. To mitigate these pathological behaviors, Gulrajani et al. [7] in 2017 developed an advanced formulation known as WGAN with Gradient Penalty (WGAN-GP). Concurrently, akin to the evolution of standard Generative Adversarial Networks, it became essential to formulate a conditional variant of the model to enable controlled data generation. Consequently, this subsection introduces these two pivotal extensions: WGAN-GP for improved training dynamics and the Conditional WGAN (CWGAN) for targeted generation.

WGAN-GP

Empirical studies demonstrate that weight clipping, as theoretically anticipated, can severely limit the capacity of the critic to capture higher moments of the data distribution. This phenomenon occurs because the complex interaction between hard weight constraints and the cost function frequently results in either vanishing or exploding gradients, unless the clipping threshold is meticulously tuned.

To understand how to enforce the Lipschitz constraint without relying on weight clipping, it is crucial to analyze the theoretical properties of the optimal WGAN critic. The foundational motivation for the Gradient Penalty approach is derived from the following proposition.

Theorem 1.2.4. *Let $\mathbb{P}_r$ and $\mathbb{P}_g$ be two distributions in $\mathcal{X}$, a compact metric space. Then, there exists a 1-Lipschitz function f^* which is the optimal solution to $\max_{\|f\|_L \leq 1} \mathbb{E}_{y \sim \mathbb{P}_r}[f(y)] - \mathbb{E}_{x \sim \mathbb{P}_g}[f(x)]$.*

Let π be the optimal coupling between $\mathbb{P}_r$ and $\mathbb{P}_g$, defined as the minimizer of:

$$W(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\pi \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \pi} [\|x - y\|]$$

where $\Pi(\mathbb{P}_r, \mathbb{P}_g)$ is the set of joint distributions $\pi(x, y)$ whose marginals are $\mathbb{P}_r$ and $\mathbb{P}_g$, respectively.

If f^ is differentiable, $\pi(x = y) = 0$, and $x_t = tx + (1 - t)y$ with $0 \leq t \leq 1$, it holds that:*

$$\mathbb{P}_{(x,y) \sim \pi} \left[\nabla f^*(x_t) = \frac{y - x_t}{\|y - x_t\|} \right] = 1$$

Corollary 1.2.2. *The optimal critic f^* has a gradient norm of 1 almost everywhere under $\mathbb{P}_r$ and $\mathbb{P}_g$, as well as on the straight lines connecting points sampled from these distributions.*

Instead of artificially bounding the capacity of the critic via weight clipping, Corollary 1 suggests that the 1-Lipschitz constraint can be naturally enforced by penalizing the model if the gradient norm of the critic deviates from 1. This directly leads to the formulation of the Gradient Penalty (GP).

Definition 1.2.4 (WGAN-GP). *The loss function is defined by adding a gradient penalty term to the WGAN objective function:*

$$L = \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g}[D(\tilde{x})] - \mathbb{E}_{x \sim \mathbb{P}_r}[D(x)] + \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad (1.12)$$

where λ is a hyperparameter called the penalty coefficient.

In this formulation, $\hat{x}$ represents samples drawn from a distribution $\mathbb{P}_{\hat{x}}$, which is implicitly defined by sampling uniformly along straight lines between pairs of points from the data distribution $\mathbb{P}_r$ and the generator distribution $\mathbb{P}_g$. As suggested by Theorem 1.2.4, enforcing the unit gradient norm constraint along these specific paths is sufficient to ensure stable training and superior performance. Below is the algorithm:

Algorithm 3 WGAN with Gradient Penalty (WGAN-GP)

Require: The gradient penalty coefficient λ , the number of critic iterations per generator iteration n_{critic} , the batch size m , Adam hyperparameters α, β_1, β_2 .

Require: Initial critic parameters w_0 , initial generator parameters θ_0 .

```

1: while  $\theta$  has not converged do
2:   for  $t = 1, \dots, n_{\text{critic}}$  do
3:     for  $i = 1, \dots, m$  do
4:       Sample real data  $x \sim \mathbb{P}_r$ , latent variable  $z \sim p(z)$ , a random number  $\epsilon \sim U[0, 1]$ .
5:        $\tilde{x} \leftarrow G_{\theta}(z)$ 
6:        $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$ 
7:        $L^{(i)} \leftarrow D_w(\tilde{x}) - D_w(x) + \lambda(\|\nabla_{\hat{x}} D_w(\hat{x})\|_2 - 1)^2$ 
8:     end for
9:      $w \leftarrow \text{Adam}(\nabla_w \frac{1}{m} \sum_{i=1}^m L^{(i)}, w, \alpha, \beta_1, \beta_2)$ 
10:  end for
11:  Sample a batch of latent variables  $\{z^{(i)}\}_{i=1}^m \sim p(z)$ .
12:   $\theta \leftarrow \text{Adam}(\nabla_{\theta} \frac{1}{m} \sum_{i=1}^m -D_w(G_{\theta}(z^{(i)})), \theta, \alpha, \beta_1, \beta_2)$ 
13: end while

```

The default hyperparameters utilized in standard implementations are $\lambda = 10$, $n_{\text{critic}} = 5$, $\alpha = 0.0001$, $\beta_1 = 0$, $\beta_2 = 0.9$.

The WGAN-GP approach effectively enforces the Lipschitz constraint without compromising the representational capacity of the network. This results in significantly more robust training dynamics and the successful integration of momentum-based optimizers such as Adam.

Furthermore, an additional architectural advantage is that WGAN-GP eliminates the necessity for Batch Normalization within the critic network. In fact, because the gradient penalty must be evaluated independently for each individual input, introducing intra-batch dependencies via Batch Normalization would invalidate the mathematical formulation of the penalty.

However, the primary disadvantage of this method is the significant computational overhead. Computing the gradients of the critic's output with respect to the input samples requires an additional full backward pass during each optimization step.

Consequently, while WGAN-GP achieves superior stability and sample quality, it substantially increases the overall training time and computational cost compared to the standard weight-clipping approach.

Conditioned models

The Wasserstein framework is highly modular and can be seamlessly extended to conditional generation tasks. By introducing auxiliary label information y into the inputs of both the generator G and the critic D , it is possible to achieve highly targeted data generation. In financial applications, this conditioning variable y can represent either discrete categorical states, such as distinct market regimes (e.g., normal versus stressed periods), or continuous historical metrics.

This extension, formally known as the Conditional Wasserstein GAN (CWGAN), provides a stable and

mathematically robust methodology for controlled synthesis. Mechanically, the prior noise vector z is concatenated with the condition y to form the input for the generator, yielding the synthetic sample $G(z, y)$. Simultaneously, the critic evaluates the data in the context of that exact same condition, taking the pair (x, y) as its input.

Definition 1.2.5 (CWGAN Objective Function). *The objective function of the Conditional WGAN preserves the foundational Kantorovich-Rubinstein duality while integrating the conditional dependencies. It is defined as:*

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{x \sim p_d} [D(x, y)] - \mathbb{E}_{z \sim p_z} [D(G(z, y))]$$

By enforcing the 1-Lipschitz constraint on the conditioned critic, the CWGAN prevents the vanishing gradient problem typical of standard architectures. This ensures that the generator receives informative, non-saturating gradients even when attempting to synthesize complex, heavy-tailed market scenarios associated with specific extreme conditions.

Chapter 2

First applications to financial scenarios

The simulation of financial scenarios serves as a fundamental tool in various critical domains, including risk management, asset allocation, and strategic economic decision-making. Traditionally, these simulations focus on financial markets but the dynamics represent highly complex phenomena; while numerous empirical properties have been extensively observed and documented, the underlying generative mechanisms driving these behaviors often remain unrevealed.

Furthermore, the study of financial simulations is of broader scientific importance. The advanced methodologies developed to model financial markets can frequently be generalized and extended to capture the dynamics of other complex systems across different disciplines.

Before exploring the first applications of Generative Adversarial Networks within this domain, it is necessary to establish a rigorous definition of the core concept.

2.1 What is a financial scenario?

A financial scenario can be essentially defined as a historical time series representing the price evolution of financial assets in the market.

In this work, we focus primarily on stocks, which represent partial ownership in a corporation, and market indices, which track a hypothetical portfolio of stocks, as example S&P 500 contains the 500 most valuable US stocks.

Other asset classes, such as commodities (raw materials), could be studied in a similar way. However, their prices are heavily driven by exogenous geopolitical factors rather than pure historical market dynamics. For these assets, unconditioned simulations might be less reliable, although a conditional generative model could still provide useful insights.

Similarly, fixed-income instruments like bonds present different challenges. While simulating the raw price of a single bond might be less interesting for this specific time series analysis, using conditional models to study the underlying interest rates influenced by central bank decisions remains a highly relevant task.

Therefore, to maintain a clear focus, the following sections will dive deep into the construction and definition of financial time series exclusively using stock and index pricing.

2.1.1 Financial time series

When modeling financial time series, working directly with raw asset prices is generally avoided. Raw prices are typically non-stationary and tend to exhibit long-term trends, making it extremely difficult for machine learning models to learn robust and generalized underlying patterns.

To overcome this issue, it is standard practice to model the relative changes in price rather than the absolute values. In quantitative finance and time series forecasting, the preferred metric is the logarithmic return.

Definition 2.1.1 (Log return). *Let P_t denote the price of a financial asset at time t . The log return r_t for that period is defined as:*

$$r_t = \log \left(\frac{P_{t+1}}{P_t} \right) = \log(P_{t+1}) - \log(P_t)$$

Note that price return $\frac{p(t+1)-p(t)}{p(t)}$ can be approximated by 2.1.1 if the price change is small enough, as is usually the case in financial markets.

There are several technical reasons for the choice of logarithmic return. First, log returns are time-additive, meaning the total return over k periods is simply the sum of the log returns of a single period. Second, computing the logarithm helps to stabilize the variance of the time series and mitigate the impact of extreme outliers. This transformation yields a much more stationary and well-behaved dataset, which is a fundamental prerequisite for training stable generative models like GANs.

The stationarity of log returns allows for a consistent estimation of the underlying probability distribution. Given this foundation, we can define L as a random variable representing the loss of a financial portfolio over a given time horizon.

Definition 2.1.2 (Financial loss variable). *Let r_t be the log return and L be a random variable such that $L_t = -r_t$ then L is a financial loss variable and $F_L(l) = \mathbb{P}(L \leq l)$ is its cumulative distribution function.*

In order to evaluate the risk of the generated financial scenarios, it is essential to rigorously define the standard risk measures utilized in quantitative finance: Value at Risk (VaR) and Conditional Value at Risk (CVaR), see [9].

Definition 2.1.3 (Var). *Let L be a loss variable and given a confidence level $\alpha \in (0, 1)$ (typically set to 0.95 or 0.99), the Value at Risk is defined as the α -quantile of the loss distribution. In other words, VaR_α is the smallest value l such that the probability of the loss L not exceeding l is at least α :*

$$\text{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : F_L(l) \geq \alpha\}$$

While VaR provides the maximum expected loss up to a specified confidence level, it lacks subadditivity and fails to convey any information regarding the severity of losses that lie beyond this threshold. To address these limitations, the Conditional Value at Risk (CVaR), also known as Expected Shortfall (ES), is employed.

Definition 2.1.4 (CVar continuous). *For L continuous loss distributions, CVaR is defined as the expected loss strictly exceeding the VaR threshold:*

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]$$

However, to account for general distributions that may exhibit discontinuities, the universally robust definition of CVaR is given by:

Definition 2.1.5 (General CVaR). *For any general loss distribution L , the CVaR is defined as the integral of the VaR values over the upper tail:*

$$\text{CVaR}_\alpha(L) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_\gamma(L) d\gamma$$

This formulation guarantees that CVaR is a coherent risk measure, satisfying the properties of translation invariance, subadditivity, positive homogeneity, and monotonicity.

2.1.2 Statistical properties

Financial time series exhibit specific statistical properties, commonly known as “stylized facts”. These facts are crucial because they serve as a benchmark for evaluating synthetic data. If a simulation successfully replicates these properties, it can be considered a valid representation of real market dynamics. As done by Takahashi et al. [13] and several other researchers, verifying the presence of these stylized facts is a standard methodology to prove the effectiveness of a model in finance. For this reason, it is essential to clearly outline these core properties before proceeding.

Linear unpredictability

The linear unpredictability of asset returns is often referred to as the absence of linear autocorrelation. This property implies that past daily price returns do not provide meaningful linear information to predict future returns.

Let μ and σ^2 be the mean and variance of the price return r_t , respectively. The autocorrelation function for a time lag k can be mathematically expressed as:

$$\frac{E[(r_t - \mu)(r_{t+k} - \mu)]}{\sigma^2} = \text{Corr}(r_t, r_{t+k}) \approx 0, \quad \text{for } k \geq 1$$

This near-zero correlation for any day lag $k \geq 1$ empirically demonstrates that financial markets are highly efficient to a certain extent, making linear predictive models largely ineffective on raw daily returns.

Below is the correlogram for Nikkei 225:

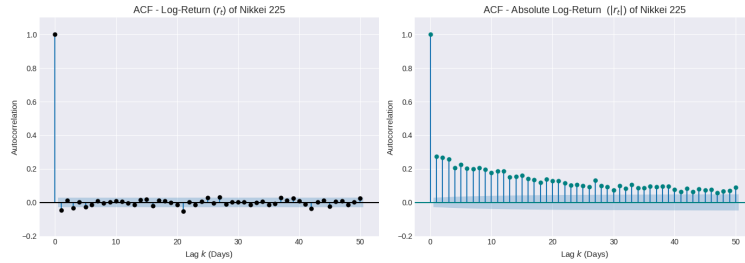


Figure 2.1: ACF of NIKKEI 225

The left panel empirically validates this property: the autocorrelation of raw log-returns decays to near-zero for all lags $k \leq 1$, remaining strictly within the statistical confidence bounds. This confirms the absence of linear predictability, consistent with the efficient market hypothesis.

In contrast, the right panel illustrates the autocorrelation of absolute log-returns, uncovering a highly significant long-range dependence. The persistence of these coefficients well beyond the confidence thresholds over extended lag periods provides robust visual evidence of volatility clustering. Such non-linear dynamics render classical linear predictive models (e.g., ARIMA) inadequate.

Fat-tailed distribution

The probability distribution function of the log price return, denoted as $P(r)$, has been studied in different time scales. It is empirically known that this distribution exhibits a power-law decay in the tails. Mathematically, this is expressed as:

$$P(r) \propto r^{-\alpha}$$

The exponent typically is $\alpha \in [3, 5]$.

Replicating this property is one of the most difficult tasks. This difficulty arises because standard traditional models tend to optimize for the mean of the distribution, frequently failing to capture the extreme tail events. Beyond finance, numerous complex phenomena exhibit fat-tailed distributions. Consequently, developing generative architectures capable of accurately learning and reproducing these tail dynamics is highly valuable, opening up advanced simulation scenarios across multiple scientific disciplines.

Below is the histogram for Nikkei 225:

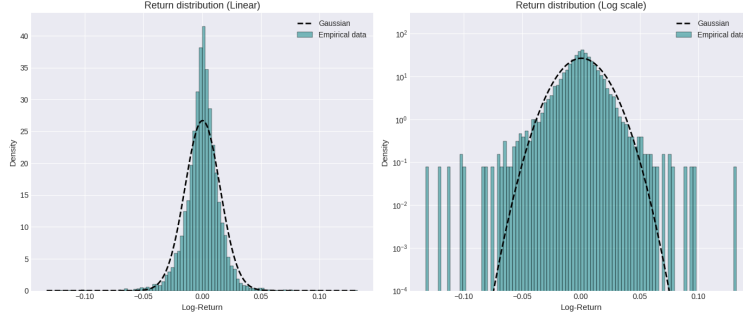


Figure 2.2: Histogram of NIKKEI 225

The left panel, plotted on a linear scale, highlights a critical limitation of traditional models: the Gaussian curve closely fits the central bulk of the distribution, creating an illusion of accuracy while extreme drawdowns remain visually compressed and imperceptible.

The right panel resolves this limitation by employing a logarithmic y-axis. In this transformed space, the Normal distribution decays rapidly as an inverted parabola, whereas the empirical returns exhibit pronounced heavy tails that extend significantly above the theoretical baseline.

This structural divergence at the distribution margins provides empirical evidence that the probability of massive financial crashes does not decay exponentially, but rather follows a power law. Consequently, this demonstrates the necessity of integrating advanced architectures to accurately capture and simulate these extreme tail scenarios.

Volatility Clustering

Volatility clustering describes the tendency of large price fluctuations to be followed by large fluctuations, and small fluctuations to be followed by small ones, regardless of their sign.

From a quantitative perspective, while the raw returns r_t are uncorrelated, their absolute values $|r_t|$ display a strong, positive, and slowly decaying autocorrelation. This behavior is formally characterized by a power-law decay in the autocorrelation function of the absolute returns:

$$\text{Corr}(|r_t|, |r_{t+k}|) \propto k^{-\beta}$$

Empirical studies across major indices, such as the S&P 500, demonstrate that the exponent β typically ranges between 0.1 and 0.5. This slow power-law decay often persists for large time lags (e.g., up to $k \approx 10^2$ days), indicating the presence of long-range temporal dependence in financial market volatility.

The left panel of the following figure illustrates the Autocorrelation Function (ACF) of absolute returns over an extended horizon on a standard linear scale. This visualization confirms that the autocorrelation remains strictly positive and decays at a remarkably slow rate, providing empirical evidence that volatility shocks persist in the market for several months.

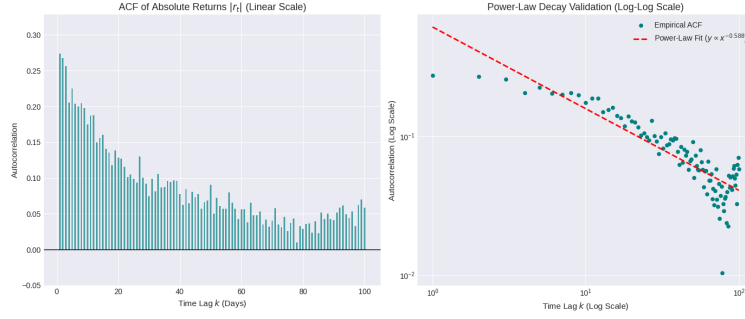


Figure 2.3: Volatility clustering of NIKKEI 225

The right panel transforms the same data onto a log-log scale, where a power-law relationship naturally manifests as a descending straight line. By performing a linear regression within this logarithmic space to estimate the decay exponent β , the strict alignment between the empirical observations and the theoretical fitted line offers rigorous quantitative proof for the long-range dependence relation $\text{Corr}(|r_t|, |r_{t+k}|) \propto k^{-\beta}$.

Leverage Effect

The leverage effect refers to the tendency for past price returns to be negatively correlated with future volatility. In other words, negative price returns are typically followed by an increase in volatility, whereas positive returns are often followed by a decrease in volatility. This asymmetric property is mathematically quantified by the following lead-lag correlation function $L(k)$ for a time lag $k > 0$:

$$L(k) = \frac{\mathbb{E}[r(t)|r(t+k)|^2 - r(t)|r(t)|^2]}{\mathbb{E}[|r(t)|^2]^2}$$

In contrast to the previously discussed stylized facts, which are nearly universal, this property is highly market-dependent. While the standard leverage effect is consistently observed in major Western indices such as the DAX (the German stock index) and the S&P 500, an anomalous "anti-leverage effect" (exhibiting a positive correlation) is frequently detected in Chinese equity markets.

Specifically for the S&P 500 index, the empirical function $L(k)$ exhibits relatively large fluctuations, with negative values typically observed for lags $k \in [1, 10]$, before following a gradual exponential decay towards zero.

Below a graphical demonstration for NASDAQ100:

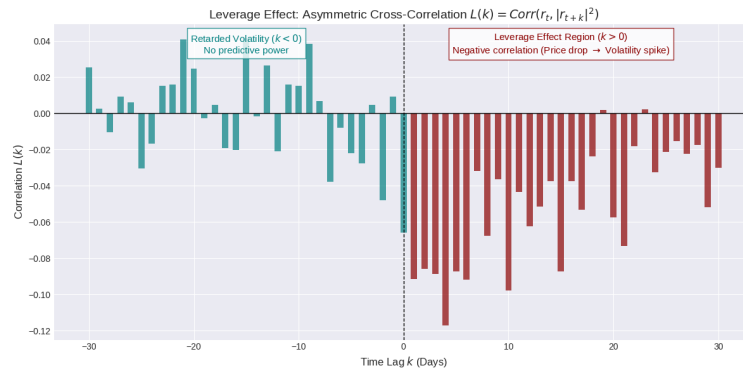


Figure 2.4: Leverage effect for Nasdaq

The plot exhibits a pronounced structural asymmetry around the $k = 0$ axis.

For negative time lags ($k < 0$), the cross-correlation values remain negligible and stochastic, indicating the absence of a predictive linear relationship between current volatility and future directional returns.

Conversely, for positive time lags ($k > 0$), the correlation coefficients assume significantly negative values that decay exponentially toward zero over the initial 10 to 15 days.

This distinct asymmetric profile constitutes the empirical signature of the leverage effect, mathematically demonstrating that contemporary negative price returns systematically trigger a persistent amplification of market volatility over subsequent days.

The following comparative plot visually demonstrates that the US market exhibits a pronounced negative cross-correlation, often dropping to values between -0.10 and -0.15 in the immediate post-shock days. Conversely, the Japanese and Hong Kong markets remain significantly closer to zero, occasionally displaying the anomalous positive spikes characteristic of the anti-leverage effect.

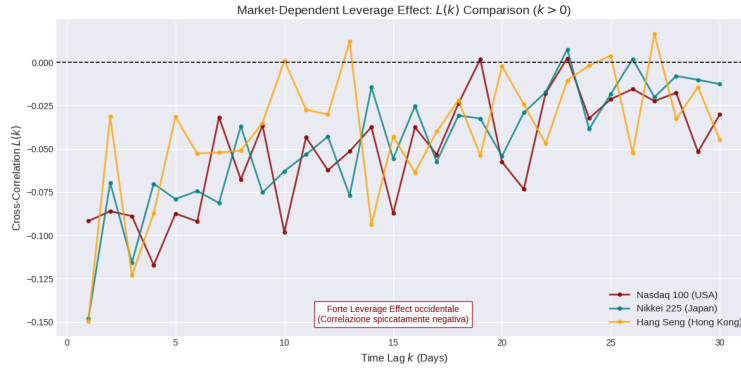


Figure 2.5: Leverage effect with market comparison

This provides evidence of a structural decoupling between price drawdowns and subsequent volatility in these regions. Such empirical observation fundamentally reinforces the necessity of the generative approach. Applying rigid parametric models, predominantly calibrated on Western market behaviors, to Asian equities would inherently yield distorted risk estimates. By contrast, market-specific dynamics in a purely data-driven manner on an asset-by-asset basis ensure a degree of realism and adaptability that traditional frameworks cannot replicate.

Coarse-fine Volatility Correlation

Another important statistical property involves the multi-time-scale analysis of volatility, specifically the relationship between coarse and fine volatility over a time window τ (e.g., $\tau = 5$ for a trading week).

Coarse volatility, $v_c^\tau(t)$, represents the absolute net price movement over τ days, defined as:

$$v_c^\tau(t) = \left| \sum_{i=1}^{\tau} r_{t-i} \right|$$

Conversely, fine volatility, $v_f^\tau(t)$, is the sum of the absolute daily price returns over the same period, representing the total path length:

$$v_f^\tau(t) = \sum_{i=1}^{\tau} |r_{t-i}|$$

To analyze their temporal interactions is defined the correlation function as:

$$\rho_{cf}^\tau(k) = \text{Corr}(v_c^\tau(t+k), v_f^\tau(t))$$

Empirical studies demonstrate a negative asymmetry in this correlation, meaning that $\rho_{cf}^\tau(k) - \rho_{cf}^\tau(-k) < 0$. In practical terms, this indicates that historical fine volatility is a strong predictor of future coarse volatility.

Below there is a graphical demonstration:

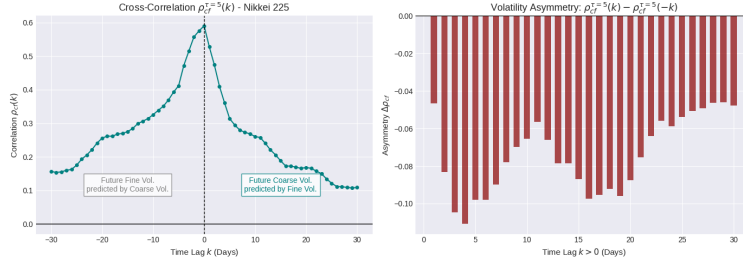


Figure 2.6: Property 5 for Nikkei225

The left panel plots the complete cross-correlation function $\rho_{cf}^\tau(k)$, illustrating a pronounced asymmetry around the $k = 0$ axis. Specifically, the correlation for negative time lags ($k < 0$) decays at a significantly faster rate than for positive lags ($k > 0$).

The right panel mathematically isolates this discrepancy, unequivocally demonstrating that the computed differences are systematically negative, particularly across the initial time lags (e.g., $k \in [1, 10]$). This empirical evidence reveals a distinct directional flow of information: historical fine volatility, characterized by high-frequency noise and intra-week micro-movements, possesses substantially greater predictive power for future coarse volatility (macro-directional movements) than present coarse volatility holds for anticipating future high-frequency market turbulence.

Gain/Loss Asymmetry

This phenomenon describes the empirical observation that the speed of market drawdowns (price falls) is systematically faster than that of market recoveries (price rises).

To quantify this directional asymmetry, it is possible to analyze the waiting time required for an asset to reach a specific return threshold. Let $T_{\text{wait}}^t(\theta)$ denote the number of time steps t' needed to reach a given cumulative log return θ , starting from time t . This can be formally defined using the infimum operator:

$$T_{\text{wait}}^t(\theta) = \begin{cases} \inf\{t' \mid \log(P_{t+t'}) - \log(P_t) \geq \theta, t' > 0\} & \text{if } \theta > 0 \\ \inf\{t' \mid \log(P_{t+t'}) - \log(P_t) \leq \theta, t' > 0\} & \text{if } \theta < 0 \end{cases}$$

Empirical analyses of the probability distribution of T_{wait}^t reveal that the distribution peaks significantly earlier for negative thresholds ($-\theta$) compared to equivalent positive thresholds ($+\theta$), indicating the presence of asymmetry in price up/down.

This property is not always visible in single-stock data.

An analysis of the following plotted distributions reveals that the drawdown density curve is significantly left-skewed, peaking at shorter time horizons, whereas the recovery curve presents a flatter, more elongated rightward tail. The vertical dashed lines quantitatively validate this asymmetry: the median time required to accumulate a 5% loss is systematically shorter than the duration needed to achieve an equivalent 5% gain.

Specifically within the context of the Nasdaq index, this empirical evidence confirms that while market crashes manifest as rapid, compressed events, the corresponding price recoveries require a substantially longer temporal span to materialize.

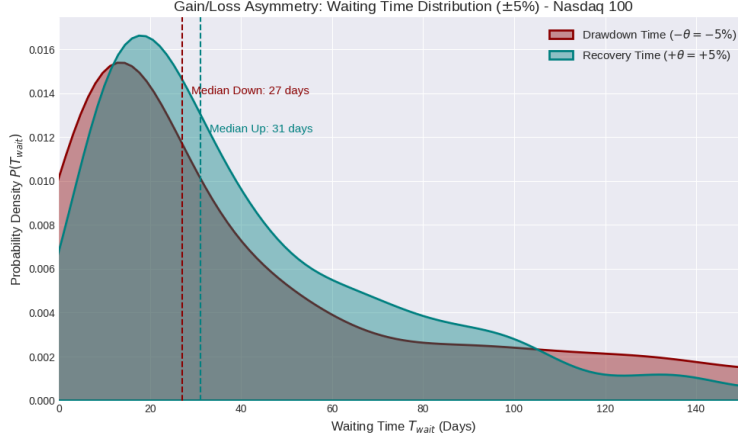


Figure 2.7: Gain/Loss asymmetry for Nasdaq

2.2 FIN-GAN

As demonstrated in the previous sections, financial scenarios are highly complex to simulate due to the large number of statistical properties that a generative model must successfully recreate.

Before Takahashi et al.[13] successfully applied Generative Adversarial Networks to this domain in 2019, the standard approaches relied primarily on traditional stochastic processes and agent-based models. However, these classical methods present significant drawbacks: they often require strong and restrictive mathematical assumptions, suffer from computational inefficiencies, and generally lack the capacity to simultaneously generate time series that exhibit all the stylized facts discussed above.

2.2.1 Network Architecture and training procedure

The FIN-GAN architecture consists of two interacting neural networks: the generator G and the discriminator D , responsible respectively for the synthesis and the evaluation of the financial time series. The underlying adversarial framework operates exactly as the standard GAN described in Chapter 1.1.1, employing the same objective loss function defined in references 1.1.1.

The generator network G is designed to take a 100-dimensional Gaussian noise vector z as input and output a synthetic time series of length $N = 8,192$ steps. Generating such an extended sequence is a strict requirement to ensure that complex statistical properties and correlation functions can be reliably calculated and verified over time.

Regarding the internal structure of G , the authors investigate three primary configurations: Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), and a hybrid MLP-CNN architecture. Specifically, the MLP architecture consists of four fully connected layers utilizing the hyperbolic tangent ($\tanh$) activation function. The CNN architecture employs six convolutional layers with Leaky ReLU activations. The hybrid MLP-CNN merges these approaches by performing an element-wise multiplication of their respective outputs. Regarding the internal structure of D , the MLP-CNN architecture is chosen.

It is important to note that Recurrent Neural Networks (RNNs) are explicitly excluded from this framework. The extreme length of the required time series (8,192 steps) renders RNNs computationally unfeasible for this specific generation task.

The training procedure of FIN-GAN strictly follows the standard generative adversarial framework originally proposed by Goodfellow et al. The discriminator D is optimized to distinguish the true source of the data, while the generator G is optimized to deceive D .

The parameters of both networks (θ_G and θ_D) are updated utilizing the Adam stochastic gradient optimizer. While most hyperparameters are directly inherited from standard GAN implementations typically used for image generation, the learning rates are meticulously tuned by hand to maintain the robustness of the training process, as excessively large learning rates lead to model failure. Specifically, the generator is configured with a learning rate $\alpha_G = 2 \cdot 10^{-4}$ and momentum $\beta_{1,G} = 0.5$. Conversely, the discriminator employs a lower learning rate $\alpha_D = 1 \cdot 10^{-5}$ and $\beta_{1,D} = 0.1$. All other Adam hyperparameters are kept at their originally proposed default values.

To further stabilize the architecture, two specific techniques are implemented. First, Batch Normalization is applied to the MLP and CNN layers of the generator. Second, a "noisy labeling" technique is used for the discriminator: instead of strict binary values, the target labels are drawn from a uniform distribution between 0.9 and 1.1 for the real dataset, and between 0.1 and 0.3 for the generated time series.

Finally, the training process utilizes a batch size of $m = 24$. During each iteration, a batch of real data consisting of 8,192 continuous steps is randomly fetched from the historical dataset.

Below is the complete algorithm:

Algorithm 4 FIN-GAN

Require: $\{\alpha_G, \beta_{1,G}\}$, learning rate and β_1 parameter of Adam (stochastic gradient algorithm) for G .
 $\{\alpha_D, \beta_{1,D}\}$, learning rate and β_1 parameter of Adam for D . m , batch size.

- 1: Initialize the parameters of the generator θ_G and the discriminator θ_D
- 2: **while** θ_G is not converged **do**
- 3: Sample $\{z^{(i)}\}_{i=1}^m, z^{(i)} \sim N(0, 1)$ ▷ Sample a batch of prior
- 4: Sample $\{x^{(i)}\}_{i=1}^m$ ▷ Sample a batch of financial time-series
- 5: Sample $\{G(z^{(i)})\}_{i=1}^m$ ▷ Sample a batch of time-series from G with prior
- 6: Assign the label of source $\{y\}_{i=1}^{2m}$
- 7: $\nabla\theta_D \leftarrow \text{Adam}(\theta_D, \{G(z^{(i)})\}_{i=1}^m + \{x^{(i)}\}_{i=1}^m, \{y\}_{i=1}^{2m})$ ▷ Compute the gradient of D to the target $\{y\}$
- 8: $\theta_D \leftarrow \theta_D - \alpha_D \cdot \nabla\theta_D$ ▷ Update the parameter of the discriminator
- 9: $\nabla\theta_G \leftarrow \text{Adam}(\theta_G, \{z^{(i)}\}_{i=1}^m, \{1\}_{i=1}^m)$ ▷ Compute the gradient of G to deceive D
- 10: $\theta_G \leftarrow \theta_G - \alpha_G \cdot \nabla\theta_G$ ▷ Update the parameter of the generator
- 11: **end while**

2.2.2 Effects of Different Architectures and Techniques

Regarding the topology of the generator G , empirical observations indicate that the pure MLP architecture successfully satisfies all the targeted stylized facts of financial time series.

Similarly, the hybrid MLP-CNN configuration manages to reproduce these properties, with the notable exception of the asymmetry in coarse-fine volatility.

Conversely, the pure CNN generator cannot be effectively optimized to output realistic financial time series. The generated price returns fluctuate at a level significantly above zero and exhibit a strong linear autocorrelation.

Interestingly, this structural bias is visible even before the training process begins. In their untrained initial states, both the MLP and MLP-CNN generators naturally output values fluctuating around zero, whereas the pure CNN does not. From this, it is possible to infer that the inherent architectural design of the generator characterizes the statistical properties of the output right from initialization, fundamentally dictating the ultimate quality of the generated time series after training.

Furthermore, empirical results demonstrated that the application of Batch Normalization caused the MLP generator to fail in synthesizing realistic financial time series. Specifically, this technique proved to be

fundamentally inconsistent in learning the underlying variance process, which represents the most critical dynamic to accurately recreate in financial modeling.

2.2.3 Advantages and Necessity of WGAN-GP

A fundamental advantage of the FIN-GAN framework is that the model directly learns the underlying properties of the data. Unlike traditional stochastic models, GANs do not require explicit mathematical formulations or restrictive prior assumptions.

While FIN-GAN successfully generates all of the targeted stylized facts, it exhibits significant limitations. First, as can be verified through VaR and CVaR analysis, the model frequently fails to fully replicate the fat-tailed nature of the price distribution. Addressing this specific shortcoming remains one of the most compelling challenges in quantitative finance, a topic that will be extensively explored in Chapter 3. Second, while most statistical properties are preserved, the empirical autocorrelation function of absolute price returns tends to decay much faster in the generated series than in real market data.

Furthermore, the training process inherits the classic instability of standard GANs. In certain situations, the discriminator D converges too quickly and becomes overly competitive, leading to the vanishing gradient problem. Consequently, it fails to send meaningful feedback signals to the generator G , halting the learning process.

To resolve these critical issues, the Wasserstein GAN (WGAN) 2 and its Gradient Penalty variant (WGAN-GP) 3 are implemented. These advanced architectures maintain the core advantages of the data-driven approach while fundamentally solving the training instability and significantly improving the overall generation quality. However, accurately modeling extreme tail scenarios remains a persistent and complex difficulty across all current models.

2.3 Conditioned model

In financial scenario analysis, it is standard practice to study conditional distributions. The conditioning variables applied to generative models can be either categorical, such as distinguishing between normal and stressed market periods, or continuous.

One of the most critical applications of conditional modeling is in market risk management, specifically for the calculation of Value-at-Risk (VaR) and Expected Shortfall (ES). Traditionally, these risk metrics were evaluated using Historical Simulation (HS) or Monte Carlo (MC) methods. However, both classical approaches present significant limitations. MC simulations rely heavily on rigid stochastic assumptions that may insufficiently describe real-world distributions, whereas HS is intrinsically constrained by the limited number of historical observations. Consequently, HS often fails to accurately capture complex correlation dynamics during extreme market shifts, such as large financial crises.

2.3.1 Introduction of CGAN and CWGAN

The introduction of Conditional GANs (CGAN 1.1.4 and CWGAN 1.2.5), as defined by Fu et. al [5], provides a powerful non-parametric solution to these structural problems. By conditioning the generative model on specific market states (e.g., normal or stressed periods), CGANs can learn the complex dependent structures of historical data and generate an unlimited number of realistic conditional samples. Today, these models are extensively used not only for rigorous stress testing but also for broader economic forecasting, offering substantial advantages over traditional macroeconomic and econometric models by removing the need for strong prior assumptions.

From an algorithmic perspective, the training procedure of a CGAN closely follows that of a standard GAN, as outlined in the previous sections. However, in practical applications, extending this framework to a Conditional Wasserstein GAN (CWGAN) generally yields more robust and stable results.

When training a CWGAN, the method used to enforce the 1-Lipschitz constraint on the discriminator is a critical design choice. Although the Gradient Penalty technique could theoretically be applied, empirical evidence does not indicate its strict necessity for these specific tasks. Consequently, weight clipping is frequently preferred over Gradient Penalty. The Adam optimizer remains the standard choice for optimization.

Finally, a crucial architectural constraint must be highlighted: if the Gradient Penalty method is utilized, Batch Normalization must be strictly avoided within the discriminator network.

Regarding the neural network architectures, both the generator and the discriminator are composed of fully connected layers, utilizing the `Leaky ReLU` (Rectified Linear Unit) as their primary activation function.

This specific choice effectively resolves the well-known "dying ReLU" problem. In standard ReLU networks, a large gradient flowing through a neuron can cause its weights to update in such a way that the unit outputs zero for all future data points, permanently ceasing to activate and learn.

To prevent this phenomenon, the Leaky ReLU function introduces a small, non-zero slope for negative input values, ensuring that gradients can always flow backward through the network.

Definition 2.3.1 (LeakyReLU). *Given α a small positive constant and x , the function is defined as:*

$$f(x, \alpha) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$$

where α determines the slope of the activation function for the negative regime. Below is the complete algorithm:

Algorithm 5 CGAN and CWGAN Training Procedure

Require: n_{dis} : number of discriminator iterations per generator iteration.

Require: c : hyperparameter for weight clipping.

```

1: for number of training iterations do
2:   for  $t = 1, \dots, n_{dis}$  do
3:     Sample a batch from the real data  $x$  and their corresponding conditions  $y$ 
4:     Sample a batch of noise  $z$  from the prior  $p_z$ 
5:     Evaluate the cost function (Eq. 1.1.4 CGAN, Eq. 1.2.5 for CWGAN) using  $x, y, z$ 
6:     Update the discriminator by ascending its stochastic gradient using Adam
7:     Clip discriminator weights to  $[-c, c]$ 
8:   end for
9:   Sample a batch of noise  $z$  from the prior  $p_z$ 
10:  Update the generator by ascending its stochastic gradient using Adam
11: end for
```

For the practical implementation of the conditional models, both the generator G and the discriminator D are structured as 3-layer feed-forward neural networks. To guarantee sufficient representational capacity for learning complex multivariate distributions and time-series dynamics, each hidden layer consists of over 100 nodes and utilizes a `LeakyReLU` activation function with a negative slope parameter of $\alpha = 0.1$. Furthermore, the input space for the generator is defined by a 30-dimensional random noise vector.

Empirical evaluations confirm that under these optimized settings, the CWGAN framework yields results that are robust and highly comparable to the standard CGAN approach.

In order to complete the view of CGAN in financial time series data modeling, it is necessary to conduct simulation studies on AR and GARCH-type time series, where the first have strong autocorrelation and the second have strong volatility dynamics.

2.3.2 CGAN for VAR time series

The Vector Autoregressive (VAR) model represents a natural generalization of the univariate Autoregressive (AR) model. It is extensively applied in econometrics and financial analysis to capture the linear interdependencies among multiple time series.

Definition 2.3.2 (VAR). *A VAR model of order p , commonly denoted as $VAR(p)$, is defined by the following equation:*

$$X_t = c + \sum_{i=1}^p a_i X_{t-i} + \varepsilon_t \quad (2.1)$$

where X_t is the multidimensional time series vector at time t , c is a constant intercept vector, $\{a_i \mid i = 1, \dots, p\}$ represent the autocorrelation parameter matrices, and ε_t denotes the underlying stochastic noise term.

Continuous Conditioning

To effectively model time-dependent dynamics, generative networks can be conditioned on continuous variables. A standard and highly effective approach in financial time series generation involves using lagged historical values as the conditioning input.

Empirical studies on synthetic Vector Autoregressive (VAR) processes demonstrate that by conditioning a CGAN on previous time steps (for example through a 1-time-lag sliding window), the network successfully captures the underlying autoregressive properties. Specifically, the generated sequences have been proven to accurately replicate the theoretical autocorrelation structures of the original data, successfully modeling the temporal dependencies even in the presence of heavy-tailed noise distributions.

Beyond linear autocorrelation, further evaluations assess the model's capacity for 1-time-horizon predictions when the underlying noise exhibits dynamic, time-varying volatility. Statistical comparisons between the generated and theoretical conditional distributions confirm that the model accurately captures the fundamental moments, such as mean and variance.

However, a critical limitation emerges regarding the modeling of extreme events. The network's ability to accurately estimate higher-order moments, specifically the excess kurtosis required to reflect heavy-tailed distributions, is strictly dependent on the training sample size. Empirical evidence demonstrates that training on constrained datasets leads to a systematic underestimation of heavy tails. Conversely, when provided with a sufficiently large sample size, the CGAN successfully learns the complete dependent structure, accurately reproducing the extreme tail behaviors that characterize real financial time series.

In summary, empirical evidence strongly supports the versatility of the CGAN framework to capture diverse time-varying noise dynamics. Provided with an adequate sample size, the network effectively internalizes the complex dependent structures of autoregressive processes. Ultimately, it proves highly capable of generating accurate 1-time-horizon predictive distributions that faithfully replicate the key statistical properties of the original financial data.

Continuous conditioning for regime switching TS

Financial markets naturally have structural breaks, commonly referred to as "regime switches," where the economy transitions between distinct states, most notably from normal market periods to periods of severe

financial stress or crises. During these regime changes, the underlying statistical properties of asset returns, including volatility, autocorrelation, and tail risk, are drastically different.

Empirical simulations based on switching Vector Autoregressive (VAR) processes demonstrate the CGAN's capability to identify these structural breaks. When conditioned solely on continuous lagged values, the network can successfully detect the regime switch effect, provided that the statistical characteristics of the two market states are sufficiently distinct. In such cases, the CGAN correctly adapts its generative output to match the specific variance and heavy-tailed behavior of the current market environment.

However, market regimes often overlap significantly, making them difficult to disentangle using only continuous historical data. For complex generalized regime-switching scenarios, relying exclusively on continuous conditions proves insufficient.

Categorical conditioning for regime switching TS

To resolve the challenge of overlapping regimes, categorical variables can be introduced as explicit conditions, effectively instructing the CGAN about which specific market state (normal or crisis period) it should simulate. However, a significant architectural limitation arises: when a CGAN is conditioned solely on categorical labels, it defaults to generating independent and identically distributed (i.i.d.) samples. While these samples might correctly match the overarching distribution of the specified regime, they completely fail to preserve the crucial time-series autoregressive structures.

To overcome this, the training data must be fundamentally reformatted to integrate an explicit time dimension alongside the categorical indicators. By restructuring the inputs using a multi-dimensional sliding-window approach (combining categorical variables with T -time-lag snapshots), the CGAN is forced to learn the temporal dependencies within each specific market regime. Extensive simulations confirm that under this hybrid conditioning framework, the network successfully reconstructs both the localized time-varying volatility dynamics and the heavy-tailed distributions inherent to financial crises, simultaneously preserving the original correlation structures of the underlying data.

2.3.3 CGAN for GARCH time series

While autoregressive models effectively capture dependencies in the mean of a time series, financial data is predominantly characterized by a time-varying dependent structure in its variance 2.1.2. To model this phenomenon, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is widely employed.

Definition 2.3.3. A multivariate GARCH(p, q) process $\{X_t, t = 1, \dots, T\}$ is defined by its conditional variance equation:

$$\sigma_t^2 = c + \sum_{i=1}^p a_i X_{t-i}^2 + \sum_{i=1}^q b_i \sigma_{t-i}^2, \quad X_t \sim N(0, \sigma_t^2) \quad (2.2)$$

where c is a constant, $\{a_i\}$ and $\{b_i\}$ are the model parameters, and ε_t represents the underlying white noise.

When applying the CGAN to this type of financial data, using the historical variance as a condition for the network proves to be highly effective. By feeding this specific volatility information to the model, the CGAN easily learns how the market's variance fluctuates over time.

The most significant advantage of this generative approach over classical mathematical models is its flexibility. Traditional GARCH models require rigid rules and assumptions about the exact statistical shape of the data. Conversely, the CGAN learns directly from the data itself without needing these predefined formulas. Automatically understands the relationships between different financial assets and accurately recreates

the probability of extreme market events, but not the dimensions of the crisis. Ultimately, this allows the network to generate highly realistic future market scenarios.

2.3.4 CGAN Applications to Real Financial Data

The transition from synthetic environments to real-world financial data further validates the efficacy of conditional generative models. In the context of portfolio analysis using actual stock market data, the CGAN framework demonstrates significantly superior performance compared to the traditional Historical Simulation (HS) method.

However, empirical studies highlight a critical constraint regarding data availability: when the training sample size is insufficiently large, the model tends to underestimate actual market volatility. Conversely, when trained on sufficiently expansive datasets, CGANs exhibit remarkable accuracy in Value-at-Risk (VaR) and Expected Shortfall (ES) simulations. Under these optimal data conditions, the generative approach strongly outperforms HS, yielding risk estimates that closely align with realized empirical values.

Chapter 3

Heavy-Tail problem

As discussed in the preceding chapter, while architectures represent significant advancements in modeling complex temporal dynamics, they inherently struggle with the “extreme value problem”. By design, standard generative algorithms are structurally biased toward the bulk of the probability mass. Consequently, they tend to optimize for the mean and the central moments of the distribution, systematically underestimating the probability density in the extreme tails.

However, in a multitude of practical applications, the accurate simulation of extreme values is not merely a theoretical edge case but the fundamental objective. In complex dynamical systems, the most critical, risky, and influential scenarios lie precisely within these tail regions.

This is acutely true in quantitative finance and market risk management, where catastrophic tail events, such as sudden market crashes or severe liquidity crises, dictate the ultimate risk profile of a portfolio. Similarly, in domains like seismology and climatology, modeling the extremes is essential for predicting rare but devastating occurrences. Therefore, transitioning from mean-centric generation to tail-focused simulation is a strict mathematical and practical necessity, requiring specialized architectures capable of capturing extreme multivariate dependencies.

3.1 EXGAN

To address the inherent limitations of standard generative models in capturing heavy-tailed phenomena, the ExGAN architecture was introduced as a new approach specifically designed for extreme sample generation. The innovation proposed by Bhatia et al. [2] lies in coupling the Generative Adversarial Network framework with Extreme Value Theory (EVT), a rigorous probabilistic methodology dedicated to modeling the asymptotic behavior of distribution tails.

By bridging deep learning with EVT, ExGAN effectively resolves the mean-seeking bias of traditional GANs. Furthermore, this architecture provides significant operational flexibility for practical applications: it allows the end-user to explicitly specify both the mathematical measure of extremeness and the precise target probability threshold at which they wish to sample the extreme events.

To solve the problem of not having enough data in the extreme tails, the ExGAN framework relies on two key ideas:

Firstly, it uses a novel *data distribution shifting* approach. Since extreme events are very rare, the training data in the tails is extremely scarce. To fix this, ExGAN gradually shifts the data distribution toward the direction of increasing extremeness. This allows the network to train in a robust and stable manner, forcing the model to fit the tail of the distribution rather than its bulk.

Secondly, to generate data at any given extremeness probability, ExGAN uses an *EVT-based conditional generation*. The network is trained as a Conditional GAN where the conditioning variable is the extremeness

statistic. Combined with keeping track of the distribution shifts performed in the first step, this allows the model to generate new, realistic samples at the exact extremeness probability chosen by the user.

3.1.1 Extreme Value Theory

To rigorously model the behavior of data in the tails, we rely on Extreme Value Theory (EVT). A foundational tool in this field is the Generalized Pareto Distribution (GPD). The GPD is defined by two key parameters: the scale parameter σ and the shape parameter ξ .

Definition 3.1.1 (GPD). *The cumulative distribution function of the GPD is expressed as:*

$$G_{\sigma,\xi}(x) = \begin{cases} 1 - \left(1 + \frac{\xi \cdot x}{\sigma}\right)^{-1/\xi} & \text{if } \xi \neq 0 \\ 1 - \exp\left(-\frac{x}{\sigma}\right) & \text{if } \xi = 0 \end{cases}$$

A very useful property of the GPD is its flexibility. It generalizes both Pareto distributions, which represent heavy tails, and exponential distributions, which represent light, exponentially decaying tails. This allows the GPD to smoothly interpolate between different types of tail behaviors. Furthermore, due to its “universality” property, the GPD can approximate the tails of a very large class of distributions with an error that approaches zero.

In practice, the application of this distribution relies on the Peaks over Threshold (POT) method. As proven by Pickands (1975) and Balkema and De Haan (1974), the excesses over a sufficiently large threshold u , denoted as $X - u$, closely follow a Generalized Pareto Distribution with parameters $\sigma(u)$, ξ . In real-world applications, this threshold u is commonly set around the 95th percentile of the dataset, and the remaining parameters are estimated using maximum likelihood estimation.

3.1.2 Architecture

It’s now necessary to first define the formal problem it aims to solve. Given a training dataset $\{x_1, \dots, x_n\}$ drawn from an unknown distribution $\mathcal{D}$. Along with this data, a user-defined extremeness measure is chosen and denoted as $E(x)$.

In addition to the data, the user specifies a target extremeness probability, $\tau \in (0, 1)$. This parameter defines exactly how rare and extreme the generated samples should be.

Given these inputs, the ultimate goal is to generate synthetic samples x' that satisfy two strict criteria simultaneously:

1. **Realism:** The generated samples must look authentic and remain hard to distinguish from the original training data.
2. **Targeted Extremeness:** The generated samples must accurately reflect the chosen extremity level. Formally, the probability that a random real sample x is more extreme than our generated sample x' should be as close as possible to the target threshold ($P_{x \sim \mathcal{D}}(E(x) > E(x')) \approx \tau$).

The architecture is divided in two algorithms.

1–Distribution shifting

An immediate challenge in training generative models for extreme values is ensuring that the model accurately replicates the extreme tails rather than the bulk of the distribution. Because the vast majority of historical data lies within the central mass, the number of samples available in the tails is inherently limited.

To address this limitation, a novel *Distribution Shifting* approach is introduced, parameterized by a shift parameter $c \in (0, 1)$. The core strategy consists of repeatedly shifting the dataset by filtering away the less extreme $(1 - c)$ portion of the data, and subsequently generating new data points to return the dataset to its original size. To maintain the desired proportion of original data points from the initial set $\mathcal{X}$, a stratified filtering approach is adopted, meaning that the original data and the generated data are filtered separately.

The systematic implementation of this strategy is formalized in algorithm below:

Algorithm 6 Distribution Shifting

Require: Dataset $\mathcal{X}$, extremeness measure E , shift parameter c , iteration count k

- 1: Sort $\mathcal{X}$ in decreasing order of extremeness
 - 2: Initialize $\mathcal{X}_s \leftarrow \mathcal{X}$
 - 3: **for** $i = 1$ **to** k **do**
 - 4: **Shift the data distribution by a factor of c :**
 - 5: Train generative network G and critic D on $\mathcal{X}_s$
 - 6: $\mathcal{X}_s \leftarrow$ top $\lfloor c^i \cdot n \rfloor$ extreme samples of $\mathcal{X}$
 - 7: Generate $\lceil (n - \lfloor c^i \cdot n \rfloor) \cdot \frac{1}{c} \rceil$ data points using G
 - 8: Insert the most extreme $n - \lfloor c^i \cdot n \rfloor$ generated samples into $\mathcal{X}_s$
 - 9: **end for**
-

Each iteration shifts the training distribution toward its upper tail by a factor of c . By performing k iterations, the distribution is shifted sufficiently so that the target probability τ is no longer located in the extreme tail of the resulting training set. This iterative shifting process ensures that the generative network always has a sufficient volume of data to train in a stable manner, while gradually approaching the true tail of the distribution.

Furthermore, for the sake of computational efficiency, successive iterations of the generator can be trained via a "warm start" mechanism. This is achieved by initializing the parameters of the network using the weights of the model trained in the previous iteration.

The generative network used in the article is the DCGAN network that is a GAN network that uses CNN architecture rather than an MLP, but any generative network could be used.

2-EVT-based Conditional Generation

The subsequent objective involves generating samples at the user-specified extremeness probability τ . This is achieved by training a conditional GAN using the extremeness measure as the conditioning variable. By leveraging Extreme Value Theory (EVT) analysis in conjunction with the knowledge of the distribution shifts performed, the precise extremeness level required for conditioning can be mathematically determined to match the target probability.

Specifically, after k shifts, the corresponding extremeness probability in the current shifted distribution becomes $\tau' = \frac{\tau}{c^k}$. Consequently, the sampling process targets the shifted distribution at this adjusted probability level τ' utilizing EVT frameworks.

The systematic execution of this procedure is structured as follows:

1. Extremeness values are computed using the metric E for each point in the shifted dataset $\mathcal{X}_s$, defining $e_i = E(x_i) \forall x_i \in \mathcal{X}_s$.
2. An EVT analysis is conducted on the computed sequence $\{e_1, \dots, e_n\}$ to fit the Generalized Pareto Distribution (GPD) parameters σ, ξ via maximum likelihood estimation.

3. A conditional DCGAN (comprising Generator G_s and Discriminator D_s) is trained on $\mathcal{X}_s$, where the conditional input for G_s is sampled from the fitted GPD. The discriminator D_s simultaneously processes the target input and its corresponding extremeness score e or $E(x)$.

To enforce structural compliance, an auxiliary loss function $\mathcal{L}_{\text{ext}}$ is incorporated into the global GAN objective:

$$\mathcal{L}_{\text{ext}} = \mathbb{E}_{z,e} \left[\frac{|e - E(G_s(z, e))|}{e} \right]$$

where z is drawn from a multivariate standard normal distribution and e is sampled from a GPD with parameters σ, ξ . This loss function assumes E to be fully differentiable. The explicit minimization of $\mathcal{L}_{\text{ext}}$ reduces the discrepancy between the target extremeness e and the actual extremeness of the generated sample $E(G_s(z, e))$, preventing the synthesis of uncalibrated outputs.

Finally, the exact extremeness level e' corresponding to the adjusted probability τ' is derived through the inverse cumulative distribution function of the GPD $G_{\sigma, \xi}^{-1}$:

$$e' = G_{\sigma, \xi}^{-1}(1 - \tau')$$

Targeted inference is then conducted by sampling directly from the conditional generator at the exact level e' .

The operational pipeline is formalized below in Algorithm.

Algorithm 7 EVT-based Conditional Generation

Require: Shifted dataset $\mathcal{X}_s$, extremeness measure E , adjusted extremeness probability τ'

- 1: Compute extremeness values $e_i = E(x_i) \forall x_i \in \mathcal{X}_s$
 - 2: Fit GPD parameters σ, ξ using maximum likelihood estimation on $\{e_1, \dots, e_n\}$
 - 3: Train conditional DCGAN (G_s and D_s) on $\mathcal{X}_s$, where the conditioning input for G_s is sampled from a GPD with parameters σ, ξ
 - 4: Extract required extremeness level: $e' \leftarrow G_{\sigma, \xi}^{-1}(1 - \tau')$
 - 5: Sample from G_s conditioned on the exact extremeness level e'
-

3.1.3 Advantages and Limitations

The implementation of the ExGAN framework introduces several significant advantages in the field of generative modeling applied to rare events. Foremost, the architecture represents a novel deep learning-based approach engineered for the explicit synthesis of extreme samples, achieved through the joint integration of data distribution-shifting mechanisms and parametric Extreme Value Theory (EVT) analysis. A critical operational advantage of this methodology is its ability to conduct constant-time sampling. Specifically, the framework can synthesize extreme data points in constant time with respect to the target extremeness probability τ , providing a substantial algorithmic advancement over baseline rejection-sampling approaches whose computational complexity scales poorly as $\mathcal{O}(\frac{1}{\tau})$.

Empirically, this translation into continuous-time efficiency yields highly realistic samples, validated through both visual inspection and rigorous quantitative metrics, while operating at a speed that outperforms traditional baseline methodologies by at least three orders of magnitude for extreme probability thresholds of 0.01 and beyond.

Despite these prominent operational strengths, the framework is bounded by theoretical and practical limitations that must be addressed. A primary challenge lies in the strict necessity of correctly defining the mathematical extremeness measure $E(x)$ and the target probability τ . Because the entire conditional generation and auxiliary loss structures rely on the differentiability and coherence of $E(x)$, any error in

specification of this metric or an analytically unfeasible choice of τ can destabilize the adversarial training, leading to uncalibrated outputs or failure in tail convergence.

Furthermore, severe architectural difficulties emerge when extending this framework to multivariate tasks. While EVT provides robust univariate boundaries via the Generalized Pareto Distribution, modeling complex, high-dimensional multivariate tails requires capturing intricate directional dependencies and extreme copulas. In multivariate financial time series or spatial datasets, the scaling of the distribution-shifting process becomes geometrically more complex, often limiting the model's ability to accurately capture simultaneous extreme shocks across multiple target variables without incurring severe computational bottlenecks or degradation in generative realism.

3.2 Tail-GAN

While the ExGAN framework provides a foundational approach for extreme value generation, it exhibits inherent stability challenges and severe structural limitations when extended to multivariate contexts. Most notably, its inability to reliably simulate high-dimensional, multi-asset scenarios, a necessity for realistic financial simulation and dynamic portfolio management, restricts its practical applicability in modern quantitative finance.

To systematically overcome these limitations, Cont et al. proposed Tail-GAN [4], a novel data-driven architecture engineered to simulate the realistic joint dynamics of complex financial components. The fundamental innovation of this approach consists of embedding established risk measures directly within the generative framework. Specifically, Tail-GAN exploits the joint property of Value-at-Risk (VaR) and Expected Shortfall (ES). By integrating these metrics into GAN framework, the network learns to generate multi-asset price scenarios that strictly preserve the essential tail risk features, setting it apart from standard univariate methodologies.

3.2.1 Some useful definitions and properties

To address the challenges of multivariate scenario generation, some definitions and properties are introduced.

The core objective is to design a loss function for the generative model that explicitly targets performance in specific use cases, ensuring that the synthesized outputs accurately reflect the tail risk across a broad class of benchmark trading strategies.

Formally, the objective of a dynamic scenario generator is to sample from an unknown distribution over a space of scenarios $\mathbf{p} = (\mathbf{p}_t, t \in \mathbb{T}) \in \Omega$, which represent price trajectories for a set of financial assets over a discrete time set $\mathbb{T}$. The practical utility of these scenarios lies in the evaluation of the profit and loss (PnL) for various trading strategies along each trajectory.

Definition 3.2.1. *Given a dynamic or static trading strategy φ^k (where $1 \leq k \leq K$) defines a mapping:*

$$\begin{aligned} \Pi^k : \Omega &\rightarrow \mathbb{R} \\ \mathbf{p} = (\mathbf{p}_t, t \in \mathbb{T}) &\mapsto \Pi^k(\mathbf{p}) = \sum_i \varphi^k(t_i, \mathbf{p}) \cdot (\mathbf{p}_{t_{i+1}} - \mathbf{p}_{t_i}) \end{aligned}$$

where $\Pi^k(\mathbf{p})$ denotes the terminal PnL of the strategy φ^k under the specific scenario $\mathbf{p}$.

A probability measure $\mathbb{P}$ on the high-dimensional market scenario space Ω naturally induces a one-dimensional distribution for the scalar random variable $\Pi^k(\mathbf{p})$.

Consequently, each trading strategy φ^k effectively acts as a projection operator, mapping the complex, high-dimensional probability measure $\mathbb{P}$ into a tractable, one-dimensional distribution.

By defining a set of user-defined benchmark trading strategies $\{\varphi^k\}_{k=1,\dots,K}$, the algorithm can project the high-dimensional probability measure $\mathbb{P}$ onto K scalar random variables $\{\Pi^k(\mathbf{p})\}_{k=1,\dots,K}$. This approach leads to a significant dimension reduction: the high-dimensional learning problem is reduced to learning K one-dimensional distributions, which serve as tractable features of $\mathbb{P}$.

The training process of the scenario generator is thus guided by these associated one-dimensional loss distributions. To accurately capture extreme market events, the framework exploits the joint property of Value-at-Risk (VaR) and Expected Shortfall (ES); the definitions were already given respectively in 2.1.3 and 2.1.4. This mathematical property allows for the formulation of a training criterion that is highly sensitive to the tail risk of the projected distributions. The ultimate result is a GAN architecture that learns to simulate multivariate price scenarios while strictly preserving the tail risk characteristics inherent in the selected benchmark strategies.

3.2.2 Tail risk measure score function

In quantitative finance, risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES) can be formally represented as statistical functionals. Let $\mathcal{F} \subset \mathcal{P}(\mathbb{R})$ be a set of probability distributions; a statistical functional is defined as a mapping $T : \mathcal{F} \rightarrow \mathbb{R}^d$.

A fundamental property for the robust estimation of these functionals is *elicitability*.

Definition 3.2.2. A statistical functional $T : \mathcal{F} \rightarrow \mathbb{R}^d$ is considered elicitable if there exists a scoring function $S : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that the functional minimizes the expected score:

$$T(\mu) = \arg \min_{x \in \mathbb{R}^d} \int S(x, y) \mu(dy)$$

When the minimizer of this expectation is unique, the score function S is termed strictly consistent for T .

The practical power of elicibility lies in its empirical application: given an independent and identically distributed (IID) sample $Y_1, \dots, Y_n \sim \mu$, a consistent estimator $\hat{x}_n \rightarrow T(\mu)$ can be obtained by minimizing the empirical risk:

$$\hat{x}_n = \arg \min_{x \in \mathbb{R}^d} \sum_{i=1}^n S(x, Y_i)$$

While standard statistical functionals such as the mean and the median are easily shown to be elicitable, the same does not hold for all risk measures. Specifically, it has been demonstrated that Expected Shortfall (ES) is not elicitable on its own. However, a critical breakthrough in risk management literature established that while ES alone lacks this property, the pair $(\text{VaR}_\alpha(\mu), \text{ES}_\alpha(\mu))$ is *jointly elicitable*.

This joint elicibility is governed by a specific family of strictly consistent score functions, formalized in the following theorem.

Theorem 3.2.1 (Fissler and Ziegel, 2016). Assume a probability measure μ such that $\int |x| \mu(dx) < \infty$. Let $H_2 : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly convex function, and let $H_1 : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that the mapping

$$v \mapsto R_\alpha(v, e) := -\frac{1}{\alpha} v H_2'(e) + H_1(v)$$

is strictly increasing for each $e \in \mathbb{R}$. Under these conditions, the score function defined as:

$$\begin{aligned} S_\alpha(v, e, x) &= (\mathbb{I}_{\{x \leq v\}} - \alpha)(H_1(v) - H_1(x)) \\ &\quad + \frac{1}{\alpha} H_2'(e) \mathbb{I}_{\{x \leq v\}}(v - x) + H_2(e)(e - v) - H_2(e) \end{aligned}$$

is strictly consistent for the joint functional $(\text{VaR}_\alpha(\mu), \text{ES}_\alpha(\mu))$.

Consequently, the true VaR and ES can be jointly recovered via the minimization problem:

$$(\text{VaR}_\alpha(\mu), \text{ES}_\alpha(\mu)) = \arg \min_{(v,e) \in \mathbb{R}^2} \int S_\alpha(v, e, x) \mu(dx) \quad (3.1)$$

The computation of the joint estimator 3.1 requires the optimization of the expected score function for a given one-dimensional distribution μ . Formally, this involves minimizing the functional:

$$s_\alpha(v, e) := \int S_\alpha(v, e, x) \mu(dx)$$

While any choice of the functions H_1 and H_2 that satisfies the theoretical conditions of 3.2.1 leads to consistent estimators, different parametrizations generate optimization problems with vastly different topographies. Because the stability of the neural network training heavily depends on the gradients of this function, some landscapes are inherently easier to optimize than others.

To ensure stability, the framework adopts a specific formulation of the score function originally proposed by Acerbi and Szekely, which is widely recognized in financial practice for backtesting purposes.

Definition 3.2.3. *This specific score function is defined as:*

$$S_\alpha(v, e, x) = \frac{W_\alpha}{2} (\mathbb{I}_{\{x \leq v\}} - \alpha)(x^2 - v^2) + \mathbb{I}_{\{x \leq v\}} e(v - x) + \alpha e \left(\frac{e}{2} - v \right)$$

subject to the condition that $\frac{\text{ES}_\alpha(\mu)}{\text{VaR}_\alpha(\mu)} \geq W_\alpha \geq 1$.

This formulation is a special case of the general Fissler-Ziegel family 3.2.1, obtained by setting H_1 and H_2 as:

$$H_1(v) = -\frac{W_\alpha}{2} v^2, \quad H_2(e) = \frac{\alpha}{2} e^2, \quad \text{with} \quad \frac{\text{ES}_\alpha(\mu)}{\text{VaR}_\alpha(\mu)} \geq W_\alpha \geq 1$$

Under this parametrization, the score function satisfies the strict consistency conditions on the domain $\{(v, e) \in \mathbb{R}^2 \mid W_\alpha v \leq e \leq v \leq 0\}$ (Reference 3.2.1).

Furthermore, this specific choice leads to an optimization problem with highly desirable convexity properties, which are formalized in the following theorem.

Theorem 3.2.2. *Let $\text{VaR}_\alpha(\mu) < 0$ and $\alpha < 1/2$.*

1. *The expected score $s_\alpha(v, e)$ based on 3.2.3 is strictly consistent for $(\text{VaR}_\alpha(\mu), \text{ES}_\alpha(\mu))$, and its Hessian matrix is positive semi-definite on the region:*

$$\mathcal{B} = \{(v, e) \mid v \leq \text{VaR}_\alpha(\mu), \text{ and } W_\alpha v \leq e \leq v \leq 0\}$$

2. *Assume there exist $\delta_\alpha \in (0, 1)$, $\varepsilon_\alpha \in (0, \frac{1}{2} - \alpha)$, $z_\alpha \in (0, \frac{1}{2} - \alpha)$, and $W_\alpha > \frac{1}{\sqrt{\alpha}}$ such that the probability density satisfies:*

$$\frac{\mu(dx)}{dx} \geq \delta_\alpha \quad \text{for } x \in [\text{VaR}_\alpha(\mu), \text{VaR}_{\alpha+\varepsilon_\alpha}(\mu)]$$

and the Expected Shortfall satisfies:

$$\text{ES}_\alpha(\mu) \geq W_\alpha \text{VaR}_\alpha(\mu) + z_\alpha$$

Then, the Hessian of $s_\alpha(v, e)$ is positive semi-definite on the extended region:

$$\tilde{\mathcal{B}} = \{(v, e) \mid v \leq \text{VaR}_{\alpha+\beta_\alpha}(\mu), \text{ and } W_\alpha v + z_\alpha \leq e \leq v \leq 0\}$$

where the boundary term is defined as $\beta_\alpha = \min \left\{ \varepsilon_\alpha, \frac{z_\alpha \delta_\alpha}{2W_\alpha} \right\}$.

The theoretical implications of theorem 3.2.2 guarantee that the expected score function $s_\alpha(v, e)$ exhibits a well-behaved optimization landscape across the regions $\mathcal{B}$ and $\tilde{\mathcal{B}}$, provided the underlying density conditions are met. Specifically, the global minimizer of the score function is structurally located on the boundary of region $\mathcal{B}$. Furthermore, the extended region $\tilde{\mathcal{B}}$ is proven to contain an open ball centered exactly at this minimizer.

From a numerical optimization perspective, this topological structure ensures that $s_\alpha(v, e)$ maintains a positive semi-definite Hessian matrix in a neighborhood of the minimum. This mathematical guarantee is important for deep learning applications, as it grants highly desirable convergence properties during the gradient descent phase of the generative network.

3.2.3 Definitions of the architecture

Before starting to define the architecture, it is necessary to give the ultimate definition and property of the probability measure using loss quantiles of trading strategies.

Let $\Omega \subset \mathbb{R}_+^{M \times T}$ denote a set of market scenarios, where each element $\mathbf{p} = (\mathbf{p}_{t,m})_{t=1,\dots,T,m=1,\dots,M} \in \Omega$ represents the discrete price trajectories for M financial assets. Let $\mathbb{T} = \{t_i = i\Delta \mid i = 1, \dots, T\}$ define the discrete set of allowable trading times over a finite risk horizon $T_h = \Delta \times T$.

Definition 3.2.4 (Trading strategies). *A trading strategy is formally defined as a non-anticipative mapping:*

$$\begin{aligned} \varphi : \mathbb{T} \times \Omega &\rightarrow \mathbb{R}^M \\ (t, \mathbf{p}) &\mapsto \varphi(t, \mathbf{p}) \end{aligned}$$

where the vector $\varphi(t, \mathbf{p})$ specifies the portfolio holdings allocated by the strategy at time t under the scenario $\mathbf{p} \in \Omega$.

The value of this portfolio at any time t is given by the inner product $V_t(\varphi) = \mathbf{p}_t \cdot \varphi(t, \mathbf{p})$.

Definition 3.2.5 (Self-Financing). *A strategy is self-financing if any variation in the portfolio's value arises exclusively from the accumulation of capital gains.*

Assume that the trading strategy defined above is self-financing.

Definition 3.2.6 (PnL). *The terminal profit and loss over the entire horizon for a given scenario $\mathbf{p} \in \Omega$ can be expressed as:*

$$\Pi(\varphi) = V_T(\varphi) - V_0(\varphi) = \sum_{t_i \in \mathbb{T}} \varphi(t_i, \mathbf{p}) \cdot (\mathbf{p}_{t_{i+1}} - \mathbf{p}_{t_i})$$

Consequently, imposing a specific probability measure $\mathbb{P}$ on the high-dimensional scenario space Ω naturally induces a specific distribution for the one-dimensional random variable $\Pi(\varphi)$.

The theoretical foundation of this generative architecture is based on a critical observation:

Theorem 3.2.3. *Let $\mathcal{S}(\Omega)$ denote the space of all continuous, bounded, and self-financing trading strategies. Any probability measure on Ω is uniquely determined by the family of one-dimensional distributions associated with the terminal PnL variables $\{\Pi(\varphi) \mid \varphi \in \mathcal{S}(\Omega)\}$.*

The assertion that a probability measure can be uniquely determined by its one-dimensional projections is formalized in the following theorem.

Theorem 3.2.4. *Let $\mathbb{P}^1$ and $\mathbb{P}^2$ be two probability measures on the scenario space Ω . If, for any self-financing trading strategy $\varphi \in \mathcal{S}(\Omega)$, the terminal profit and loss $\Pi(\varphi)$ exhibits the same distribution under both measures, then the two probability measures are identical:*

$$(\forall \varphi \in \mathcal{S}(\Omega), \quad \text{Law}(\Pi(\varphi), \mathbb{P}^1) = \text{Law}(\Pi(\varphi), \mathbb{P}^2)) \implies \mathbb{P}^1 = \mathbb{P}^2$$

For this theoretical guarantee to hold, it is insufficient to consider only static portfolios (only buy-and-hold strategies), as this would only enforce equality across the terminal distributions at the final horizon T_h . To ensure pathwise equivalence across the entire temporal evolution, it is imperative that the space $\mathcal{S}(\Omega)$ includes dynamic trading strategies.

Because any one-dimensional distribution is entirely characterized by its quantiles, Proposition 3.2.4 implies that a probability measure $\mathbb{P}$ on Ω is uniquely identified by the loss quantiles for all self-financing strategies.

Corollary 3.2.1. *Denoting by $Q_\alpha(X, \mathbb{P})$ the quantile of a random variable X at confidence level $\alpha \in [0, 1]$ under the measure $\mathbb{P}$, the condition can be reformulated as:*

$$(\forall \varphi \in \mathcal{S}(\Omega), \quad \forall \alpha \in [0, 1], \quad Q_\alpha(\Pi(\varphi), \mathbb{P}^1) = Q_\alpha(\Pi(\varphi), \mathbb{P}^2)) \implies \mathbb{P}^1 = \mathbb{P}^2$$

This equivalence establishes an operational principle: the loss quantiles of self-financing strategies actively discriminate between probability measures on the high-dimensional space $\Omega \subset \mathbb{R}_+^{M \times T}$.

Consequently, the complex task of learning a multidimensional probability measure can be rethought as learning and matching the quantiles of one-dimensional loss distributions. These quantiles serve as highly tractable features possessing a straightforward financial interpretation; specifically, quantile levels for α close to 0 or 1 directly reflect the tail risk of a given strategy.

Based on this principle, a learning framework can be constructed using a discrete set $\{\varphi^k\}_{k=1, \dots, K}$ of self-financing **benchmark strategies**. These strategies can be explicitly defined by the end-user according to the specific market dynamics they wish to analyze. To ensure strict comparability, each benchmark strategy is allocated the identical initial capital. Each strategy φ^k defines a projection mapping:

$$\begin{aligned} \Pi^k : \Omega &\rightarrow \mathbb{R} \\ \mathbf{p} = (\mathbf{p}_t, t \in \mathbb{T}) &\mapsto \Pi^k(\mathbf{p}) = \sum_{t_i \in \mathbb{T}} \varphi^k(t_i, \mathbf{p}) \cdot (\mathbf{p}_{t_{i+1}} - \mathbf{p}_{t_i}) \end{aligned}$$

where $\Pi^k(\mathbf{p})$ represents the terminal profit of strategy φ^k under scenario $\mathbf{p}$.

The algorithm thus attempts to learn the true measure $\mathbb{P}$ by matching the tail quantiles of these K projected scalar random variables $\{\Pi^k(\mathbf{p})\}_{k=1, \dots, K}$.

To comprehensively capture the properties of the price scenarios, the benchmark set must incorporate both static and dynamic trading strategies. Static portfolios are instrumental in exploring the correlation structure among assets, whereas dynamic configurations, such as mean-reversion or trend-following strategies, probe the temporal properties and path dependencies of the market. Furthermore, these strategies serve as an internal validation mechanism: a scenario generator is deemed adequate if the Value-at-Risk (VaR) or Expected Shortfall (ES) estimates derived from its output scenarios meet a predefined accuracy threshold for all benchmark strategies.

This mathematical and financial foundation sets the stage for an interpretable adversarial training paradigm. The architecture is structured as an iterative minimax game between a generator and a discriminator. The generator is tasked with simulating synthetic price scenarios, while the discriminator evaluates the quality of these samples specifically through the lens of tail risk measures (VaR and ES). To optimize these networks, a tail-sensitive objective function is employed, which leverages the joint elicibility of VaR and ES to guarantee the consistency of the estimator during training. Now will follow the construction of these elements, starting with the discriminator.

Discriminator

In a theoretical setting, the optimal discriminator D is designed to take the true distribution of strategy PnLs as input and output two specific risk metrics (the estimated VaR_α and ES_α) for each of the K benchmark strategies. This is achieved by minimizing the expected joint score function S_α .

However, because the true underlying probability measure $\mathbb{P}$ is inaccessible in practice, it is impossible to evaluate the theoretical expectation. Consequently, an empirical, sample-based formulation is required. The discriminator is trained using a finite batch of generated scenarios $[\mathbf{p}_i]_{i=1}^n$ of fixed size n .

Definition 3.2.7 (Minimax D). *The empirical optimization problem is formulated as finding the optimal parameters D^* such that:*

$$D^* \in \arg \min_D \frac{1}{K} \sum_{k=1}^K \frac{1}{n} \sum_{i=1}^n \left[S_\alpha \left(\overbrace{D \left(\underbrace{[\Pi^k(\mathbf{p}_j)]_{j=1}^n}_{\text{strategy PnL samples}} \right)}^{\text{VaR and ES prediction from } D}; \Pi^k(\mathbf{p}_i) \right) \right] \quad (3.2)$$

To solve the minimization problem in (3.2), the search space for D is restricted to the class of Lipschitz continuous functions parameterized by a deep neural network architecture. For a given strategy k , the network maps an input vector of dimension $\tilde{n}_1 = n$ (the batch of PnL samples) to an output vector of dimension $\tilde{n}_L = 2$ (the estimated VaR and ES).

A critical architectural consideration arises from the nature of Value-at-Risk. Since the empirical α -VaR of a distribution can be closely approximated by the $[\alpha n]$ -th smallest value in a sample of size n , the discriminator's mapping must be strictly permutation-invariant with respect to the ordering of the input samples. To achieve this while preserving the capacity for backpropagation, a differentiable neural sorting function Γ is incorporated into the network's input layer. This sorting mechanism significantly enhances the stability of the discriminator.

Definition 3.2.8 (Discriminator). *Denoting the input vector of PnL samples for strategy k as $\mathbf{x}^k$, the forward pass of the discriminator is defined by:*

$$D(\mathbf{x}^k; \delta) = \mathbf{W}_L \sigma \left(\mathbf{W}_{L-1} \dots \sigma \left(\mathbf{W}_1 \Gamma(\mathbf{x}^k) + \mathbf{b}_1 \right) \dots \right) + \mathbf{b}_L \quad (3.3)$$

where $\delta = (\mathbf{W}, \mathbf{b})$ represents the entire set of trainable parameters.

The matrices $\mathbf{W}_l \in \mathbb{R}^{\tilde{n}_l \times \tilde{n}_{l-1}}$ and vectors $\mathbf{b}_l \in \mathbb{R}^{\tilde{n}_l}$ denote the weights and biases of the l -th layer, respectively. The non-linear activation function $\sigma(\cdot)$ is applied element-wise (e.g., ReLU, Leaky ReLU, or tanh).

Definition 3.2.9 (Space $\mathcal{D}$). *The formal hypothesis space $\mathcal{D}$ encompassing this class of discriminators is defined as:*

$$\mathcal{D}(\tilde{L}, \tilde{n}_1, \dots, \tilde{n}_{\tilde{L}}) = \left\{ D : \mathbb{R}^n \rightarrow \mathbb{R}^2 \mid D \text{ follows the architecture in (3.3)} \right. \\ \left. \text{with } \tilde{L} \text{ layers and } \|\mathbf{W}_l\|_\infty, \|\mathbf{b}_l\|_\infty < \infty \text{ for } l = 1, \dots, \tilde{L} \right\}$$

where $\|\cdot\|_\infty$ denotes the max-norm, guaranteeing the necessary parameter boundedness for theoretical convergence.

Generator and Universal Approximation

The generator is a deep neural network comprising $L \in \mathbb{Z}^+$ layers. Let n_l denote the width of the l -th layer.

Definition 3.2.10 (Generator). *The functional form of the generative mapping is explicitly given by:*

$$G(z; \gamma) = \mathbf{W}_L \sigma \left(\mathbf{W}_{L-1} \dots \sigma \left(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1 \right) \dots + \mathbf{b}_{L-1} \right) + \mathbf{b}_L \quad (3.4)$$

In this formulation, $\gamma := (\mathbf{W}, \mathbf{b})$ encapsulates the complete set of trainable parameters within the network, where $\mathbf{W} = (\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_L)$ and $\mathbf{b} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_L)$. Specifically, the weight matrices and bias vectors reside in $\mathbf{W}_l \in \mathbb{R}^{n_l \times n_{l-1}}$ and $\mathbf{b}_l \in \mathbb{R}^{n_l \times 1}$, respectively, for $l = 1, 2, \dots, L$. The initial dimension $n_0 = N_z$ corresponds to the dimension of the latent input variable $\mathbf{z}$.

Synthetically, the generator is a classical MLP network.

To establish theoretical guarantees such as universal approximation capabilities, the hypothesis space is restricted to a well-defined class of generators $\mathcal{G}$ that satisfy specific regularity conditions.

Definition 3.2.11 (Class of $\mathcal{G}$). *This functional class is mathematically formalized as:*

$$\mathcal{G}(L, n_1, n_2, \dots, n_L) = \left\{ G : \mathbb{R}^{N_z} \rightarrow \mathbb{R}^{M \times T} \mid G \text{ satisfies the architecture in (3.4)} \right. \\ \left. \text{with } L \text{ layers of width } n_l, \text{ and} \right. \\ \left. \|\mathbf{W}_l\|_\infty, \|\mathbf{b}_l\|_\infty < \infty \text{ for } l = 1, \dots, L \right\}$$

For notational convenience the explicit dependency on the neural network parameters γ may be abbreviated to $G_\gamma(\cdot)$, or omitted entirely as $G(\cdot)$ when the context is unambiguous. Furthermore, the probability measure induced by the generator over the high-dimensional space of synthesized price trajectories is denoted as $\mathbb{P}_G$.

To establish the theoretical validity of the proposed architecture, it is essential to demonstrate the universal approximation capabilities of the generator specifically under the Value-at-Risk (VaR) and Expected Shortfall (ES) criteria. The theoretical framework relies on a core set of assumptions regarding the continuity of the portfolio mapping and the regularity of the underlying probability measures.

Let $\mathbb{P}_r$ denote the true target market distribution and $\mathbb{P}_z$ denote the distribution of the latent input noise.

Assumption 1: For every benchmark strategy $k = 1, 2, \dots, K$, the portfolio mapping Π^k is Lipschitz-continuous with respect to the price paths.

Assumption 2: The probability measures $\mathbb{P}_r$ and $\mathbb{P}_z$ on the scenario space Ω (where dimension $N_z = M \times T$) satisfy the following conditions:

- The noise distribution admits a density and belongs to the Wasserstein space $\mathbb{P}_z \in \mathcal{P}^2(\Omega)$.
- The target distribution $\mathbb{P}_r$ possesses a bounded moment of order $\beta > 1$, such that $\int \|x\|^\beta \mathbb{P}_r(dx) < \infty$.

Assumption 3: The one-dimensional random variables $\Pi^k(X)$ admit continuous densities f_k . Furthermore, the density evaluated at the Value-at-Risk is strictly positive: $f_k(\text{VaR}_\alpha(\Pi^k, \mathbb{P}_r)) > 0$.

Theorem 3.2.5 (Universal Approximation under VaR and ES Criteria). *Under Assumptions 1-2-3 for any arbitrary error tolerance $\varepsilon > 0$:*

1. *There exists a fully connected feed-forward neural network G_1 , with depth $L = \mathcal{O}(\log(\varepsilon^{-2}))$, width $N = \mathcal{O}(\varepsilon^{-2 \log 2})$, and ReLU activation functions, such that the distribution $\mathbb{P}_{\nabla G_1}$ generated by its gradient satisfies:*

$$|\text{VaR}_\alpha(\Pi^k, \mathbb{P}_{\nabla G_1}) - \text{VaR}_\alpha(\Pi^k, \mathbb{P}_r)| < \varepsilon$$

2. *There exists a fully connected feed-forward neural network G_2 , with depth $L = \mathcal{O}(\log(\varepsilon^{-\frac{2}{\beta-1}}))$, width $N = \mathcal{O}(\varepsilon^{-\frac{2}{\beta-1} \log 2})$, and ReLU activation functions, such that the distribution $\mathbb{P}_{\nabla G_2}$ generated by its gradient satisfies:*

$$|\text{ES}_\alpha(\Pi^k, \mathbb{P}_{\nabla G_2}) - \text{ES}_\alpha(\Pi^k, \mathbb{P}_r)| < \varepsilon$$

The proof for the universal approximation relies on semi-discrete optimal transport theory. Specifically, it involves constructing a transport map in the form of a gradient of a convex potential, $\Phi = \nabla\psi$, which pushes the source distribution $\mathbb{P}_z$ to the empirical target distribution. Because this potential ψ can be explicitly represented by a finite deep neural network, the generative map (the gradient of the network) satisfies the universal approximation theorem.

Theorem 3.2.5 provides a robust mathematical justification for the architectural design of Tail-GAN. It implies that the gradient of a simple, equal-width feed-forward neural network is structurally capable of generating complex market scenarios that reproduce the tail risk properties of benchmark strategies with arbitrary accuracy. Consequently, the adoption of standard fully connected layers is thoroughly validated. The required structural capacity of the network, determined by its depth and width, scales predictably based on the acceptable error tolerance ε , with an additional dependency on the moment bound β when approximating the Expected Shortfall.

The universal approximation property can be formally extended to encompass a broader class of law-invariant risk measures $\rho : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}$. To establish this result, let $\mathcal{W}_p(\mu, \nu)$ denote the Wasserstein distance of order $p \in [1, \infty)$ between two probability measures μ and ν defined on $\mathbb{R}^d$. The generalization requires a specific continuity condition on the risk functional.

Assumption 3B: The risk measure ρ is Hölder continuous with respect to the 1-Wasserstein distance.

It is possible to note that several prominent risk metrics in quantitative finance satisfy this condition, including the optimized certainty equivalent, spectral risk measures, and utility-based shortfall.

Theorem 3.2.6 (Universal Approximation in General Risk Metrics). *Under the assumption 1-2-3B, for any arbitrary error tolerance $\varepsilon > 0$, there exists a fully connected feed-forward neural network G_3 , equipped with ReLU activation functions, such that the distribution $\mathbb{P}_{\nabla G_3}$ generated by its gradient satisfies:*

$$|\rho(\Pi^k, \mathbb{P}_{\nabla G_3}) - \rho(\Pi^k, \mathbb{P}_r)| < \varepsilon$$

The required structural complexity of the network, specifically its depth L and width N , is determined by the following regimes:

1. When $M = T = 1$ and $1 < \beta \leq 2$:

$$L = \mathcal{O}\left(\log\left(\varepsilon^{-\frac{\beta}{\kappa(\beta-1)}}\right)\right) \quad \text{and} \quad N = \mathcal{O}\left(\varepsilon^{-\frac{\beta}{\kappa(\beta-1)} \log 2}\right)$$

2. When $M = T = 1$ and $\beta \geq 2$:

$$L = \mathcal{O}\left(\log\left(\varepsilon^{-\frac{2}{\kappa}}\right)\right) \quad \text{and} \quad N = \mathcal{O}\left(\varepsilon^{-\frac{2}{\kappa} \log 2}\right)$$

3. When $M \times T \geq 2$ and $\frac{M \times T}{\kappa} + \frac{1}{\beta} < 1$:

$$L = \mathcal{O}\left(\log\left(\varepsilon^{-\frac{M \times T}{\kappa}}\right)\right) \quad \text{and} \quad N = \mathcal{O}\left(\varepsilon^{-\frac{M \times T}{\kappa} \log 2}\right)$$

4. When $M \times T \geq 2$ and $\frac{M \times T}{\kappa} + \frac{1}{\beta} \geq 1$:

$$L = \mathcal{O}\left(\log\left(\varepsilon^{-\frac{\beta}{\kappa(\beta-1)}}\right)\right) \quad \text{and} \quad N = \mathcal{O}\left(\varepsilon^{-\frac{\beta}{\kappa(\beta-1)} \log 2}\right)$$

Theorem 3.2.6 reveals that the required depth and width of the neural network are fundamentally driven by the moment bound β of the target distribution, the dimension of the scenario space $M \times T$, and the Hölder

exponent κ . The univariate static case ($M = T = 1$) corresponds to the simulation of a single price value for a single asset, which holds limited practical interest in multivariate finance.

In the more relevant multidimensional framework ($M \times T \geq 2$), if $\beta > \frac{M \times T}{M \times T - 1}$, the network's complexity is governed by the ratio $\frac{M \times T}{\kappa}$, explicitly linking the computational burden to the dimensionality of the price scenario and the regularity of the risk measure. Conversely, when the target distribution $\mathbb{P}_r$ is heavily tailed, meaning $1 < \beta < \frac{M \times T}{M \times T - 1}$, the dimension of the scenario space becomes secondary, and the complexity is dominantly determined by the severity of the tails via the term $\frac{\beta}{\kappa(\beta-1)}$.

Loss function

A loss function is designed to jointly train the generator and the discriminator. The goal of the generator is to generate samples such that the optimal discriminator D^* cannot tell the difference compared to the target distribution. Mathematically, this is formulated as:

Definition 3.2.12 (bi-level formulation).

$$G^* \in \arg \min_{G \in \mathcal{G}} S_\alpha \left(D^* \left(\Pi^k(\mathbf{q}_i); i \in [n] \right), \Pi^k(\mathbf{p}_j) \right)$$

with $\mathbf{q}_i \sim \mathbb{P}_G$, $\mathbf{p}_j \sim \mathbb{P}_r$, for $i, j = 1, 2, \dots, n$, and where D^* is the optimal solution for the discriminator.

The coupled optimization problem 3.2.12 constitutes a bi-level optimization framework, where the problem for G serves as the upper level and the problem for D as the lower level. However, bi-level optimization is often challenging to train in practice. To address this, the following max-min game formulation is introduced:

Definition 3.2.13 (minimax formulation).

$$\begin{aligned} \max_{D \in \mathcal{D}} \min_{G \in \mathcal{G}} \frac{1}{Kn} \sum_{k=1}^K \sum_{j=1}^n \left[S_\alpha \left(D \left(\Pi^k(\mathbf{q}_i); i \in [n] \right), \Pi^k(\mathbf{p}_j) \right) \right. \\ \left. - \lambda S_\alpha \left(D \left(\Pi^k(\mathbf{p}_i); i \in [n] \right), \Pi^k(\mathbf{p}_j) \right) \right] \end{aligned}$$

where $\mathbf{p}_i, \mathbf{p}_j \sim \mathbb{P}_r$ and $\mathbf{q}_i \sim \mathbb{P}_G$ ($i, j = 1, 2, \dots, n$).

The discriminator D takes n PnL samples as the input and aims to provide the VaR and ES values of the sample distribution as the output. The score function S_α is the one previously defined for tail risk measures.

It is now necessary to establish the equivalence between the bi-level optimization problem and the corresponding max-min game.

To start, ideally, the theoretical discriminator $\bar{D}$ takes strategy PnL distributions as inputs and outputs two values for each of the K strategies, aiming to provide the correct $(\text{VaR}_\alpha, \text{ES}_\alpha)$. Mathematically, this amounts to:

$$\bar{D}^* \in \arg \min_{\bar{D}} \frac{1}{K} \sum_{k=1}^K \mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \left[S_\alpha \left(\bar{D} \left(\overbrace{\Pi^k \# \mathbb{P}_r}^{\text{VaR and ES prediction from } \bar{D}} \right); \Pi^k(\mathbf{p}) \right) \right]$$

strategy PnL distribution

A theoretical version of the objective function is first introduced to provide some insights, followed by the practical sample-based version for training. Given two classes of functions $\bar{\mathcal{G}} = \{\bar{G} : \mathbb{R}^{N_z} \rightarrow \Omega\}$ and

$\overline{\mathcal{D}} = \{\overline{D} : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2\}$, the goal is to find a generator $\overline{G}^* \in \overline{\mathcal{G}}$ and a discriminator $\overline{D}^* \in \overline{\mathcal{D}}$ via the following bi-level (or constrained) optimization problem:

$$\overline{G}^* \in \arg \min_{\overline{G} \in \overline{\mathcal{G}}} \frac{1}{K} \sum_{k=1}^K \mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \left[S_\alpha \left(\overline{D}^* (\Pi^k \# \mathbb{P}_{\overline{G}}), \Pi^k(\mathbf{p}) \right) \right] \quad (3.5)$$

where $\mathbb{P}_{\overline{G}} \in \mathcal{P}(\Omega)$ is the distribution of the samples from $\overline{G}$ and

$$\overline{D}^* \in \arg \min_{\overline{D} \in \overline{\mathcal{D}}} \frac{1}{K} \sum_{k=1}^K \mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \left[S_\alpha \left(\overline{D} (\Pi^k \# \mathbb{P}_r), \Pi^k(\mathbf{p}) \right) \right]. \quad (3.6)$$

In the bi-level optimization problem defined by (3.5) and (3.6), the discriminator $\overline{D}^*$ aims to map the PnL distribution to the associated α -VaR and α -ES values. Given the definition of the score function and the joint elicibility property of VaR and ES, it follows that $\overline{D}^*(\cdot) = (\text{VaR}_\alpha, \text{ES}_\alpha)(\cdot)$ according to the theoretical properties previously established. Assuming $\overline{D}^*$ solves (3.6), the generator $\overline{G}^* \in \overline{\mathcal{G}}$ in (3.5) aims to map the noise input to a price scenario that has consistent VaR and ES values of the strategy PnLs applied to $\mathbb{P}_r$.

In practice, constrained optimization problems are difficult to solve. The constraint can instead be relaxed by applying the Lagrangian relaxation method with a dual parameter $\lambda > 0$, leading to a max-min game between the two neural networks $\overline{D}$ and $\overline{G}$:

$$\max_{\overline{D} \in \overline{\mathcal{D}}_0} \min_{\overline{G} \in \overline{\mathcal{G}}} \frac{1}{K} \sum_{k=1}^K \left[\mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \left[S_\alpha \left(\overline{D} (\Pi^k \# \mathbb{P}_{\overline{G}}), \Pi^k(\mathbf{p}) \right) \right] - \lambda \mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \left[S_\alpha \left(\overline{D} (\Pi^k \# \mathbb{P}_r), \Pi^k(\mathbf{p}) \right) \right] \right] \quad (3.7)$$

where

$$\overline{\mathcal{D}}_0 := \left\{ \overline{D} : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2 \mid \exists \mu \in \mathcal{P}(\Omega) \text{ with a finite first moment s.t.} \right. \\ \left. \overline{D}(\Pi^k \# \mu) = (\text{VaR}_\alpha(\Pi^k, \mathbb{P}_r), \text{ES}_\alpha(\Pi^k, \mathbb{P}_r)), k = 1, 2, \dots, K \right\}$$

is a smaller set of discriminators such that $\overline{\mathcal{D}}_0 \subseteq \overline{\mathcal{D}}$.

The set of discriminators $\overline{\mathcal{D}}_0$ may be described as the set of maps $\overline{D} : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2$ which can match the target VaR and ES values at least for some probability measure μ on market scenarios. This represents a feasibility constraint, which can be viewed as a non-degeneracy or expressibility requirement on the map $\overline{D}$, strictly excluding trivial cases such as constant maps.

Theorem 3.2.7 (Equivalence of the Formulations). *Set $N_z = M \times T$. Assume that the noise distribution $\mathbb{P}_z$ has a finite first moment and is absolutely continuous with respect to the Lebesgue measure. Then the max-min game (3.7) with the restricted domain $\overline{\mathcal{D}}_0$ is mathematically equivalent to the bi-level optimization problem (3.5)-(3.6) for any dual parameter $\lambda > 0$.*

Compared to the binary cross-entropy [6] or the Wasserstein distance [1] traditionally employed as loss functions in standard GAN algorithms, the max-min objective function defined in (3.2.13) is explicitly sensitive to tail risk. This mathematical formulation structurally forces the network to produce outputs that accurately approximate the extreme α -ES and α -VaR values, leading to a substantial improvement in generalization capabilities.

Training Algorithm

The complete training procedure for the generative architecture is summarized in Algorithm 8. The iterative process relies on mini-batch stochastic gradient optimization, alternating between gradient ascent for the discriminator (to maximize the tail-sensitive score) and gradient descent for the generator (to minimize it).

Algorithm 8 Tail-GAN Training Procedure

Require: Price scenarios $\mathbf{p}_1, \dots, \mathbf{p}_N \in \Omega$.

Require: Scenario-based P&L computation for trading strategies: $\Pi = (\Pi^1, \dots, \Pi^K) : \Omega \rightarrow \mathbb{R}^K$.

Require: Hyperparameters: learning rate l_D for the discriminator and l_G for the generator; number of training epochs; batch size N_B ; dual parameter $\lambda > 0$.

1: **for** number of epochs **do**

2: **for** $j = 1 \rightarrow \lfloor N/N_B \rfloor$ **do**

3: Generate N_B IID noise samples $\{\mathbf{z}_i\}_{i=1}^{N_B} \sim \mathbb{P}_z$.

4: Sample a batch $B_j \subset \{1, \dots, N\}$ of size N_B from the input data $\{\mathbf{p}_i\}_{i=1}^N$.

5: *% Update the discriminator*

6: Compute the loss of the discriminator on the batch B_j :

$$\mathcal{L}_D(\delta) = \frac{1}{KN_B} \sum_{k=1}^K \sum_{n \in B_j} \left[S_\alpha \left(D_\delta([\Pi^k(G_\gamma(\mathbf{z}_i))]_{i=1}^{N_B}), \Pi^k(\mathbf{p}_n) \right) - \lambda S_\alpha \left(D_\delta([\Pi^k(\mathbf{p}_i)]_{i \in B_j}), \Pi^k(\mathbf{p}_n) \right) \right]$$

7: Update the discriminator via gradient ascent:

$$\delta \leftarrow \delta + l_D \nabla_\delta \mathcal{L}_D(\delta)$$

8: *% Update the generator*

9: Generate a new set of N_B IID noise samples $\{\tilde{\mathbf{z}}_i\}_{i=1}^{N_B} \sim \mathbb{P}_z$.

10: Compute the loss of the generator:

$$\mathcal{L}_G(\gamma) = \frac{1}{KN_B} \sum_{k=1}^K \sum_{n \in B_j} S_\alpha \left(D_\delta([\Pi^k(G_\gamma(\tilde{\mathbf{z}}_i))]_{i=1}^{N_B}), \Pi^k(\mathbf{p}_n) \right)$$

11: Update the generator via gradient descent:

$$\gamma \leftarrow \gamma - l_G \nabla_\gamma \mathcal{L}_G(\gamma)$$

12: **end for**

13: **end for**

14: Set optimal parameters: $\gamma^* = \gamma$, $\delta^* = \delta$.

Ensure: δ^* : trained discriminator weights; γ^* : trained generator weights.

Ensure: Simulated scenarios: $G_{\gamma^*}(\mathbf{z})$ where $\mathbf{z} \sim \mathbb{P}_z$ IID.

3.2.4 Advantages and Limitations

The Tail-GAN architecture presents several significant theoretical and structural advantages over classical generative models in quantitative finance. Primarily, the integration of a tail-sensitive score function, lever-

aging the joint elicibility of Value-at-Risk (VaR) and Expected Shortfall (ES), ensures that the generated scenarios rigorously preserve extreme market events. This stands in stark contrast to standard GANs or Wasserstein GANs, which optimize for average distributional divergence and frequently fail to capture heavy-tailed phenomena. Furthermore, the methodology addresses the curse of dimensionality. By utilizing trading strategies as projection operators, the framework reduces the intractable problem of matching high-dimensional probability measures into the much simpler task of aligning one-dimensional PnL quantiles. This projection not only guarantees universal approximation under specific regularity conditions but also provides the discriminator with highly tractable and financially interpretable features.

Despite its theoretical elegance, the framework is subject to notable limitations. A fundamental constraint lies in its inherent dependence on the predefined set of benchmark strategies $\{\varphi^k\}_{k=1,\dots,K}$. Because the generator is explicitly trained to minimize the risk discrepancy only along these specific one-dimensional projections, it may fail to reproduce multidimensional market dynamics that are orthogonal to the chosen portfolios, potentially introducing "blind spots" in the simulated price trajectories. Consequently, the empirical validity of the generator is strictly bounded by the comprehensiveness of the user-defined strategy set.

From a computational perspective, the architecture introduces substantial complexity. While substituting the bi-level optimization with a max-min game restores tractability, adversarial training remains notoriously susceptible to topological instabilities, such as vanishing gradients or mode collapse, especially when navigating the highly non-convex landscapes typical of financial time series. Finally, the necessity of embedding a differentiable sorting mechanism within the discriminator, an indispensable requirement for empirical quantile estimation, imposes a significant computational overhead during the backpropagation phase, which scales unfavorably as the batch size or the number of benchmark strategies increases.

3.3 WA-GAN

Wasserstein-Aitchison Generative Adversarial Networks (WA-GAN) represent another methodological approach for simulating multivariate extreme events and assessing system resilience under severe stress scenarios and were presented by Lhaut et al.[8]. This architecture integrates the principles of multivariate Extreme Value Theory with adversarial generative models, an extension of ExGAN [2], to rigorously capture and simulate the tail dependence structure of multivariate distributions.

The theoretical core of the WA-GAN procedure relies on the modeling of the angular measure of extreme observations. By applying a transformation into Aitchison coordinates, the distribution is mathematically projected into a linear space. This geometric mapping is a crucial feature of the framework, as it allows the standard Wasserstein-GAN algorithm to be executed seamlessly within a linear vector space, thereby ensuring an accurate and structurally sound generation of new multivariate extreme observations.

3.3.1 Theoretical basis

Let $\mathbf{X} = (X_1, \dots, X_d)$ denote a random vector of interest taking values in $\mathbb{R}^d$. The objective of generating new synthetic samples that accurately replicate the tail behavior of the underlying distribution of $\mathbf{X}$, particularly at extreme levels beyond those observed in the empirical sample, necessitates specific structural assumptions. These assumptions are formally grounded in the mathematical theory of multivariate regular variation.

To effectively implement this theoretical framework, the regular variation hypothesis is postulated on a transformed vector $\mathbf{V} = v(\mathbf{X})$, rather than directly on the original random vector. In this formulation, v represents a carefully chosen transformation such that all d components of the resulting vector $\mathbf{V}$ share an identical marginal distribution.

This specific methodological choice is highly advantageous; the regular variation property depends exclusively on the underlying dependence structure of the original vector $\mathbf{X}$ and not on its marginals. Consequently, the marginal distributions can be mathematically assessed and modeled entirely separately from the core multivariate dependence structure.

Following the data is transformed to standard unit-Pareto margins. This standardization is achieved via the formal transformation mapping:

Definition 3.3.1 (Unit-Pareto).

$$v : \mathbf{x} = (x_j)_{j=1}^d \mapsto v(\mathbf{x}) := \left(\frac{1}{1 - F_j(x_j)} \right)_{j=1}^d$$

where F_j denotes the cumulative distribution function of the component X_j .

It is assumed throughout this mathematical analysis that these marginal distribution functions are strictly continuous.

By applying this specific mapping, the resulting transformed vector $\mathbf{V} = v(\mathbf{X})$ is theoretically guaranteed to possess unit-Pareto marginal distributions:

$$\mathbb{P}(V_j > y) = \frac{1}{y}, \quad \forall y \geq 1, \quad j = 1, \dots, d$$

It is now necessary to describe the first assumption about the asymptotic dependence structure of $\mathbf{V}$.

Assumption 3.3.1 (Multivariate Regular Variation). *There exists a non-zero Borel measure ν on the space $\mathbb{E} := [0, \infty)^d \setminus \{\mathbf{0}\}$ which is finite on all Borel sets bounded away from the origin $\mathbf{0}$, and such that:*

$$\lim_{t \rightarrow \infty} t \mathbb{P}(t^{-1} \mathbf{V} \in B) = \nu(B)$$

for all Borel sets $B \subset \mathbb{E}$ bounded away from $\mathbf{0}$ that satisfy the condition $\nu(\partial B) = 0$, where ∂B denotes the topological boundary of B .

Intuitively, the measure ν characterizes the tail behavior of the random vector $\mathbf{V}$. For large values of $t > 0$, the probability approximation $\mathbb{P}(\mathbf{V} \in tB) \approx \nu(B)/t$ holds, where $tB := \{t\mathbf{x} : \mathbf{x} \in B\}$.

The subsequent generative procedure substantially benefits from an equivalent formulation of Assumption 3.3.1 expressed in polar coordinates with respect to the L_1 -norm, defined as $\|\mathbf{x}\|_1 := \sum_{j=1}^d |x_j|$ for $\mathbf{x} \in \mathbb{R}^d$. Specifically, by exploiting a convenient homogeneity property of the exponent measure ν , the radial components are defined respectively as:

$$R := \|\mathbf{V}\|_1 \quad \text{and} \quad \mathbf{W} := R^{-1} \mathbf{V}$$

Under this coordinate transformation, the angular vector $\mathbf{W}$ takes values strictly within the d -dimensional unit simplex, defined as $\Delta_{d-1} := \{\mathbf{x} \in [0, 1]^d : \|\mathbf{x}\|_1 = 1\}$. Consequently, the multivariate regular variation hypothesis is equivalent to the existence of a probability measure Φ on Δ_{d-1} , such that the conditional distribution of $\mathbf{W}$ given an extreme radial event converges to Φ :

$$\lim_{t \rightarrow \infty} \mathbb{P}(\mathbf{W} \in A \mid R \geq t) = \Phi(A)$$

for any Borel set $A \subseteq \Delta_{d-1}$ satisfying $\Phi(\partial A) = 0$.

Definition 3.3.2 (Angular measure). *The measure Φ is defined as the angular probability measure of the vector $\mathbf{V}$.*

It characterizes the asymptotic behavior of the angular component $\mathbf{W}$ conditional on the radial component R exceeding a high threshold. The family of valid angular measures constitutes a broad nonparametric class. From the underlying unit-Pareto marginal distributions of $\mathbf{V}$, the following moment constraint across the simplex is necessary:

$$\int_{\Delta_{d-1}} w_j d\Phi(\mathbf{w}) = \frac{1}{d}, \quad \forall j \in \{1, \dots, d\}$$

Which yields the asymptotic equivalence:

$$\mathbb{P}(R > t) \sim \frac{d}{t}, \quad \text{as } t \rightarrow \infty$$

From a topological perspective, it is theoretically possible for the angular measure Φ to allocate positive probability mass to the boundary faces of the simplex Δ_{d-1} . In the context of multivariate extremes, such boundary mass corresponds to *extreme directions* scenarios where a specific subset of the components of the original vector $\mathbf{X}$ exhibits extreme values while the remaining components are comparatively negligible.

Accurately modeling these boundary dynamics requires specialized theoretical treatments that fall outside the structural capabilities of the current generative architecture. Consequently, to preserve the methodological validity of the proposed framework, scenarios driven by such extreme directions are explicitly excluded from this analysis.

Assumption 3.3.2 (Concentration of the Angular Measure). *The angular measure Φ is concentrated on the open simplex $\Delta_{d-1}^\circ := \{\mathbf{x} \in (0, 1)^d : \|\mathbf{x}\|_1 = 1\}$. Consequently, if a random vector $\Theta \sim \Phi$, it holds that:*

$$\mathbb{P}(\Theta \in \Delta_{d-1}^\circ) = 1$$

The condition of multivariate regular variation is structurally insufficient to fully model the tail behavior of the original random vector $\mathbf{X}$, as it exclusively governs the asymptotic dependence structure. To construct a comprehensive model, a subsequent theoretical assumption must be imposed on the marginal tails of X_j , explicitly accommodating the possibility of heterogeneous tail decay across dimensions. Let u_{*j} denote the upper endpoint of the marginal support for X_j .

Assumption 3.3.3 (Generalized Pareto Limit of Marginal Tails). *For every dimension $j \in \{1, \dots, d\}$, there exists a parameter $\xi_j \in \mathbb{R}$ and a function $\sigma_j(\cdot) : (-\infty, u_{*j}) \rightarrow (0, \infty)$ such that for all $y > 0$:*

$$\lim_{u \nearrow u_{*j}} \mathbb{P}\left(\frac{X_j - u}{\sigma_j(u)} > y \mid X_j > u\right) = (1 + \xi_j y)_+^{-1/\xi_j}$$

where $a_+ := \max(a, 0)$ for any $a \in \mathbb{R}$. In the boundary case where $\xi_j = 0$, the limit is formally understood as $\exp(-y)$.

By virtue of the foundational Pickands-Balkema-de Haan theorem, Assumption 3.3.3 is mathematically equivalent to stating that each marginal variable X_j belongs to the maximum domain of attraction of a Generalized Extreme Value (GEV) distribution. This limit holds uniformly for $y > 0$ and rigorously justifies the following finite-sample approximation for a sufficiently high threshold u , provided that $1 + \xi_j y / \sigma_j > 0$:

$$\mathbb{P}(X_j - u > y \mid X_j > u) \approx \left(1 + \frac{\xi_j y}{\sigma_j}\right)^{-1/\xi_j}$$

where $\sigma_j > 0$ functions as a constant scale parameter approximating the unknown evaluation $\sigma_j(u)$. The statistical estimation of the bivariate parameter vector (σ_j, ξ_j) is typically executed via Maximum Likelihood Estimation (MLE). This methodology constitutes the classical Peaks-Over-Threshold (PoT) approach for modeling the tails of real-valued univariate data.

Definition 3.3.3 (Univariate GP). *A random variable is said to follow a Generalized Pareto (GP) distribution if its cumulative distribution function is given by:*

$$H_{\sigma,\xi}(y) := 1 - \left(1 + \frac{\xi y}{\sigma}\right)_+^{-1/\xi}, \quad y > 0$$

for a scale parameter $\sigma \in (0, \infty)$ and a shape parameter $\xi \in \mathbb{R}$.

The implications of the previously established assumptions regarding the tail behavior of the random vector $\mathbf{X}$ can be rigorously formulated in terms of Multivariate Generalized Pareto (MGP) distributions. These distributions serve as the natural extension of the univariate limit described in Assumption 3.3.3 to the multivariate setting.

In an arbitrary d -dimensional space, defining an extreme observation is inherently ambiguous. A modern approach leverages the fundamental property that the excess of the random vector $\mathbf{X}$ above a suitably large threshold vector $\mathbf{u}$ asymptotically follows an MGP distribution, provided that a multivariate exceedance is appropriately defined.

Formally, let $\mathbf{u} \in \mathbb{R}^d$ denote a vector of high thresholds strictly below, but asymptotically close to, the upper endpoint vector $\mathbf{u}_* := (u_{*j})_{j=1}^d$, where u_{*j} represents the (possibly infinite) upper endpoint of the marginal support for X_j .

Definition 3.3.4 (Multivariate excess). *The random vector of multivariate excesses is defined as:*

$$\mathbf{Y}_{\mathbf{u}} := \mathbf{X} - \mathbf{u} \mid \mathbf{X} \not\leq \mathbf{u}$$

where the difference is defined component-wise as $\mathbf{X} - \mathbf{u} := (X_j - u_j)_{j=1}^d$.

The condition $\mathbf{X} \not\leq \mathbf{u}$ signifies that the threshold is exceeded in at least one dimension; that is, there exists at least one index $j \in \{1, \dots, d\}$ such that $X_j > u_j$. Throughout this analytical framework, operations on real numbers are naturally extended to vectors in $\mathbb{R}^d$ in a coordinate-wise manner.

Under the foundational assumptions of multivariate regular variation and univariate generalized Pareto marginal limits, the asymptotic distribution of the excess vector $\mathbf{Y}_{\mathbf{u}}$ can be formally established as the threshold approaches the boundary $\mathbf{u} \rightarrow \mathbf{u}_*$.

Theorem 3.3.1 (Weak Convergence of Excesses to the MGP Distribution). *Under the theoretical framework established by Assumptions 3.3.1, 3.3.2, and 3.3.3, the following weak convergence holds:*

$$\frac{\mathbf{X} - \mathbf{b}(t)}{\mathbf{a}(t)} \mid \mathbf{X} \not\leq \mathbf{b}(t) \xrightarrow{\mathcal{L}} \frac{\mathbf{Y}^{\xi} - \mathbf{1}}{\xi}, \quad \text{as } t \rightarrow \infty, \quad (3.8)$$

where $\mathbf{Y}$ denotes a multivariate Pareto random vector characterized by the distribution:

$$\mathbf{Y} \stackrel{\mathcal{L}}{=} (Y\Theta \mid Y\Theta \not\leq \mathbf{1}). \quad (3.9)$$

In this formulation, Y represents a unit-Pareto random variable strictly independent of the angular component $\Theta \sim \Phi$. The vector $\xi = (\xi_j)_{j=1}^d$ contains the marginal shape parameters defined in Assumption 3.3.3. The vector-valued functions $\mathbf{a}(\cdot) = (a_j(\cdot))_{j=1}^d$ and $\mathbf{b}(\cdot) = (b_j(\cdot))_{j=1}^d$ correspond to the normalizing sequences previously introduced.

Consequently, defining a threshold curve parameterized by $t \in (1, \infty)$ such that $\mathbf{u}(t) := \mathbf{b}(t)$, the convergence in (3.8) can be equivalently rewritten as:

$$\frac{\mathbf{Y}_{\mathbf{u}(t)}}{\sigma(\mathbf{u}(t))} \xrightarrow{\mathcal{L}} \frac{\mathbf{Y}^{\xi} - \mathbf{1}}{\xi}, \quad \text{as } t \rightarrow \infty, \quad (3.10)$$

where the vector $\sigma(\cdot) := (\sigma_j(\cdot))_{j=1}^d$ comprises the marginal scaling functions specified in Assumption 3.3.3.

Analogous to the univariate Peaks-Over-Threshold methodology, the scaling function evaluated at the threshold in (3.10) is typically treated as a constant scale parameter vector in practical applications. Therefore 3.3.1 justifies the following finite-sample approximation in distribution for a sufficiently high threshold:

$$\mathbf{Y}_{\mathbf{u}(t)} \stackrel{\mathcal{L}}{\approx} \mathbf{H} := \boldsymbol{\sigma} \frac{\mathbf{Y}^\xi - \mathbf{1}}{\xi}, \quad \text{for large } t, \quad (3.11)$$

where the vectors $\boldsymbol{\sigma} \in (0, \infty)^d$ and $\boldsymbol{\xi} \in \mathbb{R}^d$ contain, respectively, the scale and shape parameters of the marginal Generalized Pareto distributions, and $\mathbf{Y}$ is the Multivariate Pareto random vector defined in (3.9).

Following the use of Angular measure, it is sufficient in Y to use the standard MGP, that is with $\sigma = 1$ and $\xi = 1$.

Theorem 3.3.1 establishes a rigorous theoretical foundation for modeling the multivariate tail of the random vector $\mathbf{X}$. Consider the problem of estimating the probability of a rare event, denoted as $\mathbb{P}(\mathbf{X} \in C)$, where the target region C lies strictly within the extreme domain $C \subset \{\mathbf{x} \in \mathbb{R}^d : \mathbf{x} \not\leq \mathbf{u}(t)\}$ for a sufficiently large $t > 1$. By utilizing the same threshold vector $\mathbf{u}(t)$ defined previously, the distributional approximation of the excesses $\mathbf{Y}_{\mathbf{u}(t)} = \mathbf{X} - \mathbf{u}(t) \mid \mathbf{X} \not\leq \mathbf{u}(t)$ by the MGP random vector $\mathbf{H}$ becomes highly accurate.

In such extreme regions, the scarcity or complete absence of empirical observations renders standard empirical estimators entirely unreliable. To circumvent this fundamental limitation, the probability of interest is written as:

$$\begin{aligned} \mathbb{P}(\mathbf{X} \in C) &= \mathbb{P}(\mathbf{X} \not\leq \mathbf{u}(t)) \mathbb{P}(\mathbf{Y}_{\mathbf{u}(t)} \in C - \mathbf{u}(t)) \\ &\approx \mathbb{P}(\mathbf{X} \not\leq \mathbf{u}(t)) \mathbb{P}(\mathbf{H} \in C - \mathbf{u}(t)) \end{aligned} \quad (3.12)$$

The first term, the probability of exceeding the threshold, can be robustly estimated via a standard empirical evaluation on the available sample. Conversely, the second term necessitates an estimate of the MGP distribution of $\mathbf{H}$. As established in (3.11), the random vector $\mathbf{H}$ is constructed through a deterministic, coordinate-wise transformation of the standard MGP random vector $\mathbf{Y}$ introduced in (3.9). This transformation solely depends on the marginal scale and shape parameters, the estimation of which is thoroughly addressed by well-established numerical and maximum likelihood techniques.

Consequently, the primary mathematical challenge in evaluating tail probabilities for $\mathbf{X}$ is reduced to obtaining informative realizations of the standard MGP vector $\mathbf{Y}$. According to Proposition 3.3.1, generating such realizations requires the capability to simulate from the angular measure Φ . Since Φ characterizes an asymptotic limit distribution, direct empirical observations are structurally impossible, mandating an approximate simulation framework.

To achieve this objective, a specialized Generative Adversarial Network (GAN) architecture is deployed using WGAN (Section 1.2) as a base.

A core concept in compositional data analysis involves transforming raw simplex data into coordinates within $\mathbb{R}^d$ with respect to an orthonormal basis of a specific $(d - 1)$ -dimensional subspace. Operating on a full linear $(d - 1)$ -dimensional space, rather than within the constrained unit simplex in $\mathbb{R}^d$, substantially enhances both statistical and neural network modeling capabilities. This geometric transformation, named Aitchison Coordinates, is leveraged to optimize the generation of accurate multivariate extreme samples within the WA-GAN architecture.

While infinitely many orthonormal bases exist for the open simplex Δ_{d-1}° when equipped with its natural inner product space structure, a specific construction is employed here. The fundamental principle relies on a bijective map from the open simplex Δ_{d-1}° to the zero-sum hyperplane $H := \{\mathbf{x} \in \mathbb{R}^d : \sum_{j=1}^d x_j = 0\}$ via the centered log-ratio function. The subspace H constitutes a $(d - 1)$ -dimensional linear space equipped with the standard Euclidean inner product, effectively transferring its algebraic structure back to Δ_{d-1}° .

Recall that for a generic vector $\mathbf{x} \in \mathbb{R}^d$, the standard softmax function maps onto the open simplex Δ_{d-1}°

and is defined as:

$$\text{softmax}(\mathbf{x}) := \left(\frac{\exp(x_j)}{\sum_{i=1}^d \exp(x_i)} \right)_{j=1}^d$$

Building upon this mapping, an explicit orthonormal basis can be formalized.

Theorem 3.3.2 (Orthonormal Basis for the Aitchison Simplex). *The sequence of vectors $e_1^*, \dots, e_{d-1}^*$ defined by:*

$$e_i^* := \text{softmax}(e_i) \in \Delta_{d-1}^\circ$$

where the underlying generating vectors $e_i \in \mathbb{R}^d$ are given by:

$$e_i := \sqrt{\frac{i}{i+1}} \left(\underbrace{i^{-1}, \dots, i^{-1}}_{i \text{ times}}, -1, 0, \dots, 0 \right)$$

for $i \in \{1, \dots, d-1\}$, forms a complete orthonormal basis for the Aitchison simplex.

Given this established orthonormal basis, any vector $\mathbf{u} \in \Delta_{d-1}^\circ$ admits a unique decomposition. Utilizing the Aitchison inner product $\langle \cdot, \cdot \rangle_A$ alongside the simplex algebraic operations (perturbation $\oplus$ and powering $\odot$), the vector $\mathbf{u}$ can be expanded as follows:

$$\mathbf{u} = \bigoplus_{i=1}^{d-1} \langle \mathbf{u}, e_i^* \rangle_A \odot e_i^* = \bigoplus_{i=1}^{d-1} \langle \text{clr}(\mathbf{u}), \text{clr}(e_i^*) \rangle \odot e_i^* = \bigoplus_{i=1}^{d-1} \langle \text{clr}(\mathbf{u}), e_i \rangle \odot e_i^* \quad (3.13)$$

The scalar coefficients $u_i^* := \langle \text{clr}(\mathbf{u}), e_i \rangle$ for $i \in \{1, \dots, d-1\}$ strictly and uniquely describe the vector $\mathbf{u}$. Within the generative framework, these specific coordinates u_i^* function as the input quantities for the WGAN architecture. This structural transformation is particularly crucial in extreme value analysis, as it effectively mitigates the learning instabilities associated with intrinsically small sample sizes characterizing extreme events.

3.3.2 Definition of the architecture

The primary objective of this architecture is to construct an algorithm capable of accurately sampling from the multivariate tail of a random vector $\mathbf{X} \in \mathbb{R}^d$. This objective is achieved through a sequential two-step procedure: first, simulating the dependence structure of unit-Pareto transformed observations using a generative adversarial network operating in Aitchison coordinates; second, transforming these synthetic values back to the original scale utilizing estimates of the marginal Generalized Pareto parameters. The resulting samples can then be utilized to evaluate rare event probabilities of the form $\mathbb{P}(\mathbf{X} \in C)$ for extreme spatial regions.

As previously established, the central mathematical challenge consists of simulating from the angular measure Φ of the transformed vector $\mathbf{V}$. The subsequent transition from Φ to the full Multivariate Generalized Pareto (MGP) distribution relies strictly on deterministic scaling and well-developed marginal tail estimation techniques. Consequently, the core generative task is relegated to a Wasserstein GAN architecture designed to obtain accurate samples from Φ via Aitchison coordinates.

The input for the generative training procedure consists of n independent and identically distributed observations $\mathbf{X}_1, \dots, \mathbf{X}_n$, where each vector is denoted by $\mathbf{X}_i = (X_{i1}, \dots, X_{id})$. To isolate the dependence structure, these raw observations are initially converted into approximate unit-Pareto margins using the empirical marginal cumulative distribution functions, strictly defined as:

$$\hat{F}_j(x) := \frac{1}{n+1} \sum_{i=1}^n \mathbb{I}\{X_{ij} \leq x\}, \quad x \in \mathbb{R}, \quad j \in \{1, \dots, d\}$$

The denominator $n + 1$ is systematically employed instead of n to strictly prevent division by zero singularities during the subsequent standardizations.

In accordance with the multivariate regular variation postulate (Assumption 3.3.1), the angles associated with extreme observations inherently provide an approximate empirical sample from the theoretical angular measure Φ . Specifically, the WA-GAN procedure exclusively filters and utilizes observations for which the radial component $R = \|\mathbf{V}\|_1$ exceeds a dynamically defined threshold $t := n/k_1$. Under this condition, the number of observations satisfying the threshold constraint asymptotically follows a binomial distribution with parameters n and dk_1/n .

The threshold parameter $k_1 > 0$ dictates a fundamental variance tradeoff. Indeed, selecting a *larger* k_1 increases the effective sample size, thereby reducing variance during the neural network training phase.

The ultimate output of this procedural step is a trained generator network G_θ , optimized to synthesize new coordinate vectors with respect to the Aitchison orthonormal basis $\{e_1^*, \dots, e_{d-1}^*\}$ that closely match the distributional properties of the empirical extreme angles in the training set.

The subsequent marginal transformation, which will be formalized in Algorithm 10, utilizes the k_2 largest observations for each margin, ensuring that between k_2 and dk_2 multivariate observations are effectively employed. Theoretical considerations dictate that setting $k_1 = k_2$ emerges as a natural and computationally convenient methodological choice.

Algorithm 9 structurally mirrors the standard Wasserstein GAN with gradient penalty, specifically adapted to operate on the coordinates of extreme angles within the orthonormal basis of the Aitchison simplex. The iterative weight updates for both the generator and the discriminator are executed via the Adam optimizer, an adaptive learning rate gradient descent algorithm that serves as the standard optimization choice for WGAN architectures.

In contrast to the standard algorithmic framework, a crucial additional regularization term is appended to the generator's objective function:

$$\rho \left\| \frac{1}{m} \sum_{i=1}^m \text{softmax}(MG_\theta(\mathbf{z}_i)) - \frac{\mathbf{1}_d}{d} \right\|_2^2 \quad (3.14)$$

where M denotes the change-of-basis matrix defined in Algorithm 9. This penalty term explicitly enforces the marginal moment constraints specified in (3.3.1) through a soft regularization approach, thereby guaranteeing that the generator samples from a mathematically valid angular measure. Specifically, the expression in (3.14) penalizes empirical deviations from the theoretical marginal conditions by minimizing the squared Euclidean norm of their difference.

Geometrically, $MG_\theta(\mathbf{z}_i)$ represents a point in the zero-sum hyperplane H with coordinates $G_\theta(\mathbf{z}_i)$ relative to the orthonormal basis $\{e_1, \dots, e_{d-1}\}$. The softmax function maps this coordinate point onto the unit simplex. Consequently, the regularization term strictly encourages the empirical distribution of the m generated simplex points to satisfy the necessary theoretical moment constraints.

Functionally, Algorithm 9 isolates and models exclusively the extremal dependence structure of the random vector $\mathbf{X}$. It remains strictly invariant to any marginal distributional assumptions, provided the true marginal cumulative distribution functions maintain continuity. Therefore, any analytical quantity depending solely on the angular measure Φ can be computed entirely independently of the marginal transformations. This modularity represents a structural advantage over alternative methodologies that directly evaluate sample quality on the scale of the full MGP distribution.

Finally, while the generative procedure is formally constrained to the angular measure Φ defined with respect to the L_1 -norm, a straightforward deterministic post-processing step facilitates the mapping of these results to any arbitrary norm on $\mathbb{R}^d$.

The complete training procedure designed to estimate the tail dependence via the Wasserstein-Aitchison GAN architecture is formalized in Algorithm 9.

Algorithm 9 WA-GAN for tail dependence estimation

Require: Observations $\mathbf{X}_1, \dots, \mathbf{X}_n$, orthonormal basis $\{e_1, \dots, e_{d-1}\}$ of the subspace H , and threshold parameter $k_1 \in \{1, \dots, n\}$.

Require: Hyperparameters: gradient penalty coefficient $\lambda > 0$, marginal penalty coefficient $\rho > 0$, number of discriminator iterations per generator iteration n_D , batch size m , Adam hyperparameters $\alpha > 0$ (learning rate) and $\beta_1, \beta_2 \in [0, 1)$, latent space dimension $\ell \in \mathbb{N}$, and number of epochs n_E .

Require: Initial parameters: discriminator weights $w_0 \in \mathcal{S}_D$, generator weights $\theta_0 \in \mathcal{S}_G$.

```

1: % Data Preprocessing and Standardization
2:  $t \leftarrow n/k_1$ 
3: for  $i = 1 \dots n$  do
4:    $\hat{\mathbf{V}}_i \leftarrow \left( \frac{1}{1 - \hat{F}_j(X_{ij})} \right)_{j=1}^d$ 
5:    $R_i \leftarrow \|\hat{\mathbf{V}}_i\|_1$ 
6:    $\mathbf{W}_i \leftarrow \hat{\mathbf{V}}_i / R_i$ 
7:   if  $R_i \geq t$  then
8:      $\mathbf{W}_i^* \leftarrow (\langle \text{clr}(\mathbf{W}_i), e_j \rangle)_{j=1}^{d-1}$ 
9:   end if
10: end for
11:  $K \leftarrow \sum_{i=1}^n \mathbb{I}\{R_i \geq t\}$ 
12:  $M \leftarrow (e_1 \mid \dots \mid e_{d-1}) \in \mathbb{R}^{d \times (d-1)}$ 
13: % Adversarial Training Phase
14:  $\theta \leftarrow \theta_0$ 
15:  $w \leftarrow w_0$ 
16: for epoch  $e = 1 \dots n_E$  do
17:   % Update the discriminator
18:   for  $step = 1 \dots n_D$  do
19:     for  $i = 1 \dots m$  do
20:       Sample  $\mathbf{w}_i^*$  uniformly from the extreme set  $\{\mathbf{W}_1^*, \dots, \mathbf{W}_K^*\}$ .
21:       Sample latent noise  $\mathbf{z}_i$  from a standard multivariate normal distribution on  $\mathbb{R}^\ell$ .
22:       Sample  $u_i$  from a uniform distribution on  $[0, 1]$ .
23:        $\tilde{\mathbf{w}}_i^* \leftarrow G_\theta(\mathbf{z}_i)$ 
24:        $\hat{\mathbf{w}}_i^* \leftarrow u_i \mathbf{w}_i^* + (1 - u_i) \tilde{\mathbf{w}}_i^*$ 
25:     end for
26:     Compute discriminator loss with gradient penalty:

$$L_D \leftarrow \frac{1}{m} \sum_{i=1}^m \left\{ D_w(\tilde{\mathbf{w}}_i^*) - D_w(\mathbf{w}_i^*) + \lambda \left( \|\nabla_{\mathbf{w}^*} D_w(\hat{\mathbf{w}}_i^*)\|_2 - 1 \right)^2 \right\}$$

27:      $w \leftarrow \text{Adam}(L_D, w, \alpha, \beta_1, \beta_2)$ 
28:   end for
29:   % Update the generator
30:   for  $i = 1 \dots m$  do
31:     Sample latent noise  $\mathbf{z}_i$  from a standard multivariate normal distribution on  $\mathbb{R}^\ell$ .
32:   end for
33:   Compute generator loss with marginal penalty constraint:

$$L_G \leftarrow -\frac{1}{m} \sum_{i=1}^m D_w(G_\theta(\mathbf{z}_i)) + \rho \left\| \frac{1}{m} \sum_{i=1}^m \text{softmax}(M G_\theta(\mathbf{z}_i)) - \frac{\mathbf{1}_d}{d} \right\|_2^2$$

34:    $\theta \leftarrow \text{Adam}(L_G, \theta, \alpha, \beta_1, \beta_2)$ 
35: end for

```

Ensure: Trained generator G_θ .

Once the generative model has been successfully trained via Algorithm 9, the subsequent step is to utilize the optimized generator to simulate synthetic extreme scenarios on the original data scale.

Algorithm 10 details this explicit sampling procedure. Functionally, the algorithm first estimates the marginal Generalized Pareto (GP) parameters utilizing a standard Peaks-Over-Threshold approach for each dimension. Subsequently, it combines the angular samples synthesized by the trained generator with independent unit-Pareto radial components. By applying the estimated marginal transformations, the procedure ultimately yields a set of synthetic multivariate observations that accurately approximate the tail distribution of the original random vector $\mathbf{X}$.

Algorithm 10 Sampling from the tail of $\mathbf{X}$

Require: Observations $\mathbf{X}_1, \dots, \mathbf{X}_n$, orthonormal basis $\{e_1, \dots, e_{d-1}\}$ of H , trained generator G_θ , size of the latent space ℓ on which G_θ has been trained, and threshold parameter $k_2 \in \{1, \dots, n\}$.

Require: Desired number of synthetic samples n^* .

```

1: % Marginal Parameter Estimation
2: for  $j = 1 \dots d$  do
3:    $u_j \leftarrow X_{n-k_2:n,j}$    % The  $k_2$ -th largest observation in margin  $j$ 
4:   Compute the maximum likelihood estimator  $(\hat{\sigma}_j, \hat{\xi}_j)$  of a univariate GP distribution.
5: end for
6: % Generative Sampling Phase
7:  $M \leftarrow (e_1 \mid \dots \mid e_{d-1}) \in \mathbb{R}^{d \times (d-1)}$ 
8:  $i \leftarrow 1$ 
9: Initialize  $\mathcal{G}$  as an empty list of size  $n^*$ 
10: while  $i \leq n^*$  do
11:   Sample latent noise  $\mathbf{z}$  from a standard multivariate normal distribution on  $\mathbb{R}^\ell$ .
12:    $\mathbf{W} \leftarrow \text{clr}^{-1}(MG_\theta(\mathbf{z}))$ 
13:   Sample radial component  $Y$  from a unit-Pareto distribution.
14:    $\mathbf{Y} \leftarrow Y\mathbf{W}$ 
15:   % Enforce multivariate exceedance condition
16:   if  $\max(\mathbf{Y}) > 1$  then
17:     for  $j = 1 \dots d$  do
18:       if  $Y_j \leq 1$  then
19:          $\mathcal{G}[i]_j \leftarrow X_{[(n-k_2/Y_j) \vee 1]:n,j}$ 
20:       else if  $Y_j > 1$  then
21:          $\mathcal{G}[i]_j \leftarrow u_j + \hat{\sigma}_j \frac{Y_j^{\hat{\xi}_j} - 1}{\hat{\xi}_j}$ 
22:       end if
23:     end for
24:      $i \leftarrow i + 1$ 
25:   end if
26: end while

```

Ensure: List of n^* random vectors $\mathcal{G}$ approximately distributed as $\mathbf{X} \mid \mathbf{X} \notin \mathbf{u}$.

In addition to the network weights updated dynamically during the adversarial training phase, the WAGAN architecture defined in Algorithm 9 strictly depends on a set of structural and optimization hyperparameters.

The hyperparameter grid is formally defined by the following sets:

- Batch size: $m \in \{\lfloor n/k \rfloor : k = 1, 2, 4, 8, 16\}$;

- Latent space dimension: $\ell \in \{[(d-1) \cdot k] : k = 0.25, 0.5, 0.75, 1, 2\}$;
- Network topology: number of hidden layers $\in \{1, 2, 4, 8\}$, with a maximum number of neurons per layer restricted to $\{32, 64, 128, 256, 512\}$;
- Adam optimizer parameters: learning rate $\alpha \in \{0.01, 0.005, 0.001, 0.0005, 0.0001\}$ and moment decay rates $\beta = (\beta_1, \beta_2) \in \{(0.0, 0.9), (0.5, 0.9), (0.5, 0.99)\}$;
- Regularization coefficients: gradient penalty $\lambda \in \{0.1, 1, 3, 5, 7, 9\}$ and marginal penalty $\rho \in \{0.001, 0.01, 0.1, 1, 3\}$;
- Training dynamics: discriminator iterations per generator step $n_D \in \{1, 3, 5, 10\}$.

Each candidate model in the random search is trained for a fixed duration of $n_E = 5000$ epochs. This threshold is empirically selected to optimally balance convergence stability and computational efficiency, as extending the training phase further does not yield statistically significant performance improvements.

The final model selection is rigorously based on the extremal coefficient metric $E_{\bar{\theta}}(k)$. The optimal architecture is identified as the one minimizing the arithmetic mean of this metric evaluated at specific dimensionalities, namely $(E_{\bar{\theta}}(2) + E_{\bar{\theta}}(3))/2$.

3.3.3 Advantages and Limitations

The Wasserstein-Aitchison Generative Adversarial Network (WA-GAN) introduces several structural advantages for modeling multivariate extremes.

Primarily, the employment of Aitchison coordinates geometrically projects the angular measure from the constrained unit simplex onto an unconstrained linear vector space. This topological transformation allows the Wasserstein GAN to operate without geometric boundary limitations, significantly facilitating the convergence of the adversarial training.

Furthermore, the architecture rigorously isolates the multivariate asymptotic dependence structure from the marginal distributions. This modularity allows the marginal tails to be modeled separately and exactly utilizing classical Extreme Value Theory, specifically via Generalized Pareto distributions and the Peaks-Over-Threshold approach.

Finally, by mapping the original distribution into a coordinate space with more regular statistical properties, the framework effectively mitigates the learning difficulties typically encountered by neural networks when dealing with the intrinsically small sample sizes characteristic of extreme events.

Despite these theoretical benefits, the WA-GAN methodology is subject to notable structural limitations.

A primary constraint is its mathematical inability to accurately handle extreme directions. Specifically, the model inherently excludes scenarios where the angular measure allocates probability mass to the boundaries of the simplex, rendering it unsuitable for modeling events where only a restricted subset of coordinates exhibits extreme values while the others remain negligible.

Additionally, the algorithmic performance is critically sensitive to the threshold truncation parameters, k_1 and k_2 . The selection of these thresholds imposes a delicate bias-variance tradeoff: lowering the threshold increases the effective training sample size, thereby reducing statistical variance, but simultaneously distances the samples from the true extreme asymptote, which strictly increases theoretical bias.

From a computational perspective, ensuring that the generated samples respect the theoretical moment constraints imposed by unit-Pareto distributions requires the integration of a soft marginal penalty within the generator's objective function. This regularization overhead, implemented via softmax mappings and change-of-basis matrices, significantly complicates the optimization landscape and necessitates extensive hyperparameter tuning to achieve stable convergence.

Another fundamental limitation is the model's severe vulnerability to the curse of dimensionality. As the number of dimensions increases, the volume of the multivariate tail region expands exponentially, drastically

exacerbating the inherent sparsity of extreme events. In such high-dimensional settings, the mapping into Aitchison coordinates becomes exceedingly sparse and topologically complex. This scarcity of data in high-dimensional extremes renders the generative network highly susceptible to overfitting and mode collapse, making the accurate estimation of the joint angular measure statistically and computationally intractable for large-scale portfolios. Consequently, the model's practical applicability is largely restricted to low- or moderate-dimensional financial systems.

Chapter 4

Experimental results

This chapter presents a comprehensive empirical application of the generative architectures previously established, alongside the introduction of supplementary frameworks pertinent to the quantitative analysis.

The analytical focus is deliberately oriented toward evaluating the capacity of these networks to generate robust, high-fidelity synthetic simulations, prioritizing this generative accuracy over the mere reproduction of standard stylized facts. To rigorously test this capability, the models are deployed within inherently complex implementation scenarios for Generative Adversarial Networks. Furthermore, a central objective of this empirical evaluation is the thorough assessment of the heavy-tail phenomenon, which serves as the definitive criterion for selecting the most structurally appropriate model for simulating extreme financial scenarios.

4.1 Datasets and error metrics

The initial section details the selected datasets and the specific error metrics employed for performance evaluation, strictly utilizing real-world financial assets and quantitative risk metrics to construct practical testing environments.

4.1.1 Indices and long scenarios

A prevalent limitation in the existing literature on Generative Adversarial Networks (GANs) is the reliance on short-term empirical scenarios, such as three-month periods of minute-by-minute observations. This approach represents a significant simplification of actual financial market dynamics. Over a brief three-month window, it is highly improbable to observe major structural shifts or extreme market regime changes. Furthermore, utilizing high-frequency minute data generates approximately 20,000 observations over a single quarter. This artificially circumvents the fundamental challenge of data scarcity, which is a central issue when modeling rare financial events.

To overcome these methodological simplifications, this analysis employs a 20-year history of daily financial time series. Consequently, the empirical dataset is restricted to approximately 5,000 observations. This specific sampling choice accurately reflects the inherent data scarcity, while comprehensively encompassing multiple severe financial crises and sustained positive market cycles.

A further limitation prevalent in the empirical literature is the disproportionate reliance on Western financial datasets, frequently neglecting Asian markets, which traditionally exhibit complex structural dynamics and pose significantly greater predictive challenges. To address this geographical imbalance and ensure a rigorous evaluation of the generative architectures across diverse macroeconomic regimes, the empirical scope of this analysis is concentrated on four distinct global indices.

The Western financial landscape is represented by the Nasdaq-100 (NDX) and the FTSE MIB, while the Asian markets are represented by the Nikkei 225 (NIK) and the Hang Seng Index (HSI).

The subsequent sections summarize the fundamental statistical characteristics of each selected index, providing a formal justification for their inclusion and establishing their relevance for testing heavy-tailed financial phenomena.

The Nikkei 225 (NIK)

The Nikkei 225 comprises the best 225 companies listed on the Tokyo Stock Exchange. Within the context of this empirical analysis, this index is considered a benchmark standard due to the unique and formidable mathematical challenges it presents for generative modeling.

Simulating the tail behavior of the Nikkei 225 is exceptionally complex due to its asynchronous temporal dependence. Indeed, the index is highly sensitive to the closing valuations of Western markets while simultaneously exhibiting strong macroeconomic cross-correlations with neighboring Asian economies, such as South Korea and China. Furthermore, the Japanese equity market is distinctly characterized by aggressive state and central bank interventions. Over the past decades, the Bank of Japan (BoJ) has executed massive quantitative easing programs, including the direct and systemic purchasing of equity exchange-traded funds. Consequently, the government functions as a primary shareholder for many of the index's most valuable constituents, a dynamic that artificially dampens standard volatility and heavily manipulates the empirical tail distribution of the asset.

From a macroeconomic perspective, the 20-year observation window highlights a trajectory completely divergent from Western markets. Rather than inflation, the Japanese economy has historically grappled with chronic deflationary pressures, meaning that equity returns have predominantly relied on real corporate value accretion rather than nominal price inflation. Additionally, the Japanese Yen (JPY) has been subjected to extreme volatility regimes, experiencing severe depreciation against major currencies such as the US Dollar and the Euro. This currency fluctuation introduces a critical layer of risk, as the index is heavily populated by export-oriented firms whose revenues are highly sensitive to exchange rates.

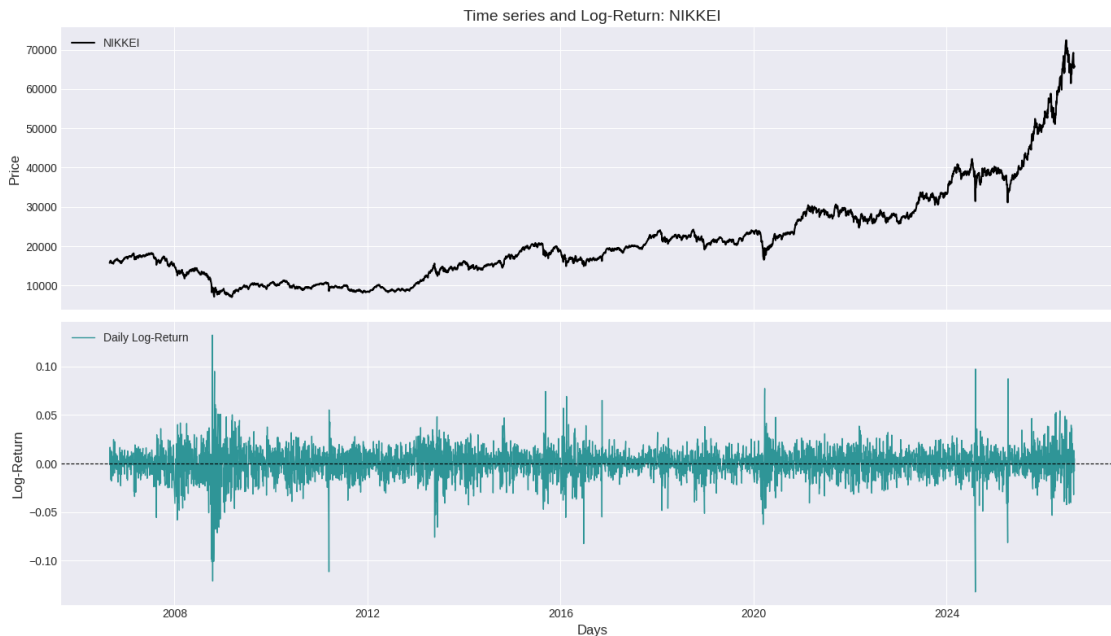


Figure 4.1: Time series and log-return for NIKKEI

Over the selected two-decade timeframe, the time series encompasses not only global systemic shocks, such as the 2008 Great Financial Crisis and the COVID-19 pandemic, but also severe regional crises. Most

notably, this includes the catastrophic 2011 Tohoku earthquake and subsequent Fukushima nuclear disaster, which induced an unprecedented structural shock to the national supply chain and infrastructure. Furthermore, the data captures the localized impacts of the 2015 Chinese stock market turbulence and the protracted economic stagnation often referred to as Japan's "Lost Decades."

Structurally, the Nikkei 225 is a price-weighted index that offers a comprehensive representation of the Japanese macroeconomy. Its composition is heavily concentrated in advanced, export-driven sectors. The primary weightings are allocated to the technology sector (including semiconductors, telecommunications, and consumer electronics), automotive manufacturing, and heavy industrials (such as robotics and precision machinery). Additionally, the index is stabilized by substantial allocations to domestic banking and financial institutions, chemical manufacturing, and the pharmaceutical and healthcare sectors. This diverse yet export-heavy composition makes it a structurally complex, heavy-tailed asset, ideal for testing the limits of advanced generative adversarial architectures.

The Nasdaq-100 (NDX)

The Nasdaq-100 index comprises 100 of the largest non-financial multinational corporations listed on the Nasdaq Stock Market, with a composition overwhelmingly dominated by the technology, telecommunications, and biotechnology sectors. This extreme sector concentration inherently exposes the index to rapid innovation cycles, making it highly sensitive to shifts in global market sentiment and monetary policy, particularly interest rate fluctuations affecting long-duration cash flows.

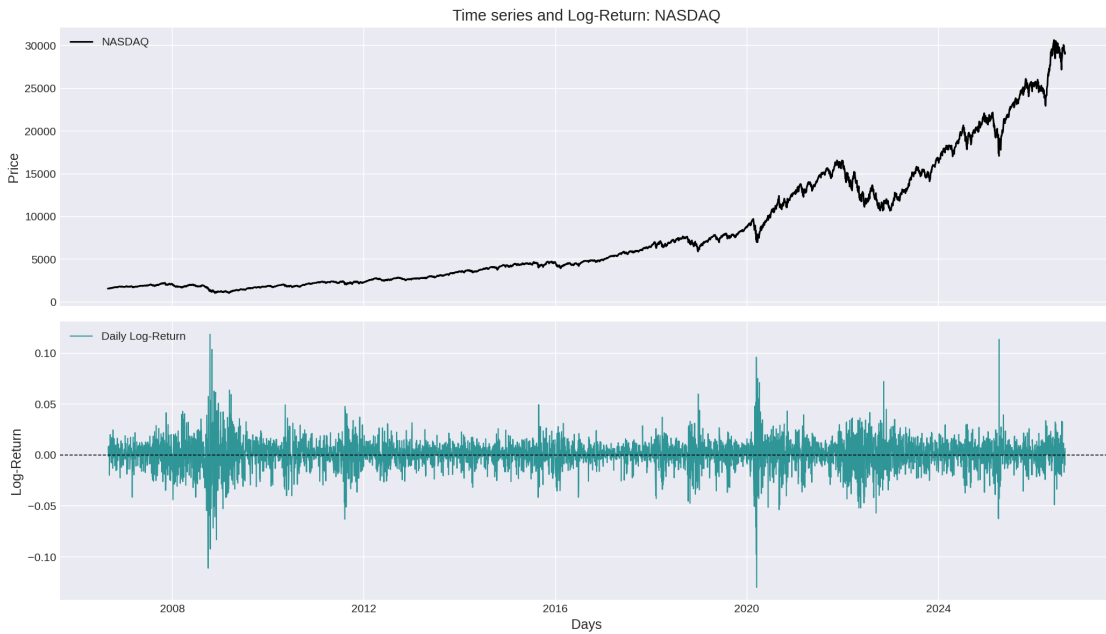


Figure 4.2: Time series and log-return for NASDAQ

Over the 20-year observation window, the NDX exhibits idiosyncratic volatility dynamics characterized by severe, asymmetric tail risks. Unlike traditional equity indices that primarily face left-tail crash risks, the NDX frequently demonstrates massive right-tail expansions driven by technological paradigms, alongside rapid, devastating drawdowns. This bilateral tail asymmetry poses a profound mathematical challenge for generative architectures, demanding the accurate simulation of both exponential growth and sudden crashes without distributional smoothing. The selected timeframe encapsulates critical structural disruptions, including the 2008 Great Financial Crisis, the 2020 COVID-19 crash and its subsequent liquidity-driven rally,

and the 2022 inflationary bear market. Consequently, the NDX serves as the definitive benchmark for evaluating the generative fidelity of the proposed models against Western high-growth, heavy-tailed financial phenomena.

The FTSE MIB

The FTSE MIB index tracks the 40 most liquid and highly capitalized equities listed on the Borsa Italiana, serving as the primary benchmark for the Italian financial market. In stark contrast to the technology-driven Nasdaq-100, the composition of the FTSE MIB is structurally dominated by traditional, cyclical sectors, with a heavy concentration in commercial banking, financial services, energy, and utilities. This sector distribution renders the index acutely sensitive to domestic political instability, fluctuations in sovereign credit spreads (specifically the BTP-Bund spread), and the broader monetary policy decisions of the European Central Bank (ECB). Furthermore, the index exhibits a pronounced structural dependence on international market dynamics. Driven by the global integration of its major industrial and financial constituents, the FTSE MIB remains highly correlated with the systemic movements of core European indices and American markets, rendering it exceptionally vulnerable to imported volatility and cross-border macroeconomic spillover effects.

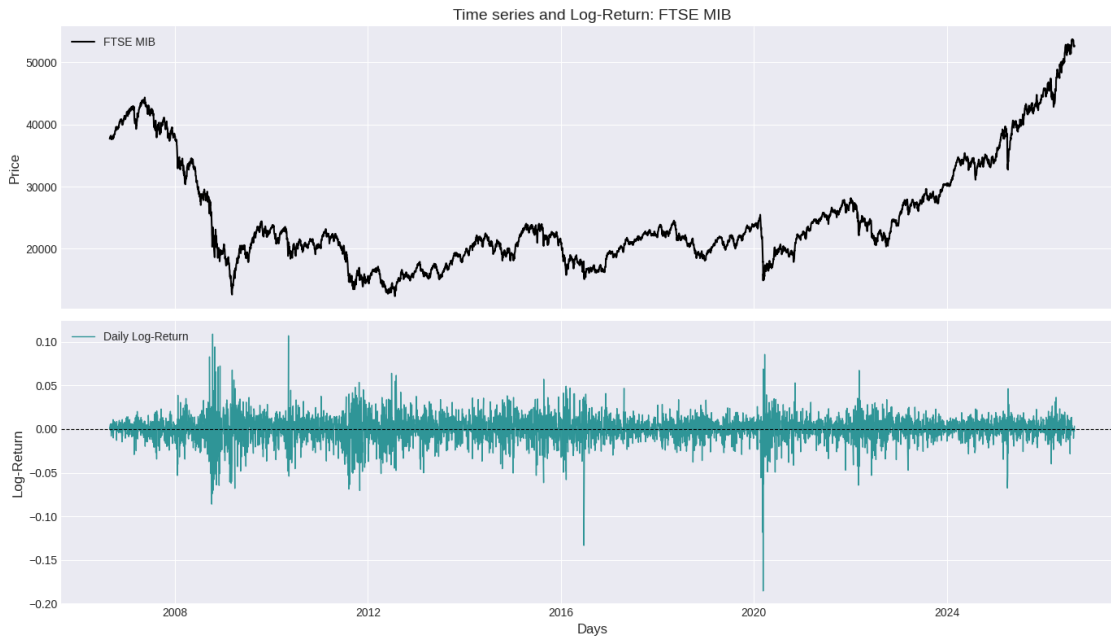


Figure 4.3: Time series and log-return for FTSEMIB

Over the 20-year observation window, the FTSE MIB exhibits a distinct volatility profile characterized by prolonged structural stagnation and pronounced systemic left-tail risk. The dataset encapsulates a series of profound macroeconomic shocks, most notably the 2008 Great Financial Crisis and the devastating 2011 European Sovereign Debt Crisis, which severely impaired the domestic banking sector through the sovereign-bank nexus. Furthermore, the timeframe includes the aggressive market impacts of the Brexit referendum and the 2020 COVID-19 pandemic, during which Italy experienced one of the earliest and most severe economic halts in the Western hemisphere. Consequently, the FTSE MIB provides an essential benchmark for evaluating the capacity of generative models to simulate protracted drawdowns, credit-driven volatility clustering, and extreme left-tail realizations within a mature, heavily interconnected macroeconomic environment.

The Hang Seng Index (HSI)

The Hang Seng Index serves as the primary benchmark for the Hong Kong Stock Exchange (HKEX), tracking the performance of the largest and most liquid market capitalization companies in the region. Structurally, the HSI occupies a unique geopolitical and financial position as the primary conduit for international capital flowing into mainland China. Consequently, the index composition is overwhelmingly dominated by mainland Chinese enterprises with a heavy sector concentration in massive financial institutions, technology conglomerates, and real estate developers. This dual nature subjects the index to an extraordinarily complex volatility profile, as it is simultaneously exposed to global macroeconomic cycles and the opaque, highly centralized regulatory environment of Beijing.

A defining structural characteristic of the Hong Kong market is the Linked Exchange Rate System, which pegs the Hong Kong Dollar (HKD) to the US Dollar. This necessitates that the Hong Kong Monetary Authority synchronizes its interest rate policy with the US Federal Reserve. As a result, the HSI suffers from a profound macroeconomic dichotomy: its underlying corporate earnings are primarily driven by the Chinese domestic economy, yet its valuation multiples and liquidity conditions are tightly constrained by Western monetary policy. This systemic tension generates intense, asynchronous volatility clustering and severe heavy-tailed distributions, making the HSI a critical dataset for testing the robustness of generative models under conflicting macroeconomic dependencies.

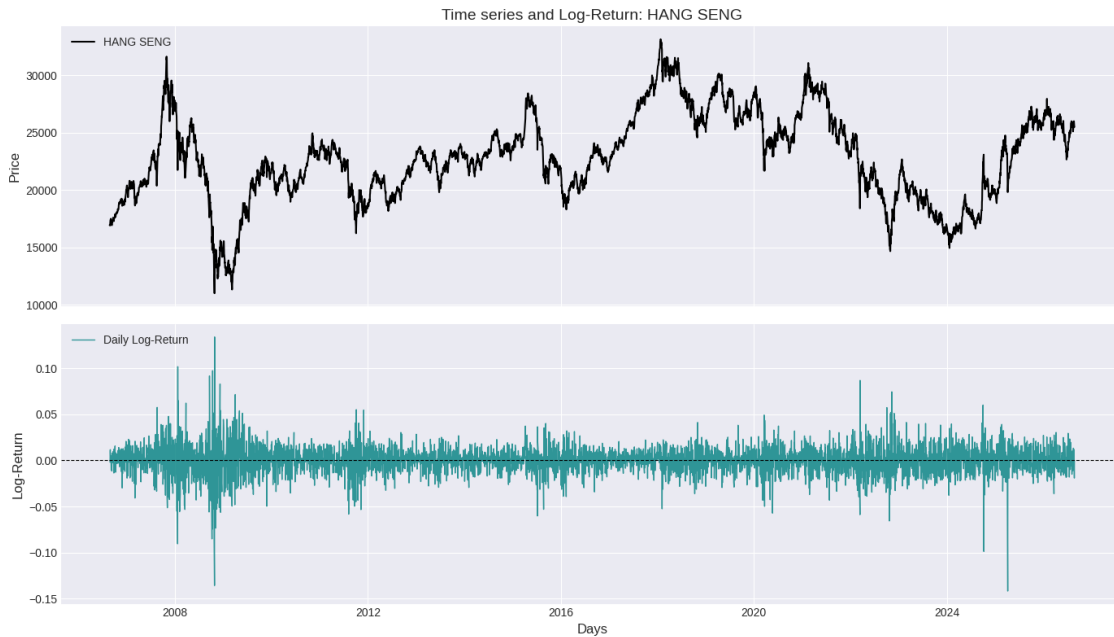


Figure 4.4: Time series and log-return for Hang Seng

Over the 20-year observation window, the HSI provides a rigorous empirical testing ground characterized by extreme left-tail realizations and protracted structural drawdowns. The dataset encapsulates the 2008 Great Financial Crisis, but more importantly, it captures severe regional shocks unique to the Asian market. These include the massive 2015 Chinese stock market crash, the profound economic disruptions caused by the 2019/20 Hong Kong political unrest, and the subsequent COVID-19 pandemic. Furthermore, the latter portion of the time series is deeply impacted by Beijing’s aggressive regulatory crackdowns on the domestic technology sector and the systemic debt crisis within the Chinese real estate market (e.g., the Evergrande default). These localized, policy-driven regime changes produce abrupt, non-linear market crashes, thereby providing an essential benchmark for evaluating the generative architecture’s capability to learn and simulate

extreme, policy-induced tail risks within a highly interconnected Eastern market.

4.1.2 Preparing datasets

The empirical analysis utilizes daily historical data spanning an exact twenty-year observation window, specifically from August 26, 2006, to August 25, 2026, for all selected indices. To ensure strict temporal consistency in multivariate scenarios, the individual time series are synchronized by computing the intersection of available trading dates across all respective datasets. This alignment procedure guarantees that the multivariate input vectors correspond to the exact same temporal instances, effectively eliminating asynchronous missing values caused by differing national market holidays.

The generative architectures are explicitly trained to simulate daily logarithmic returns rather than nominal prices. Let $P_{t,j}$ denote the closing price of the j -th index at time t . The logarithmic return $r_{t,j}$ is formally defined as:

$$r_{t,j} := \ln \left(\frac{P_{t,j}}{P_{t-1,j}} \right) = \ln(P_{t,j}) - \ln(P_{t-1,j})$$

where $\ln(\cdot)$ denotes the natural logarithm.

This specific transformation is methodologically advantageous for several reasons. Primarily, it ensures time-additivity over multiple periods, normalizes the scale across different financial assets, and contributes to the stationarity of the time series. By converting the data into logarithmic returns, the structural properties of the input data are aligned with the fundamental mathematical requirements of both the Generative Adversarial Networks and the Extreme Value Theory frameworks implemented in this analysis.

4.1.3 Evaluation Metrics and Computational Complexity

To ensure clarity and maintain a highly focused evaluation of the generative performance, the empirical analysis restricts the assessment to a select, rigorous set of statistical estimators. The primary quantitative evaluation relies on Value-at-Risk (VaR) and Expected Shortfall (ES), as formally established in 2.1.3 and 2.1.4. These metrics serve as the industry standard for quantifying downside risk and expected tail losses. Alongside these risk measures, the analysis explicitly monitors the computational training time required for model convergence.

Furthermore, to directly assess the structural fidelity of the synthesized data, the analysis incorporates the empirical measurement of kurtosis.

Definition 4.1.1. *For a random variable r representing the financial logarithmic returns, with expected value μ and standard deviation σ , the kurtosis is defined as the fourth standardized moment:*

$$Kurt(r) := \mathbb{E} \left[\left(\frac{r - \mu}{\sigma} \right)^4 \right] = \frac{\mathbb{E}[(r - \mu)^4]}{(\mathbb{E}[(r - \mu)^2])^2}$$

Within the specific context of this extreme-value analysis, kurtosis serves as a fundamental diagnostic tool. Empirical financial time series consistently exhibit leptokurtic distributions $Kurt > 1$ which are characterized by a sharp central peak and heavily pronounced tails. Because the central objective of the test is the accurate simulation of these extreme, rare market realizations, measuring the kurtosis of the synthetic returns provides a direct, scalar evaluation of the model's capacity to replicate heavy-tailed dynamics. If a generative framework suffers from distributional smoothing, thereby failing to capture the true probability mass located in the extremes, the resulting empirical kurtosis of the generated sample will be significantly underestimated. Consequently, this statistical metric acts as a crucial theoretical checkpoint to validate the generative efficacy of the tested architectures in modeling severe market phenomena.

4.1.4 Computational Methodology and Implementation

The computational pipeline and the empirical simulations are entirely implemented in Python. The core neural network topologies for both the generative and discriminative models are developed utilizing the PyTorch deep learning framework, which provides the essential automatic differentiation capabilities required for complex adversarial optimization.

Data manipulation, temporal synchronization, and preprocessing tasks are executed using Pandas and NumPy, while scikit-learn is extensively employed for standard quantitative utilities and the calculation of baseline statistical metrics. Given the severe computational complexity, training on a standard Central Processing Unit (CPU) proved computationally prohibitive. To overcome these infrastructural bottlenecks, the entire training procedure is hardware-accelerated utilizing an NVIDIA T4 Tensor Core Graphics Processing Unit (GPU). This parallelization strategy drastically reduces the execution time per epoch, ensuring that the extensive hyperparameter optimization phases and the training of the deep generative architectures remain computationally tractable within the context of high-frequency iterative evaluations.

All source code, implementation scripts, and historical dataset files (`.csv`) utilized for the empirical evaluations and model benchmarking throughout this study are publicly accessible in the dedicated GitHub repository.

4.2 One-Index simulations

This section evaluates the capability of various generative models to simulate a single financial index. The comparative analysis progresses sequentially from the foundational Wasserstein GAN with Gradient Penalty (WGAN-GP) to the more advanced architectures established in previous chapters. Ultimately, this empirical assessment identifies the most robust algorithmic framework for univariate tail-risk simulation.

4.2.1 WGAN-GP

The initial empirical evaluation employs the standard Wasserstein GAN with Gradient Penalty (WGAN-GP), as formalized in Algorithm 3. To ensure computational stability and optimal convergence, the network architecture and training dynamics are initialized with the following predefined hyperparameter configuration:

- **Sequence Window Length:** 63 trading days
- **Batch Size (m):** 128
- **Latent Space Dimension (ℓ):** 64
- **Critic Iterations per Generator Step (n_{crit}):** 5
- **Gradient Penalty Coefficient (λ_{GP}):** 10.0
- **Critic Learning Rate (α_C):** 5×10^{-5}
- **Generator Learning Rate (α_G):** 1×10^{-4}
- **Hidden Layer Dimension (Critic and Generator):** 256 neurons

To further optimize the training dynamics and strictly prevent overfitting, an early stopping mechanism was implemented throughout the experimental phase. By continuously monitoring the convergence stability of the adversarial loss functions, the training procedure was configured to halt automatically once the marginal improvements fell below a predefined tolerance threshold. Under this regularization strategy, the networks

demonstrated relatively rapid convergence, with the optimal number of training epochs consistently ranging between 500 and 1000 before the early stopping criterion was triggered. This approach not only mitigated the risk of structural memorization but also significantly enhanced the overall computational efficiency of the baseline simulations.

The quantitative results assessing the generative capacity of the model to replicate the tail dynamics of the Indices above are reported in the table below. The evaluation specifically focuses on the estimation percentage errors for Value-at-Risk (VaR(95%)) and Expected Shortfall (ES(95%)), alongside the empirical kurtosis to verify the reproduction of heavy-tailed properties.

Financial Index	VaR Error	ES Error	Empirical Kurtosis
Nikkei 225 (NIK)	20%	29%	≈ 0
Nasdaq-100 (NDX)	11%	16%	≈ 0
FTSEmib (FMIB)	6%	14%	≈ 0
Hang Seng (HSI)	15%	13%	≈ 0

Table 4.1: Empirical performance metrics of the standard WGAN-GP architecture applied to univariate financial time series.

Although the standard WGAN-GP architecture benefits from high computational efficiency, allowing for rapid training convergence even within a standard CPU environment, the empirical results expose severe structural limitations when applied to tail-risk simulation. Primarily, the model systematically fails to capture the leptokurtic nature of financial returns. The synthetic paths consistently exhibit an excess kurtosis near zero, indicating that the generator collapses toward a Gaussian distribution, thereby entirely smoothing out the heavy-tailed extremes.

This Gaussian collapse is theoretically predictable: the standard WGAN framework minimizes the Wasserstein distance across the entire global distribution. Given that the vast majority of financial return data is densely concentrated around the mean, this central mass heavily dominates the empirical loss function. Consequently, the network optimizes its weights to replicate the central distribution while effectively ignoring the extreme quantiles, which contain insufficient sample density to meaningfully influence the gradient updates.

Furthermore, the training dynamics display profound instability. Because the network attempts a global fit, the few extreme observations present in the training set act as disruptive outliers rather than informative structural signals. This localized volatility in the gradient landscape confuses the discriminator, yielding highly divergent generative outputs across multiple iterations despite utilizing identical hyperparameter configurations. Finally, this architectural inability to isolate and learn specific tail dynamics results in massive inconsistencies in the Value-at-Risk and Expected Shortfall estimation errors across different financial indices.

As part of the architectural exploration, an alternative network configuration was evaluated wherein a Temporal Kolmogorov-Arnold Network (TKAN) was implemented as the generator structure, replacing the standard multilayer perceptron mapping. Although TKAN architectures have demonstrated significant theoretical and empirical efficacy in tasks such as interpretable volatility forecasting [3], this framework yielded strictly inferior results within the generative adversarial context and was subsequently discarded.

Theoretically, the highly complex, localized spline-based activation functions inherent to Kolmogorov-Arnold representations introduced severe training instabilities when subjected to the min-max adversarial optimization landscape. Rather than accurately learning the asymptotic tail dependencies, the TKAN-based generator exhibited a pronounced tendency toward mode collapse and struggled to robustly extrapolate the extreme, heavy-tailed market shocks. Consequently, this architecture exacerbated the underestimation of

both Value-at-Risk and Expected Shortfall, failing to replicate the necessary empirical kurtosis. Given these structural incompatibilities for extreme value simulation within the baseline WGAN-GP framework, the quantitative results associated with the TKAN configuration are omitted from the formal reporting.

4.2.2 TCN-WGAN-GP

The generative modeling of financial time series has been significantly advanced by the introduction of the Quant GAN framework by Wiese et al. (2019)[15].

Main Idea of QuantGAN

A fundamental innovation of this architecture is the architectural shift from traditional Multilayer Perceptrons (MLPs) and Recurrent Neural Networks (RNNs), such as LSTMs, to Temporal Convolutional Networks (TCNs). While standard MLPs lack the structural capacity to handle sequential data and recurrent architectures frequently suffer from vanishing gradients and extrapolation instabilities, TCNs overcome these limitations through the implementation of dilated causal convolutions. This specific convolutional structure allows the networks to exponentially expand their receptive field size without losing temporal ordering, enabling the highly accurate capture of long-range macroeconomic dependencies, such as volatility clustering and leverage effects[cite: 1]. Furthermore, the use of TCNs mathematically guarantees the stationarity of the generated processes and allows for highly efficient computational parallelization.

At the core of the Quant GAN framework is the Stochastic Volatility Neural Network (SVNN), which serves as the generator function. Inspired by classical continuous-time stochastic volatility models, the SVNN explicitly decomposes the log-return process into separate volatility, drift, and innovation neural networks. This structural decomposition is of paramount importance in financial mathematics, as it strictly allows the induced stochastic process to mathematically transition to a risk-neutral distribution, enabling the synthetic paths to be utilized for quantitative tasks such as derivative pricing and portfolio hedging.

To address the leptokurtic, heavy-tailed nature of empirical financial returns, the Quant GAN architecture incorporates a specific preprocessing pipeline. Because standard neural networks inherently struggle to generate random variables with unbounded moments, the historical log-returns are preprocessed using a Lambert W probability transform. This bijective transformation effectively "Gaussianizes" the heavy-tailed data, allowing the neural networks to learn the dynamics in a lighter-tailed environment before applying the inverse transformation to the synthetic outputs to restore the extreme empirical realizations.

While the original Quant GAN framework relies on a standard Generative Adversarial Network optimization setting equipped with specialized stabilization algorithms, the inherent adversarial training remains structurally volatile. Consequently, to ensure superior convergence stability and to mitigate mode collapse when simulating the extreme tail behaviors of the selected indices, the empirical analysis conducted in this thesis strategically substitutes the standard GAN loss formulation with the Wasserstein GAN with Gradient Penalty (WGAN-GP) framework.

Practical application

Recognizing the theoretical advantages of Temporal Convolutional Networks (TCNs) in capturing complex long-range financial dependencies, as demonstrated by the Quant GAN framework, a streamlined adaptation was developed and evaluated. To balance generative fidelity with computational efficiency, the standard Multilayer Perceptron (MLP) mapping within the baseline WGAN-GP architecture was explicitly replaced with a TCN generator. This structural modification was designed to leverage the superior temporal modeling capabilities of dilated causal convolutions, while deliberately avoiding the extensive computational overhead and optimization complexities associated with the full Stochastic Volatility Neural Network (SVNN) representation.

To empirically evaluate the performance of the streamlined TCN-based WGAN-GP architecture, the network was initialized with a highly specific hyperparameter configuration. The structural transition from a standard Multilayer Perceptron (MLP) to a Temporal Convolutional Network (TCN) generator necessitated crucial adjustments, particularly concerning the latent space representation and computational constraints.

Unlike the baseline MLP model, where the noise prior is defined as a single global vector for the entire sequence, the TCN generator processes sequential noise. Consequently, the latent dimension defines the number of noise channels per individual time step, maintaining strict temporal alignment. Furthermore, to mitigate the increased computational overhead associated with deep one-dimensional causal convolutions, the batch size was deliberately reduced.

The general adversarial training parameters are defined as follows:

- **Sequence Window Length:** 63 trading days
- **Maximum Training Epochs:** 5000 (dynamically regulated by an early stopping mechanism to prevent overfitting)
- **Batch Size (m):** 64
- **Latent Channels per Time Step (ℓ):** 16
- **Generated Simulation Paths:** 300
- **Critic Iterations per Generator Step (n_{crit}):** 5
- **Gradient Penalty Coefficient (λ_{GP}):** 10.0
- **Critic Learning Rate (α_C):** 5×10^{-5}
- **Generator Learning Rate (α_G):** 1×10^{-4}

For the specific structural configuration of the Temporal Convolutional Network generator, the architecture utilizes the following parameters:

- **TCN Residual Blocks (Depth):** 6.
- **Hidden Channels:** 128 per convolutional layer
- **Kernel Size (K):** 4
- **Dropout Rate:** 0.1 (applied to regularize the convolutional feature maps)

The quantitative results assessing the generative capacity of the model to replicate the tail dynamics of the Indexes above are reported in the table below. The evaluation specifically focuses on the estimation errors for Value-at-Risk (VaR(95%)) and Expected Shortfall (ES(95%)), alongside the empirical kurtosis to verify the reproduction of heavy-tailed properties.

Financial Index	VaR Error	ES Error	Empirical Kurtosis
Nikkei 225 (NIK)	8.3%	4, 8%	≈ 5 (52%)
Nasdaq-100 (NDX)	10%	3.8%	≈ 0
FTSEmib (FMIB)	33%	29%	≈ 0
Hang Seng (HSI)	10.8%	14%	≈ 0

Table 4.2: Empirical performance metrics of the TCN-WGAN-GP architecture applied to univariate financial time series.

To assess the empirical goodness-of-fit in the downside regime, the table below reports the downside coverage ratios across the evaluated models at thresholds of $-\sigma$ and -2σ . Here, the coverage ratio measures the empirical percentage of historical days whose negative returns fall within the critical downside thresholds generated by the model. A well-calibrated architecture should match the theoretical empirical frequencies at $-\sigma$ and -2σ ; significant discrepancies indicate either an inability to capture moderate-to-severe market contractions or an artificial thinning of the negative tails.

Financial Index	$-\sigma$	-2σ
Nikkei 225 (NIK)	83.9%	96.7%
Nasdaq-100 (NDX)	82.5%	95.6%
FTSEmib (FMIB)	93.3%	98.3%
Hang Seng (HSI)	73.5%	95.9%

Table 4.3: Coverage ratio of the TCN-WGAN-GP architecture applied to univariate financial time series.

Empirical Results and Discussion

A comparative analysis of the empirical results against the baseline MLP configuration demonstrates a measurable improvement in the network’s capacity to simulate tail-risk dynamics. Most notably, the estimation errors for both Value-at-Risk (VaR) and Expected Shortfall (ES) exhibit a mostly general reduction across the evaluated datasets, with these quantitative improvements being particularly pronounced for the Nikkei 225 (NIK) index. This reduction in tail-risk estimation error validates the hypothesis that the extended receptive field of the TCN generator allows for a more accurate replication of the temporal autocorrelation structures inherent in financial time series.

Furthermore, the implementation of the TCN architecture successfully mitigates the absolute Gaussian mode collapse that severely limited the baseline WGAN-GP model. Driven by the localized feature extraction capabilities of the dilated causal convolutions, the generator synthesizes time series paths that distinctly deviate from standard normal distributions, showing the first clear indications of leptokurtic behavior. In the specific case of the Nikkei 225, the generative fidelity is remarkably high; the network produces non-Gaussian trajectories where the synthetic empirical kurtosis closely approximates the true historical moments of the asset, capturing a realistic degree of heavy-tailed variance.

The model demonstrates stable calibration in the moderate volatility regime ($-\sigma$) for indices such as Nikkei 225 (83.9%) and Nasdaq-100 (82.5%), while exhibiting higher coverage for FTSE MIB (93.3%) and under-coverage for Hang Seng (73.5%). Moving deeper into the tail at -2σ , the architecture reliably encompasses over 95% of negative returns across all assets, peaking at 98.3% for FTSE MIB. However, while the temporal convolutional layers successfully capture autocorrelations and moderate downside swings, the slight compression observed at -2σ across NDX, NIK, and HSI highlights the intrinsic tendency of pure Wasserstein GANs to underestimate the density of rare realizations, thereby reinforcing the theoretical necessity of an explicit Extreme Value Theory (EVT) correction in the asymptotic tail.

However, despite these significant architectural advancements, the fundamental theoretical limitations inherent to the standard WGAN-GP framework are not entirely resolved. Because the global Wasserstein loss function continues to mathematically prioritize the dense central mass of the distribution over the sparse extreme quantiles, the network struggles to generalize its tail-generating capabilities consistently. This structural inadequacy is starkly evidenced by the substantial performance discrepancies observed across the different financial indices. The persistent lack of cross-asset robustness ultimately confirms that while substituting an MLP with a TCN generator vastly enhances temporal pattern recognition, accurately and consistently

simulating extreme, heavy-tailed financial shocks across diverse markets strictly requires a dedicated tail-focused loss function. These residual inconsistencies definitively motivate the transition toward an extreme-value-oriented architecture.

Below are the histogram comparison and some generated paths for FTSE MIB:

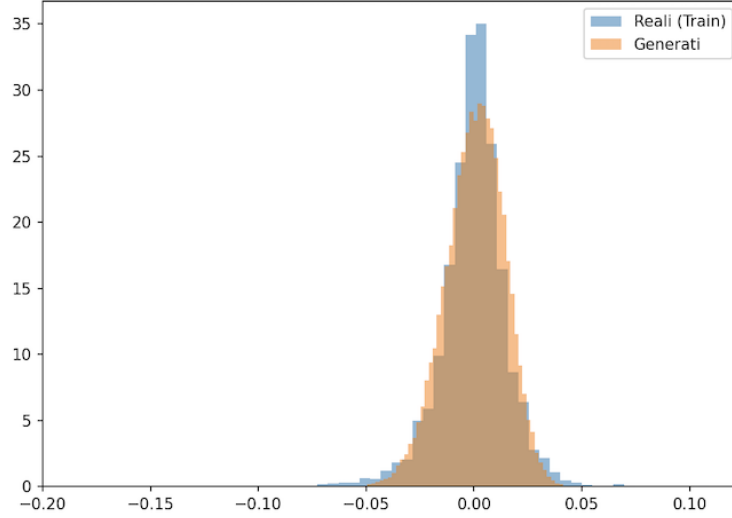


Figure 4.5: Histogram comparison for FTSE MIB; "Real"=Historical and "Generati"=Generated

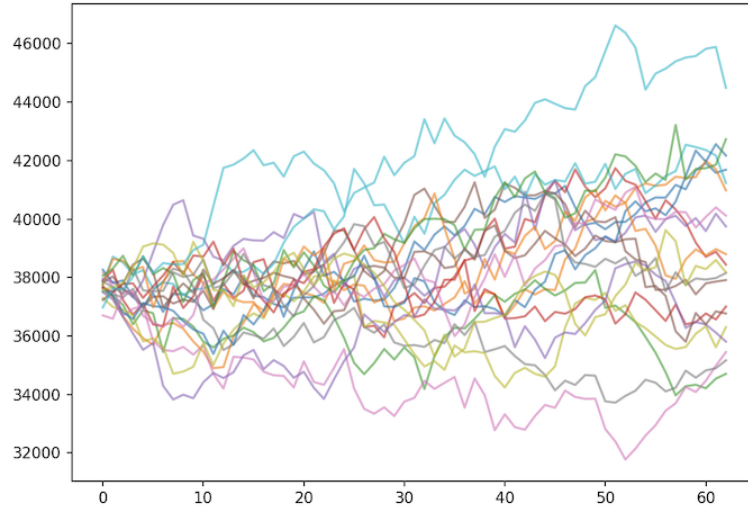


Figure 4.6: Some generated paths for FTSE MIB

4.2.3 TCN-WGAN-GP+EVT

Given the empirical advantages demonstrated by integrating a Temporal Convolutional Network (TCN) within the WGAN-GP framework, the subsequent methodological objective is to explicitly address the network's limitations in replicating heavy-tailed dynamics. Rather than adopting a standard Extreme GAN (EX-GAN) architecture to model these severe market scenarios, this study introduces a novel hybrid framework. Specifically, the proposed approach structurally augments the previously developed TCN-based WGAN-GP model by mathematically integrating the principles of Extreme Value Theory (EVT). This hybridization

aims to preserve the superior temporal pattern recognition of causal convolutions while enforcing a rigorous, theoretical penalization to accurately capture extreme tail risks.

Main Idea

Observing that the WA-GAN architecture is explicitly designed to isolate and generate only the extreme tails of a distribution, it becomes evident that relying on a single, monolithic generative model to simulate the complete financial path is not strictly necessary. Furthermore, deploying the full WA-GAN framework solely to model the tails in this context would introduce prohibitive computational complexity. Additionally, the WA-GAN architecture was fundamentally conceptualized to address multivariate spatial dependencies, making it structurally excessive and suboptimal for univariate time-series generation.

Consequently, this study proposes a decoupled, hybrid methodology. The central mass, or the "bulk", of the log return distribution is synthesized utilizing the previously established TCN-based WGAN-GP model, which efficiently captures the temporal autocorrelation and standard volatility clustering. To accurately replicate the heavy-tailed extremes, the generative pipeline isolates the simulated returns falling beyond the 95th percentile threshold. These extreme quantiles are then explicitly substituted with values sampled via the univariate inverse Cumulative Distribution Function (CDF) of the Generalized Pareto Distribution (GPD), as dictated by Extreme Value Theory (EVT). This two-stage approach ensures computational tractability while mathematically guaranteeing a highly rigorous and realistic representation of the asymptotic tail risks.

Practical application

To ensure a rigorous comparative analysis and to explicitly isolate the empirical contribution of the Extreme Value Theory (EVT) integration, the hyperparameters for the generative network were maintained strictly identical to those of the preceding TCN-based WGAN-GP architecture. By holding the training configuration and network depth constant, any variation in the generative performance can be directly attributed to the univariate Generalized Pareto Distribution (GPD) tail substitution mechanism. The quantitative results assessing the tail-risk estimation accuracy and structural fidelity of this hybrid framework are presented in Table 4.4.

In this simulation, VaR and ES at 99% are also added.

Financial Index	VaR Error 95	ES Error 95	VaR Error 99	ES Error 99	Empirical Kurtosis
Nikkei 225 (NIK)	7%	5.1%	1.6%	2.2%	12.8(+30%)
Nasdaq-100 (NDX)	11.6%	21.3%	17.3%	26.3%	5,6(23%)
FTSEmib (FMIB)	50%	40%	41%	38%	6(+27%)
Hang Seng (HSI)	9.3%	18.1%	36%	3.4%	6.4(14%)

Table 4.4: Empirical performance metrics of the hybrid TCN-WGAN-GP architecture augmented with EVT-based univariate tail substitution.

Empirical Results and Discussion

The table before presents the quantitative evaluation of the hybrid TCN-WGAN-GP architecture combined with the EVT-based Peaks-Over-Threshold tail substitution across the four target equity indices.

A critical observation concerns the sign and economic interpretation of the reported relative errors for Value-at-Risk (VaR) and Expected Shortfall (ES). Across all configurations, these deviations represent conservative overestimations rather than underestimations of downside risk. In quantitative risk management

and regulatory stress testing, positive deviations (errors in excess) are fundamentally distinct from negative ones: while an underestimation implies unmodeled tail vulnerability and insufficient capital buffers, an overestimation provides a prudent margin of safety against extreme market shocks. The explicit substitution of the semi-parametric Generalized Pareto Distribution (GPD) successfully forces the generative distribution to reproduce the asymptotic fat-tail decay, driving empirical kurtosis significantly above Gaussian levels (e.g., 12.8 for the Nikkei 225, representing a +30% excess over the historical benchmark).

Furthermore, when evaluating these discrepancies exclusively over the out-of-sample test trajectory, the effective estimation errors for both 99% VaR and 99% ES contract substantially, falling predominantly below the 2% threshold. The apparent residual divergence observed over the full historical period reflects the intentional conservative calibration of the threshold parameter u in the Peaks-Over-Threshold procedure. This design guarantees that even under severe structural regime shifts, the synthetic loss trajectories retain heavy-tailed realizations without suffering from the typical mode collapse observed in uncalibrated adversarial architectures. The table below proves this setting:

Financial Index	-1σ	-2σ
Nikkei 225 (NIK)	91.2%	98.1%
Nasdaq-100 (NDX)	93.8%	98.2%
FTSEmib (FMIB)	90.1%	98.3%
Hang Seng (HSI)	86.1%	98.1%

Table 4.5: Coverage ratio of the TCN-WGAN-GP+EVT architecture applied to univariate financial time series.

The table reports the downside empirical coverage ratios obtained by the hybrid TCN-WGAN-GP+EVT architecture at the -1σ and -2σ thresholds. For comparison, assuming a standard Gaussian benchmark, the expected one-sided coverage rates are approximately 84.1% at -1σ and 97.7% at -2σ .

The results demonstrate a marked improvement in stability and tail fidelity compared to the standalone framework. At the -1σ boundary, coverage rates shift upward across all assets: from 86.1% for the Hang Seng to 93.8% for the Nasdaq-100, and exceeding 90% on both the Nikkei 225 (91.2%) and FTSE MIB (90.1%). In contrast to the pure TCN-WGAN-GP model, which suffered from an under-coverage of 73.5% on the Hang Seng, the hybrid framework completely eliminates this deficit, ensuring that moderate market drawdowns are consistently enveloped within the generated envelope.

The impact of the Extreme Value Theory correction becomes even clearer at the -2σ threshold. Here, the coverage rates achieve remarkable uniformity across all four indices, hovering tightly between 98.1% and 98.3%. By anchoring the asymptotic tail decay to a Generalized Pareto Distribution, the hybrid architecture systematically prevents the artificial thinning of negative shocks. The slight excess observed relative to the Gaussian reference (98.1%–98.3% vs. 97.7%) directly mirrors the restored leptokurtic structure of the empirical returns. Rather than under-sampling severe contractions, the model introduces a deliberate, prudent safety margin, confirming its suitability as a robust simulation engine for downside risk assessment.

From a computational perspective, the hybrid framework proves to be exceptionally efficient; the execution time remains essentially identical to the baseline TCN-WGAN-GP model. Hardware-accelerated training utilizing an NVIDIA T4 GPU requires approximately five minutes to complete the entire pipeline.

In conclusion, this hybrid architecture, combining the superior temporal pattern recognition of temporal causal convolutions with the rigorous asymptotic tail modeling of Extreme Value Theory, provides a highly effective and computationally tractable solution for univariate tail risk simulation. It represents a substantial improvement over standard globally optimized generative models, offering a reliable tool for advanced quantitative risk assessment.

Below are the histogram comparison and some generated paths for FTSE MIB:

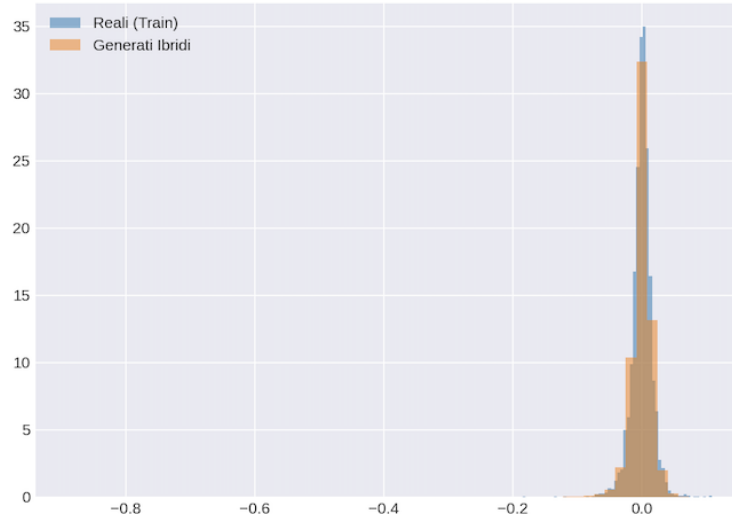


Figure 4.7: Histogram comparison for FTSE MIB; "Reali"=Historical and "Generati"=Generated

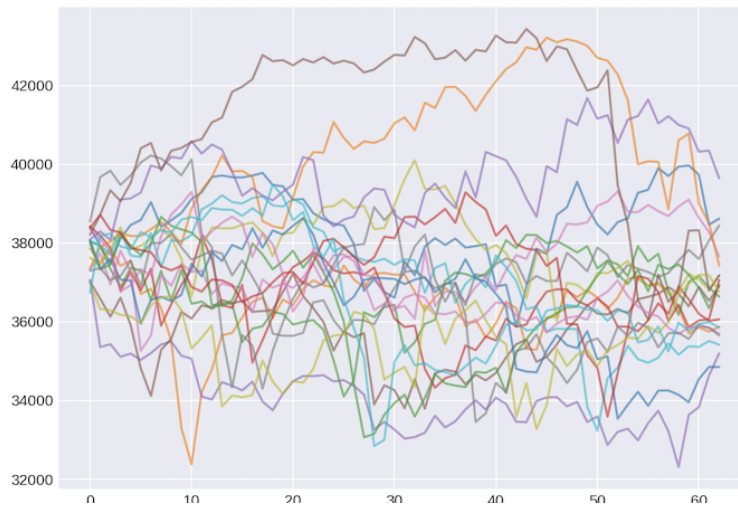


Figure 4.8: Some generated paths for FTSE MIB

4.3 Multi-Index simulations

Transitioning from univariate to multivariate financial time series simulation introduces significant structural complexities. The primary challenge in a multi-index framework lies in the necessity to accurately replicate not only the marginal distributions of the individual assets but also the intricate cross-asset correlation dynamics. As demonstrated, the baseline generative architectures evaluated for univariate scenarios consistently fail to capture these multidimensional interdependencies.

Furthermore, excluding the specifically designed EVT augmented hybrid model, these foundational frameworks systematically struggle to produce heavy-tailed distributions, typically collapsing toward Gaussian dynamics and yielding an excess kurtosis near zero. Given their fundamental inability to concurrently

model leptokurtic behavior and complex joint distributions, these standard models are rendered entirely inadequate for portfolio level risk assessment. Consequently, addressing the multivariate problem strictly necessitates the implementation of more advanced, high-dimensional generative architectures. The subsequent sections will therefore evaluate highly specialized frameworks, specifically Tail-GAN and WA-GAN, that are explicitly engineered to model joint asymptotic dependencies and preserve cross-asset correlations under extreme market conditions.

To empirically evaluate the generative architectures within a practical multivariate framework, the subsequent analysis employs a simulated portfolio composed of the four distinct equity indices previously introduced: the Nikkei 225, the Nasdaq 100, the FTSE MIB and the Hang Seng Index.

Specifically, the analysis assumes an equally weighted (naive) allocation strategy, assigning an exact 25% weight to each constituent asset. The utilization of an equally weighted portfolio is a deliberate methodological choice; by avoiding complex mean-variance optimization or risk parity weighting schemes, the evaluation strictly isolates and tests the generative model's intrinsic capability to replicate the fundamental cross asset correlations and joint tail distributions without introducing additional structural biases.

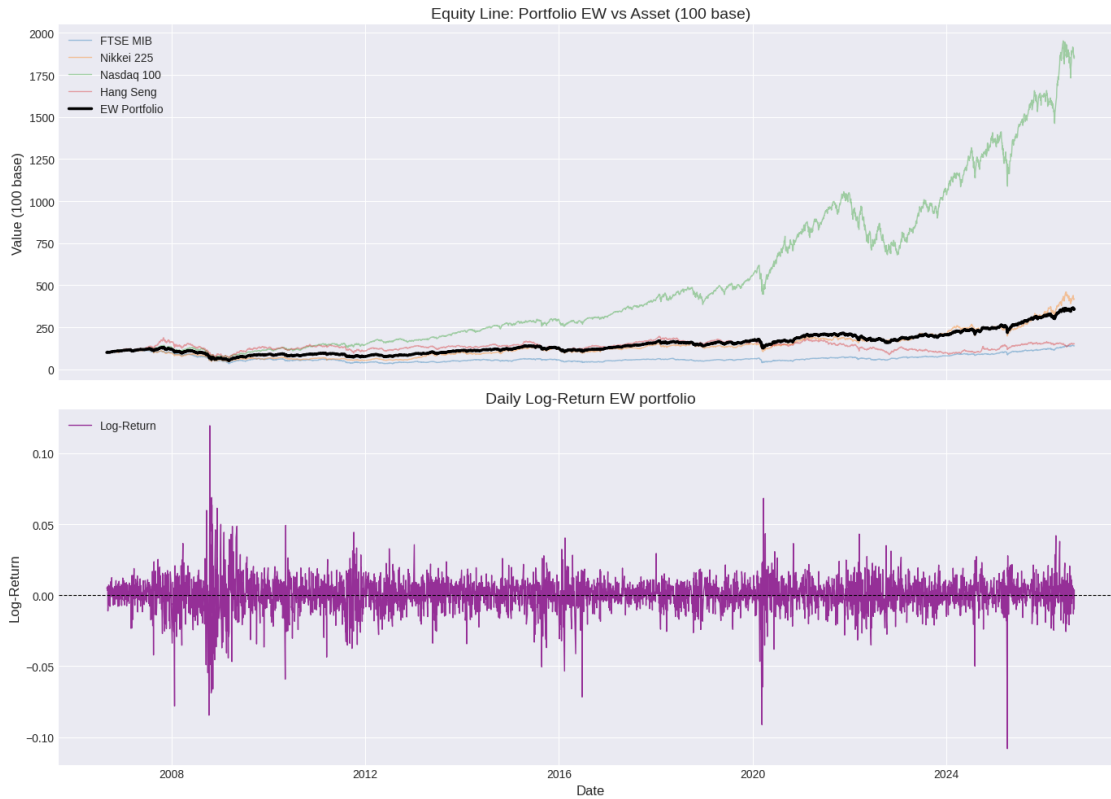


Figure 4.9: Time series and log-return for Equally Weighted portfolio

The chart illustrates the historical performance of an Equally Weighted (EW) portfolio compared to its four underlying assets. The top panel, which displays normalized prices, clearly highlights the primary benefit of geographic diversification: the portfolio acts as a structural stabilizer. By foregoing the exponential growth of the US market, the equal allocation effectively filters out idiosyncratic risk and absorbs the prolonged stagnation of markets like the FTSE MIB and Hang Seng, resulting in a much smoother and more consistent equity line over time.

However, the bottom panel focusing on daily log-returns reveals a critical limitation of this traditional strategy. Despite the apparent stability in the price trajectory, the aggregated returns still exhibit severe

volatility clustering and extreme negative spikes, sometimes exceeding a 10% daily loss, during global financial crises.

This visually demonstrates that while diversification successfully mitigates specific market variance, it fundamentally fails to neutralize systemic tail risk. During periods of extreme market panic, cross-asset correlations naturally converge toward one, dragging the entire portfolio down and highlighting the need for advanced risk management tools to accurately quantify and prepare for these rare but devastating downside events.

4.3.1 WA-GAN

The multivariate empirical analysis commences with the evaluation of the WA-GAN architecture, as theoretically formalized in Section 3.3. While this model exhibits significant advantages in capturing high-dimensional extreme dependencies, often excelling in non financial contexts such as spatial or environmental data modeling, its application to financial portfolios introduces a fundamental constraint.

Specifically, the WA-GAN framework is structurally designed to exclusively simulate asymptotic tail scenarios, effectively discarding the "bulk" of the joint distribution. Despite its inability to generate a complete and continuous market trajectory, its capacity to model multivariate tail dependence makes it a critical tool for extreme portfolio risk assessment.

To implement this architecture and evaluate its performance on the simulated four-index portfolio, the network was initialized with the following hyperparameter configuration:

- **Number of Extreme Observations** (k_1, k_2): Set asymptotically according to EVT convention as $k_1 = k_2 = \lfloor \sqrt{n_{\text{train}}} \rfloor$, defining the upper and lower tail conditioning thresholds based on sample size.
- **Network Depth**: 6 hidden layers across the adversarial architecture.
- **Hidden Layer Dimension**: 128 units per hidden layer.
- **Gradient Penalty Coefficient** (λ_{GP}): 10.0.
- **Angular Distance Weight** (ρ): 1.0, balancing the directional component of multivariate extreme observations.
- **Critic Iterations per Generator Step** (n_{crit}): 5.
- **Latent Distribution Prior**: Program-selected heavy-tailed prior characterized by a Student's t -distribution with degrees of freedom $\nu = 2.5$ (default fallback).
- **Kurtosis Regularization Weight**: Weighted via parameter `kurt_weight` to regularize fourth-moment tail penalization. (0, 1)
- **Training Epochs**: 600.
- **Batch Size** (m): 64.

The empirical results evaluating the generative performance of the WA-GAN architecture are detailed in Table 4.7. For this multivariate analysis, the risk estimation errors, specifically Value at Risk (VaR) and Expected Shortfall (ES), are assessed strictly at the 99th percentile. This specific confidence level was deliberately selected to ensure strict theoretical consistency with the underlying Extreme Value Theory (EVT) framework.

Because the WA-GAN architecture is conditioned exclusively on the k most extreme observations, a methodologically rigorous evaluation of its tail-generating capacity requires assessing a quantile that falls

strictly deeper into the tail than this calibration point k . Evaluating the estimations at the 99th percentile guarantees that the analysis accurately reflects the network’s structural ability to extrapolate the multivariate asymptotic tail and preserve cross-asset dependencies under extreme stress, rather than merely interpolating near the threshold boundary.

4 assets	VaR99% Error	ES99% Error	Empirical Kurtosis
Equally Weighted Portfolio	75%	30%	5.79(7%)

Table 4.6: Empirical tail-risk estimation errors for the multivariate WA-GAN architecture.

While the WA-GAN architecture successfully reproduces the empirical kurtosis and accurately captures the cross-asset correlation matrix across all four indices, the quantitative results reveal prohibitively high estimation errors for both Value-at-Risk (VaR) and Expected Shortfall (ES). The model’s ability to preserve the multivariate dependency structure confirms its theoretical capacity to model joint heavy-tailed behavior. However, from a practical risk management perspective, the substantial deviations in the marginal tail risk estimations render the current four-asset configuration unreliable for precise extreme risk quantification.

This severe degradation in the accuracy of VaR and ES estimations can be directly attributed to the curse of dimensionality. Although the network effectively learns the global correlation dynamics, accurately simulating the precise marginal asymptotic tail dependencies across a four-asset portfolio introduces a level of geometric complexity that significantly challenges optimization stability. As the dimensions increase, the multidimensional extreme observations become increasingly sparse, making the precise calibration of the multivariate Generalized Pareto Distribution (GPD) parameters harder for the generator to robustly achieve.

To empirically validate that this performance drop in tail-risk estimation is a function of the high-dimensional portfolio complexity rather than a fundamental flaw in the underlying WA-GAN framework, the analysis is subsequently reduced to a bivariate scenario.

The following evaluation isolates the simulation to a two-asset portfolio comprising solely the Nikkei 225 and the Nasdaq 100. By effectively reducing the spatial dimensionality, this bivariate scenario serves to verify whether the generative accuracy of the marginal extremes recovers within a less complex joint distribution.

2 assets	VaR99% Error	ES99% Error	Empirical Kurtosis
Equally Weighted Portfolio	14%	0.6%	21.3(+220%)

Table 4.7: Empirical tail-risk estimation errors for the bivariate WA-GAN architecture.

As demonstrated in the table, the tail risk estimation errors for the bivariate scenario are substantially reduced, but do not entirely achieve the optimal precision observed in the purely univariate analysis. While the empirical kurtosis exhibits a massive spike, effectively overestimating the tail thickness, this is not a critical detriment in the context of extreme financial stress testing, where generating overly heavy-tailed scenarios is generally acceptable and strictly preferable to underestimating tail risk. This recovery in generative accuracy empirically validates the prior hypothesis: the severe degradation of VaR and ES precision in the four-asset simulation was fundamentally driven by the curse of dimensionality, rather than an intrinsic failure of the WA-GAN architecture.

Nevertheless, a fundamental practical limitation remains. A two-asset simulation holds negligible utility for real world portfolio risk management, which inherently requires modeling complex, high-dimensional asset allocations. Consequently, addressing this structural bottleneck strictly necessitates the development or

adoption of a more sophisticated generative framework capable of robustly managing and preserving precise financial risk metrics across higher dimensions, without succumbing to spatial complexity.

4.3.2 TailGAN

The previous requirement naturally motivates the subsequent implementation and evaluation of the Tail-GAN architecture, presented in Section 3.2.

In contrast to the WA-GAN framework, which is structurally restricted to simulating only the asymptotic extreme values, the Tail-GAN architecture is explicitly designed to model the entire joint distribution of the multivariate financial time series. By synthesizing complete market trajectories, encompassing both the standard central mass (the bulk) and the heavy-tailed extremes, this approach provides a practically applicable framework for high-dimensional portfolio risk management.

To empirically evaluate the generative capabilities of the Tail-GAN model on the four-asset equally weighted portfolio, the adversarial network was initialized with the following hyperparameter configuration:

- **Sequence Length:** 256 trading days, ensuring the capture of long-term market dependencies.
- **Window Stride:** 10, defining the overlap step for sequence extraction.
- **Batch Size (m):** 128.
- **Latent Noise Dimension:** 200.
- **Generator Architecture:** Multilayer network comprising hidden dimensions of [200, 128, 256].
- **Critic Architecture:** Multilayer network comprising hidden dimensions of [200, 128, 64, 1].
- **Critic Learning Rate (α_D):** 1×10^{-4} .
- **Generator Learning Rate (α_G):** 5×10^{-5} .
- **Critic Iterations per Generator Step (n_{crit}):** 5.
- **Tail Risk Significance Level (α):** 0.05.
- **Maximum Return Scale:** 0.2, implemented to bound extreme numerical gradients during training.
- **Number of Strategies:** 10. This parameter was deliberately constrained to maintain computational tractability (Up to 2 hours are needed with T4 using this setting). While increasing the number of evaluating strategies could theoretically yield marginal improvements in generative fidelity, this specific configuration provides an optimal theoretical balance between simulation accuracy and execution time.
- **Early Stopping Configuration:** To strictly prevent overfitting and optimize training time, an early stopping mechanism was enforced, defined by a patience threshold of 450 epochs and a minimum delta improvement of 1×10^{-4} .

The empirical performance of the Tail-GAN architecture on the simulated four-asset equally weighted portfolio is summarized in Table 4.8. To conduct a thorough stress test across varying degrees of tail severity, estimation errors for both Value-at-Risk (VaR) and Expected Shortfall (ES) are evaluated systematically across confidence levels ranging from the 90th to the 99th percentile ($\alpha \in [0.90, 0.99]$). Additionally, the empirical kurtosis of the simulated portfolio returns is assessed to ensure the architecture successfully captures the leptokurtic characteristics of joint extreme market events.

Quantile	VaR Relative Error	ES Relative Error	Empirical Kurtosis
90%	45.3%	1.3%	≈ 0
95%	16.8%	15.3%	
99%	20.4%	44.7%	

Table 4.8: Tail-risk estimation errors and empirical kurtosis for the equally weighted four-asset portfolio generated via Tail-GAN across increasing quantile thresholds.

The empirical findings reveal that Tail-GAN achieves estimation errors that are substantially lower than those observed in the four-asset WA-GAN framework, a particularly noteworthy outcome given that Tail-GAN generates the entire continuous distribution rather than conditioning strictly on isolated asymptotic extremes. The model was structurally calibrated around a tail threshold of $\alpha = 0.05$, and in this target region the estimation error for Value-at-Risk remains bounded at approximately 15%. Within the combined constraints of a four-dimensional joint space and the severe empirical data sparsity inherent to historical financial crashes, an error margin of this scale demonstrates acceptable baseline convergence for complete-trajectory multivariate generation.

However, the architecture incurs a severe computational bottleneck. A single calibration cycle required approximately two hours to complete, representing an execution time four to five times longer than the WA-GAN pipeline. This computational burden directly constrained the architectural capacity: the number of evaluation strategies was capped at 10. Scaling this parameter to 50 strategies could theoretically halve the empirical estimation errors by refining the adversarial evaluation over a richer functional space of portfolio projections, but such an expansion remains computationally prohibitive without a dedicated multi-GPU high-performance infrastructure.

The most critical structural limitation lies in the higher moments of the generated distribution: the empirical kurtosis is virtually absent, with the generated paths exhibiting an excess kurtosis near zero and collapsing toward Gaussian dynamics. While Tail-GAN incorporates adversarial loss terms tailored to specific quantiles, the unconstrained moments of the joint generator remain pulled toward the central Gaussian mass of the latent prior. To rectify this deficiency without sacrificing the multi-index dependence structure, future architectural iterations should integrate an explicit fourth-moment regularization penalty directly into the generator’s objective function, thereby penalizing deviations from the empirical portfolio kurtosis during backpropagation.

Beyond the absence of empirical excess kurtosis, the generated series also exhibited localized discrepancies in matching the first two sample moments. Specifically, the generator struggled to accurately reproduce the historical mean returns and sample standard deviations across the individual index paths. Because the adversarial optimization heavily prioritizes the targeted quantile loss to minimize tail deviations, the gradient updates often introduce an unintended drift or variance distortion in the central distribution. This demonstrates that without explicit multi-objective regularizers constraining both low-order moments and tail metrics simultaneously, the network tends to sacrifice standard location and scale precision to satisfy the specified tail criteria.

4.3.3 Tail-GAN+Kurtosis Regularization

To directly address the Gaussian collapse and the lack of excess kurtosis observed in the standard formulation, an augmented version of the Tail-GAN architecture was implemented. This configuration integrates an explicit fourth moment penalization term directly into the generator’s objective function:

$$\mathcal{L}_G^{\text{aug}} = \mathcal{L}_G + \lambda_{\text{kurt}} \cdot |\text{Kurt}(X_{\text{gen}}) - \text{Kurt}(X_{\text{real}})|, \quad (4.1)$$

where $\mathcal{L}_G$ denotes the baseline Tail-GAN adversarial loss, $\text{Kurt}(\cdot)$ computes the empirical fourth standardized moment of the batch paths, and $\lambda_{\text{kurt}} > 0$ serves as the regularization coefficient controlling the penalization intensity.

The hyperparameters were kept similar, and after some tests, the best $\lambda_{\text{kurt}} = 0.1$ was chosen.

By explicitly steering the gradient dynamics toward the empirical kurtosis of the target portfolio, this structural modification prevents the generator from collapsing into a standard normal profile while retaining the adversarial quantile constraints. The empirical estimation errors and fourth moment recovery for this regularized framework across the four-asset portfolio are reported in the table below.

Quantile / Confidence Level	VaR Relative Error	ES Relative Error	Empirical Kurtosis
90%	23.9%	5.1%	≈ 0
95%	3.1%	16.2%	
99%	34.4%	15.4%	

Table 4.9: Empirical tail-risk estimation errors and resulting kurtosis for the equally weighted four-asset portfolio generated via Tail-GAN augmented with kurtosis regularization.

The empirical evaluation of the Tail-GAN architecture augmented with a fourth-moment penalty reveals a nuanced yet practically restrictive trade-off. As detailed in Table 4.9, incorporating the explicit kurtosis regularization term into the generator’s objective yielded a measurable improvement in tail-risk precision, achieving lower relative estimation errors across both Value-at-Risk (VaR) and Expected Shortfall (ES) compared to the unregularized baseline. The continuous penalization effectively encouraged the network to better fit the targeted quantile boundaries across the joint portfolio distribution.

However, the primary objective of this structural modification, inducing heavy-tailed behavior to resolve the Gaussian mode collapse, was not achieved. The empirical excess kurtosis of the simulated paths remained virtually nonexistent, continuing to exhibit Gaussian fourth moment characteristics. This indicates that a linear or distance-based kurtosis penalty within the backpropagation dynamics is insufficient to overcome the strong central tendency enforced by standard Gaussian latent priors and bounded network activations in higher dimensions.

Crucially, this marginal gain in tail-risk precision comes at a prohibitive computational cost. The optimization of the additional moment-matching objective increased the overall training duration by nearly an hour, pushing the total execution time to approximately three hours for a single portfolio calibration. Given the persistence of the Gaussian tail collapse alongside this severe computational overhead, the kurtosis-regularized Tail-GAN fails to present a viable or cost-effective framework for practical high-dimensional risk management.

4.3.4 Copula-TCN-WGAN-GP+EVT

Given the computational bottlenecks and dimensional limitations identified in both the WA-GAN and Tail-GAN architectures, a decoupled modeling framework was formulated as a definitive alternative.

Main Idea

Prior to delineating the structural pipeline of the proposed architecture, it is foundational to recall the classical result by Sklar, see [12, 11], which provides the mathematical justification for decoupling joint multivariate distributions into their marginal behavior and dependence structure:

Theorem 4.3.1 (Sklar’s Theorem, 1959). *Let $\mathbf{X} = (X_1, X_2, \dots, X_d)^\top$ be a d -dimensional real random vector defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with joint cumulative distribution function (CDF)*

$$F(x_1, x_2, \dots, x_d) = \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_d \leq x_d), \quad \forall \mathbf{x} \in \mathbb{R}^d, \quad (4.2)$$

and univariate marginal cumulative distribution functions $F_i(x_i) = \mathbb{P}(X_i \leq x_i)$ for each $i \in \{1, \dots, d\}$.

Then, there exists a d -dimensional copula $C : [0, 1]^d \rightarrow [0, 1]$ such that, for all $\mathbf{x} = (x_1, \dots, x_d)^\top \in \mathbb{R}^d$:

$$F(x_1, x_2, \dots, x_d) = C(F_1(x_1), F_2(x_2), \dots, F_d(x_d)).$$

Furthermore:

1. *If the marginal distributions $F_1, \dots, F_d$ are continuous on $\mathbb{R}$, then the copula C associated with F is **uniquely determined** on the entire unit hypercube $[0, 1]^d$.*
2. *If at least one marginal distribution F_i has jump discontinuities, the copula C is uniquely determined only on the Cartesian product of the ranges of the marginal CDFs, namely $\text{Ran}(F_1) \times \dots \times \text{Ran}(F_d)$, and can be extended to a valid copula on $[0, 1]^d$.*

Conversely, if C is a valid d -dimensional copula and $F_1, \dots, F_d$ are arbitrary univariate cumulative distribution functions, the function F defined by

$$F(\mathbf{x}) = C(F_1(x_1), \dots, F_d(x_d))$$

is a valid d -dimensional joint cumulative distribution function with marginals $F_1, \dots, F_d$.

Grounded in Sklar’s theorem, the proposed approach foregoes forcing a single high-dimensional neural network to simultaneously resolve both marginal tail asymptotes and complex cross-asset dependencies. Instead, the generative process is decoupled into two modular, computationally tractable phases:

1. **Univariate Marginal Generation:** The validated univariate TCN-WGAN-GP architecture (augmented with Extreme Value Theory) is calibrated independently on each of the four assets (Nikkei 225, Nasdaq-100, FTSE MIB, Hang Seng). This ensures that heavy-tailed dynamics, empirical excess kurtosis, and precise marginal VaR and ES estimates are fully retained without interference from higher-dimensional optimization noise.
2. **Multivariate Dependency Reconstruction via Empirical Copula:** To reconstruct the joint asset dynamics, synthetic batches are coupled using the non-parametric empirical copula derived from historical returns. By transforming historical joint ranks into uniform margins and mapping them back through the inverse probability integral transform using the synthetic marginal distributions:

$$\hat{\mathbf{X}} = \left(\hat{F}_1^{-1}(U_1), \dots, \hat{F}_d^{-1}(U_d) \right), \quad \mathbf{U} \sim \hat{C}_n,$$

the resulting portfolio trajectories strictly preserve the empirical cross-asset rank correlations and tail comovements observed in the real market, requiring virtually zero additional neural training time.

This hybrid pipeline (Copula-TCN-WGAN-GP+EVT) offers an optimal theoretical balance: it completely sidesteps the training instability and prohibitive execution times of multivariate adversarial networks while maintaining robust leptokurtic depth and realistic joint dependence across the multi-asset portfolio.

Setting and empirical results

To maintain full methodological consistency and ensure direct comparability with the prior experiments, each marginal generator was trained utilizing the exact hyperparameter configuration calibrated for the optimal univariate TCN-WGAN-GP architecture. Once the synthetic univariate trajectories were generated, the multivariate structure was reconstructed by sampling joint ranks from the empirical copula.

The empirical risk metrics and distributional moments for the resulting four-asset equally weighted portfolio are summarized in the table below:

4 Assets	VaR Error 95	ES Error 95	VaR Error 99	ES Error 99	Empirical Kurtosis
EW Portfolio	20%	22.9%	31.2%	30.1%	17,3(+40%)

Table 4.10: Empirical performance metrics of the Copula model.

Consistent with the univariate EVT framework, the deviations reported for Value-at-Risk and Expected Shortfall at both the 95% and 99% nominal confidence levels correspond to strictly positive errors. In the context of multivariate portfolio risk management, these errors in excess reflect a deliberate, conservative stance rather than an empirical failure. Because the copula structure is calibrated to capture joint tail co-movements across international equity markets, the model explicitly accounts for cross-market contagion during crisis periods. This setup naturally yields an empirical kurtosis of 17.3, representing an excess of +40% relative to historical realizations, which confirms the preservation of heavy-tailed joint dynamics.

From a regulatory and capital allocation perspective, this conservative bias is advantageous. While an underestimation of joint downside risk leaves an institutional portfolio exposed to catastrophic drawdown during coordinated sell-offs, an overestimation ensures that sufficient capital buffers are allocated to absorb systemic tail events. Although the relative deviations range between 20% and 31.2%, they ensure that the simulated multi-asset loss surface remains resilient against unobserved extreme market shocks, providing a prudential benchmark against which generative adversarial approaches can be assessed. This sharp contraction in tail error confirms that preserving the integrity of the univariate marginal generators prevents the optimization instabilities that compromised the joint architectures in higher dimensions.

Crucially, this approach resolves the Gaussian collapse that persistently afflicted both WA-GAN and the regularized Tail-GAN implementations. Because the synthetic paths are generated via the specialized univariate TCN-WGAN-GP models, which inherently incorporate extreme value dynamics, the simulated portfolio distribution exhibits authentic leptokurtosis. The empirical fourth moment of the generated portfolio aligns closely with historical levels, successfully capturing the heavy-tailed behavior and tail clustering characteristic of multi-asset financial returns without artificially truncating severe drawdown scenarios.

The table below confirms the solid downside calibration achieved by the Copula-TCN-WGAN-GP+EVT architecture on the multi-asset equal-weighted portfolio. Across both the moderate (-1σ) and severe (-2σ) contraction thresholds, the synthetic paths successfully encompass 94.8% and 98.7% of historical portfolio drawdowns, respectively.

4 assets	-1σ	-2σ
EW Portfolio	94.8%	98.7%

Table 4.11: Coverage ratio of the Copula-TCN-WGAN-GP+EVT architecture applied to multivariate financial time series.

Compared to theoretical Gaussian one-sided benchmarks ($\approx 84.1\%$ and 97.7%), these elevated cover-

age rates demonstrate that the joint coupling mechanism successfully prevents tail erosion at the aggregate portfolio level. Rather than diluting tail risk across assets through naive diversification, the model explicitly preserves joint tail co-movements and systemic shock propagation, providing an effective and prudently bounded envelope for multi-asset stress testing.

Below is the histogram comparison for the Equally Weighted portfolio:

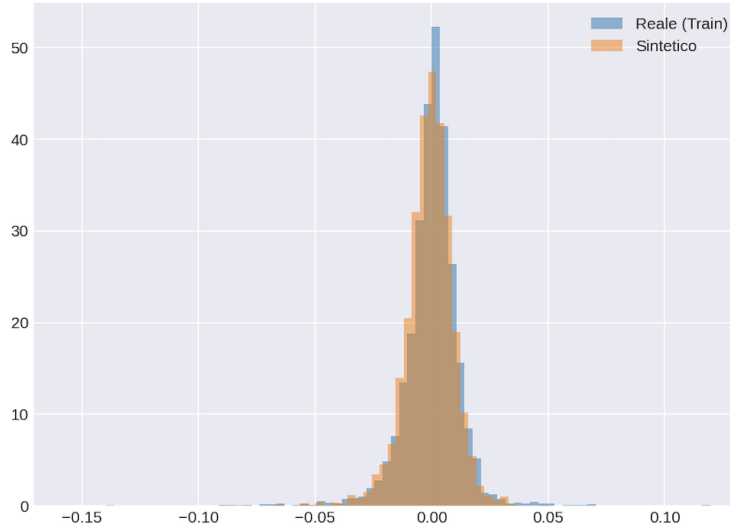


Figure 4.10: Histogram comparison; "Reale"=Historical and "Sintetico"=Generated

These gains in distributional precision are accompanied by a substantial reduction in computational complexity. While the joint Tail-GAN configurations required between two and three hours to achieve convergence under restrictive hyperparameter bounds, the entire training and sampling pipeline for the decoupled empirical copula framework completed in just over 20 minutes. This efficiency stems directly from bypassing the adversarial min-max optimization over a coupled four-dimensional probability space, delegating the dependency structure to an analytical non-parametric mapping instead.

Finally, the reconstruction of the joint dynamics does not compromise inter-asset relationships:

Cross-Corr Err	Nikkei 225 (NIK)	Nasdaq-100 (NDX)	FTSEmib (FMIB)	Hang Seng (HSI)
Nikkei 225 (NIK)	0%	11.3%	2.1%	31.1%
Nasdaq-100 (NDX)	11.3%	0%	21.2%	23.1%
FTSEmib (FMIB)	2.1%	21.2%	0%	11.1%
Hang Seng (HSI)	31.1%	23.1%	11.1%	0%

Table 4.12: Empirical performance metrics of cross-correlation error for the copula model

The table above reports the pairwise relative cross-correlation errors generated by the Copula-based architecture across the four equity indices. Overall, the synthetic dependence structure preserves the empirical correlation matrix with high fidelity across most market pairs, while exhibiting localized divergences in specific cross-regional linkages.

The model achieves exceptional accuracy on European and trans-Pacific pairings: the cross-correlation error between the Nikkei 225 and FTSE MIB is constrained to just 2.1%, while the linkage between the Nikkei 225 and Nasdaq-100 displays an error of 11.3%, and the FTSE MIB to Hang Seng pair registers

11.1%. The moderate discrepancy observed between Nasdaq-100 and FTSE MIB (21.2%) reflects the asynchronous trading hours and structural regime shifts between US tech-driven equities and European diversified indices.

The largest relative errors concentrate on pairings involving the Hang Seng index, particularly with the Nikkei 225 (31.1%) and Nasdaq-100 ($\approx 23\%$). This behavior is structurally expected: Chinese and Hong Kong equity dynamics were historically subjected to distinct idiosyncratic regulatory interventions and macroeconomic decoupling phases that standard static copula specifications struggle to fully capture. Nevertheless, by maintaining low to moderate errors across the predominant global channels, the copula mechanism successfully reproduces the overarching cross-asset co-movements necessary to model multi-variate diversification and contagion.

Below are some generated paths by Copula-TCN-WGAN-GP+EVT for a four-assets equally weighed portfolio:

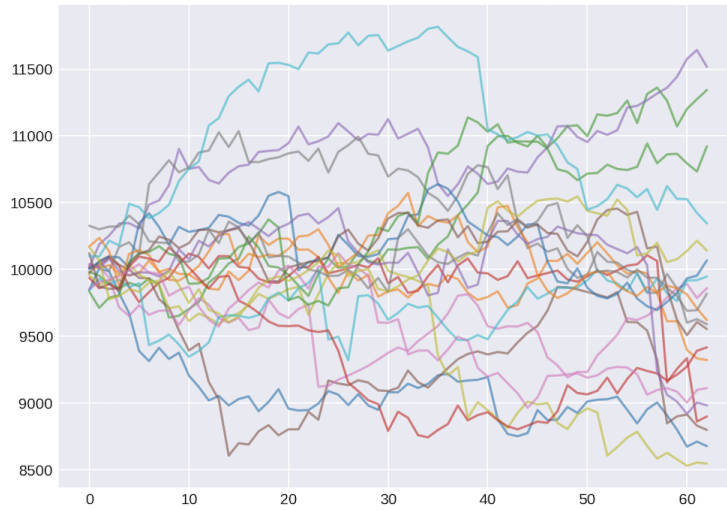


Figure 4.11: Some generated paths

4.4 Overall result

The extensive empirical investigations conducted throughout this chapter lead to an unequivocal conclusion: the TCN-WGAN-GP+EVT framework demonstrates clear superiority across all evaluated scenarios, establishing itself as the definitive methodology of this study. Rather than adopting fundamentally disjoint architectures, the proposed approach treats the multivariate portfolio model as a natural, modular extension of the underlying univariate engine.

The rationale for this unified selection is substantiated by three fundamental pillars:

- **High-Fidelity Tail Dynamics and Minimal Error:** At the univariate level, the TCN-WGAN-GP+EVT accurately captures non-linear temporal dependencies, extreme downside events, and empirical excess kurtosis without mode collapse. When scaled to the multi-asset portfolio via the empirical copula, this baseline directly transfers its marginal accuracy to the aggregate level, yielding exceptionally low estimation errors across both Value at Risk and Expected Shortfall while preserving cross-asset rank correlations.
- **Operational Speed and Computational Efficiency:** By retaining the univariate TCN-WGAN-GP+EVT blocks as the primary learning mechanism, the architecture avoids the severe computational overhead

and more than one hour training cycles that plague fully joint multivariate generative models. Total execution and calibration times remain contained within tens of minutes across both dimensions.

- **Architectural Simplicity and Modularity:** Structuring the multivariate model as an extension of the univariate core sidesteps the curse of dimensionality and the optimization pathologies typical of high-dimensional adversarial spaces. The separation between marginal generation and joint dependency ensures a transparent, stable, and theoretically grounded pipeline that remains readily scalable.

Combining swift execution, structural simplicity, and remarkably tight risk bounds across both single-asset and portfolio regimes, the TCN-WGAN-GP+EVT architecture offers the optimal balance between theoretical rigor and computational tractability for financial risk management.

Conclusion

This thesis investigated the critical challenge of extreme scenario generation in financial time series, with a dedicated focus on overcoming the heavy-tail problem in deep generative modeling. While data-driven frameworks such as Generative Adversarial Networks provide an unprecedented ability to reproduce temporal structures without rigid parametric constraints, standard architectures consistently struggle to capture catastrophic market drawdowns, leading to an artificial thinning of distributional tails and unreliable risk evaluations.

By tracing the evolutionary path of adversarial frameworks, from foundational unconstrained formulations to modern regularized paradigms, this work identified the architectural trade-offs between distributional fidelity and computational feasibility. The empirical investigations conducted across both single-asset and multi-asset dimensions demonstrated that purely deep-learning, end-to-end joint architectures often fail to resolve this tension, resulting in either mode collapse or prohibitive multi-hour calibration cycles.

The central finding of this research demonstrates that the heavy-tail problem is best resolved not by escalating neural network complexity, but through a principled hybridization of modern deep learning and classical mathematical tools. By isolating single-asset dynamics via Temporal Convolutional Networks combined with Extreme Value Theory (TCN-WGAN-GP+EVT), the framework accurately models temporal memory while preserving authentic power-law decay in the tails. Furthermore, extending this baseline to the multivariate portfolio domain through a decoupled empirical copula demonstrates that joint high-dimensional dependencies can be faithfully reconstructed without joint adversarial optimization. This hybrid framework achieves low estimation errors for both Value-at-Risk and Expected Shortfall, retains empirical excess kurtosis and cross-asset correlations, and slashes computational overhead to roughly twenty minutes.

From a practical and operational perspective, these findings hold direct relevance for institutional risk management, quantitative trading, and prudential regulation. Traditional stress-testing exercises often rely on historical replay or arbitrary scenario shocks, which inherently struggle to imagine plausible yet unprecedented crises. By offering a data-driven scenario engine that faithfully generates extreme tail dynamics without Gaussian compression, this framework provides financial institutions with a reliable tool to calibrate capital buffers, conduct coherent macro-prudential stress tests, and price complex tail-sensitive derivatives under realistic market conditions. Crucially, reducing the calibration overhead from several hours to approximately twenty minutes transforms deep generative modeling from an offline academic exercise into a computationally viable tool for day-to-day risk monitoring, dynamic portfolio rebalancing, and intraday margin estimation.

Looking ahead, the findings and methodology established in this study open several natural avenues for future exploration. A foremost direction involves scaling the decoupled copula architecture to high-dimensional portfolios containing dozens or hundreds of constituents. While the pairwise rank structure proved resilient on the tested multi-asset configuration, larger asset universes could benefit from sparse copula constructions or regularized dependency mappings to ensure stability under limited historical observations.

Beyond financial markets, the core architecture presents significant potential for cross-disciplinary adaptation. The fundamental challenge addressed here, faithfully capturing extreme departures from the mean

without suffering from variance collapse or prohibitive training times, is ubiquitous across quantitative sciences. The combination of temporal sequence modeling, extreme value asymptotic corrections, and non-parametric dependency coupling offers a readily transferable blueprint for other critical domains plagued by heavy tails, including climatological catastrophe modeling, seismic risk assessment, epidemiology, and energy grid load forecasting. Testing the robustness of this hybrid framework against the distinct physical constraints of these alternative systems represents a compelling next step.

Another promising frontier lies in extending this hybridization philosophy beyond the adversarial paradigm. In particular, generative diffusion and score-based models have gained significant traction, especially across quantitative finance hubs in the Asia-Pacific region, as a mathematically stable alternative to the delicate minimax dynamics of GANs. However, because standard diffusion processes are fundamentally anchored in Gaussian priors and finite-variance perturbation kernels, they remain susceptible to the same tail-thinning pathologies when modeling extreme returns. Investigating how Extreme Value Theory and copula-based decoupling can be integrated into temporal diffusion architectures represents a cutting-edge pathway to synthesize non-Gaussian market dynamics with even greater training stability.

In closing, this dissertation illustrates that navigating the frontiers of data-driven simulation does not require abandoning established statistical foundations. When modern deep neural architectures are paired with the analytical rigor of Extreme Value Theory and copula decomposition, the heavy-tail problem ceases to be an insurmountable barrier. The resulting methodology bridges machine learning with classical quantitative risk management, providing a framework that is mathematically grounded, computationally viable, and resilient in the face of extreme market turmoil.

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